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PAPER

Displayed type theory and semi-simplicial types

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Abstract

We introduce *Displayed Type Theory (dTT)*, a multi-modal homotopy type theory with *discrete* and *simplicial* modes. In the intended semantics, the discrete mode is interpreted by a model for an arbitrary ∞-topos, while the simplicial mode is interpreted by Reedy fibrant augmented semi-simplicial diagrams in that model. This simplicial structure is represented inside the theory by a primitive notion of *display* or *dependency*, guarded by modalities, yielding a partially-internal form of unary parametricity. Using the display primitive, we then give a coinductive definition, at the simplicial mode, of a type SST of semi-simplicial types. Roughly speaking, a semi-simplicial type X consists of a type X_0 together with, for each $x : X_0$, a displayed semi-simplicial type over X. This mimics how simplices can be generated geometrically through repeated cones, and is made possible by the display primitive at the simplicial mode. The discrete part of SST then yields the usual infinite indexed definition of semi-simplicial types, both semantically and syntactically. Thus, dTT enables working with semi-simplicial types in full semantic generality.

Keywords: semi-simplicial types, parametricity, type theory, category theory, diagram models

1. Introduction

Semi-simplicial types

Homotopy Type Theory (HoTT) Univalent Foundations Program (2013) is a perspective on intensional dependent type theory that regards types as homotopical spaces or ∞-groupoids. It has proven remarkably successful as a synthetic context in which to do homotopy theory and algebraic topology, and as an internal language for $(\infty, 1)$-toposes Shulman (2019). However, an enduring frustration has been its apparent inability to define general homotopy-coherent structures. Some infinite structures can be defined in HoTT, such as globular types and spectra; but others, such as A_∞-spaces or $(\infty, 1)$-categories, have so far resisted all attempts at definition. We know no convincing explanation for why they should be impossible, but the fact that all attempts appear to fail in a similar way suggests the operation of an as-yet-unarticulated principle.

Specifically, stating an *'infinite coherence'* property generally seems to require an infinite structure within which to assemble the coherences, while defining such a structure itself seems to require infinite coherence, leading to an infinite regress. This is in contrast to the situation in classical homotopy theory where the infinite structures to describe coherence, such as operads and simplicial diagrams, can themselves be defined using *strict* point-set-level equalities, which are then automatically fully coherent. It is tempting to try to mimic this in homotopy type theory using *definitional* equalities in place of point-set ones, but this is difficult because definitional equality is not reified in the theory and we have limited tools for forcing it to hold.

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One of the more flexible ways to enforce definitional equalities is to use type dependency, moving from a *fibred* perspective to an *indexed* one. In the simplest case, this means replacing a function $p : E \to B$ with a type family $\widetilde{E} : B \to \mathsf{Type}$. This has a corresponding projection function $\pi_1 : \Sigma\ \widetilde{E} \to B$, and we can suppose a point $\widetilde{e} : \Sigma\ \widetilde{E}$ with a definitional equality $\pi_1\ \widetilde{e} \equiv b$ by supposing a point $e : \widetilde{E}\ b$ and letting $\widetilde{e} = (b,\ e)$.

Thus, it is natural to try to define infinitely coherent structures that can be expressed in a purely indexed way. The example of this sort which has attracted the most attention is that of *semi-simplicial types*, because they are well-known within homotopy theory and could be used to encode many, if not all, other infinitely coherent structures. In indexed style, a semi-simplicial type consists of type families A_0, A_1, A_2, and so on, having types that start out as follows:

$$
\begin{aligned}
&A_0 : \mathsf{Type}\\
&A_1 : (x_{01} : A_0)\ (x_{10} : A_0) \to \mathsf{Type}\\
&A_2 : (x_{001} : A_0)\ (x_{010} : A_0)\ (\beta_{011} : A_1\ x_{001}\ x_{010})\ (x_{100} : A_0)\ (\beta_{101} : A_1\ x_{001}\ x_{100})\ (\beta_{110} : A_1\ x_{010}\ x_{100}) \to \mathsf{Type}\\
&A_3 : (x_{0001} : A_0)\ (x_{0010} : A_0)\ (\beta_{0011} : A_1\ x_{0001}\ x_{0010})\ (x_{0100} : A_0)\ (\beta_{0101} : A_1\ x_{0001}\ x_{0100})\ (\beta_{0110} : A_1\ x_{0010}\ x_{0100})\\
&\quad (\mathfrak{f}_{0111} : A_2\ x_{0001}\ x_{0010}\ \beta_{0011}\ x_{0100}\ \beta_{0101}\ \beta_{0110})\ (x_{1000} : A_0)\ (\beta_{1001} : A_1\ x_{0001}\ x_{1000})\ (\beta_{1010} : A_1\ x_{0010}\ x_{1000})\\
&\quad (\mathfrak{f}_{1011} : A_2\ x_{0001}\ x_{0010}\ \beta_{0011}\ x_{1000}\ \beta_{1001}\ \beta_{1010})\ (\beta_{1100} : A_1\ x_{0100}\ x_{1000})\ (\mathfrak{f}_{1101} : A_2\ x_{0001}\ x_{0100}\ \beta_{0101}\ x_{1000}\ \beta_{1001}\ \beta_{1100})\\
&\quad (\mathfrak{f}_{1110} : A_2\ x_{0010}\ x_{0100}\ \beta_{0110}\ x_{1000}\ \beta_{1010}\ \beta_{1100}) \to \mathsf{Type}
\end{aligned}
$$

First, we have a type A_0 of *points*. Second, we have for every two points $x_{01}, x_{10} : A_0$, a type $A_1\ x_{01}\ x_{10}$ of *lines* joining x_{01} and x_{10}. Third, for every three points $x_{001}, x_{010}, x_{100} : A_0$ and three lines $\beta_{011} : A_1\ x_{001}\ x_{010}$, $\beta_{101} : A_1\ x_{001}\ x_{100}$, and $\beta_{110} : A_1\ x_{010}\ x_{100}$, a type $A_2\ x_{001}\ x_{010}\ \beta_{011}\ x_{100}\ \beta_{101}\ \beta_{110}$ of *triangles* with the given boundary. The pattern continues with *tetrahedra*, which we may also, more technically, call 3-simplices. In general, A_n describes the type of n-simplices indexed by their boundaries.

Remark 1. The binary subscripts on variables above follow a scheme that we learned from Tim Campion, although related schemas have been rediscovered many times. The $2^{n+1} - 1$ simplices constituting an n-simplex and its boundary are labelled by the numbers from 1 to $2^{n+1} - 1$ written in binary, where the number of 1s in a binary number corresponds to the dimension of the simplex, and the binary numbers corresponding to the boundary of some simplex are obtained by replacing one or more of the 1s in its binary number by 0s. (As we will see later, the binary number 0 can also be regarded as denoting the unique (–1)-simplex in the boundary of an n-simplex in an *augmented* semi-simplicial set.)

We then list these simplices in the order given by these numbers. This ordering may seem somewhat curious, compared to the more naïve approach of listing all the 0-simplices, then all the 1-simplices, and so on; but it does retain the important property that all the simplices in the boundary of some simplex are listed before it (thus, for instance, it makes the semi-simplex category into an *'ordered direct category'*; see section 4.5.5). We will see later that this ordering is what arises most naturally in (co)inductive constructions of semi-simplicial sets (which was also Campion's motivation).

In addition to subscripting simplex variables by binary numbers according to this scheme, in this paper we will use different base *letters* to indicate the *dimension* of each variable. Thus for instance x_{001}, x_{010}, and x_{100} are all 0-simplices, while β_{011}, β_{101}, and β_{110} are all 1-simplices, and $\mathfrak{f}_{111}$ is a 2-simplex. We will not have much occasion to denote 3-simplices; for (–1)-simplices we use the Cyrillic letter $\mathfrak{z}$ *(ze)*. We may also use different letters from the same alphabets for simplices in different semi-simplicial types, e.g. $\gamma_{11} : B_1\ y_{01}\ y_{10}$ and $\delta_{11} : C_1\ z_{01}\ z_{10}$. ◁

One may think of the terms A_0, A_1, A_2, and so on as defining the fields of an infinite record type. Terms of this infinite record type SST are known as *semi-simplicial types*. Thus, the problem is to define a type of semi-simplicial types within homotopy type theory, continuing the above pattern. As a correctness criterion, one would expect that when interpreted in any $(\infty, 1)$-topos

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the type SST becomes a classifier of semi-simplicial objects. However, every attempt to internally encode the combinatorics that generate the type of A_n, as a function of n, seems to lead once again to an infinite regress.

In light of this situation, an alternative approach is to formulate more expressive type theories that can solve the problem of infinitely coherent objects. One such proposal is *Two-Level Type Theory (2LTT)* Annenkov et al. (2023), which introduces an *'outer level'* of *'exo-types'* that are not homotopy-invariant. The exo-types admit a strict exo-equality type, essentially reifying definitional equality, which can then be used analogously to classical point-set equality to define infinitely coherent structures. And by the results of Uskuplu (2023), 2LTT can be interpreted in any $(\infty, 1)$-topos, so its semantics are not significantly less general than ordinary HoTT, and the type SST defined in 2LTT does interpret to the correct classifier. However, although the exo-equality is assumed to satisfy Uniqueness of Identity Proofs, to keep type-checking decidable it cannot satisfy a reflection rule making exo-equalities into definitional equalities. Thus, it can be quite cumbersome to work with in practice.

Another proposal is *Simplicial Type Theory (STT)* Riehl and Shulman (2017), which changes perspective to view individual types as *simplicial* spaces, with additional primitives for manipulating the simplicial structure. One can then simply impose conditions on one of these *'simplicial types'* to make it represent (for instance) an $(\infty, 1)$-category. This suggests a *'synthetic'* approach to higher category theory analogous to ordinary HoTT's synthetic approach to homotopy theory, which is potentially quite powerful; and the results of Riehl and Shulman (2017); Weinberger (2022) imply that it can be interpreted in the category of simplicial objects in any $(\infty, 1)$-topos. However, the strength of the synthetic approach is also its weakness: because simplicial types are postulated rather than defined, what we can do with them is limited to what is expressed by the axiomatisation.

A coinductive definition of semi-simplicial types

In this paper we propose a third enhancement of homotopy type theory, called *Displayed Type Theory (dTT)*, in which it is possible to define and work with semi-simplicial types (and many other things). This type theory is inspired by the following idea for a *coinductive definition* of a type SST of semi-simplicial types:

Idea 2. A semi-simplicial type $A : \mathsf{SST}$ consists of:

- a type $\mathsf{Z}\ A$, of ***Z**ero simplices* of A, and
- for each 0-simplex $x : \mathsf{Z}\ A$ of A, a semi-simplicial type $\mathsf{S}\ A\ x$ *displayed over* A, called the ***S**lice* of A over x.

To see why this definition is correct, let us first try to understand what a general semi-simplicial type B displayed over A consists of. The first part of this answer is that B defines a type of *0-simplices of* B *displayed over* 0*-simplices of* A, i.e. a family $\mathsf{Z}^{\mathsf{d}}\ B : \mathsf{Z}\ A \to \mathsf{Type}$. Then, every displayed 0-simplex $z : \mathsf{Z}^{\mathsf{d}}\ B\ y$ over a 0-simplex $y : \mathsf{Z}\ A$ should define $\mathsf{S}^{\mathsf{d}}\ B\ y\ z$, a doubly displayed semi-simplicial type, for whatever this means.

Now, instead of working out the definition of a doubly dependent semi-simplicial type, let's circle back and think geometrically. Semi-simplicial types should have families of n-simplex types. If $A : \mathsf{SST}$, then we write that:

$$A_0 : \mathsf{Type}$$
$$A_0 \equiv \mathsf{Z}\ A$$

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is the type of 0-simplices in A. Similarly, if $B : \mathsf{SST}^{\mathsf{d}}\, A$ is semi-simplicial type displayed over A, then for $y : A_0$, we write:

$$B_0 : A_0 \to \mathsf{Type}$$
$$B_0\, y \equiv \mathsf{Z}^{\mathsf{d}}\, B\, y$$

for the type of 0-simplices of B displayed over the 0-simplex y of A. Putting this together, if we have two 0-simplices $x_{00}\ x_{10} : A_0$ of A, then we may form:

$$A_1 : (x_{01} : A_0)\ \ (x_{10} : A_0) \to \mathsf{Type}$$
$$A_1\, x_{01}\, x_{10} \equiv \mathsf{Z}^{\mathsf{d}}\ \ (\mathsf{S}\, A\ \ x_{01})\ \ x_{10},$$

which is the type of 1-simplices in A joining x_{01} to x_{10}.

It therefore stands to reason that B should have a type of dependent 1-simplices living over the 1-simplices of A. Thus if $\beta_{11} : A_1\, y_{01}\, y_{10}$, then given dependent endpoints $z_{01} : B_0\, y_{01}$ and $z_{10} : B_0\, y_{10}$, we should get a type $B_1\, y_{01}\, z_{01}\, y_{10}\, z_{10}\, \beta_{11}$. The formula for this happens to take the following form:

$$B_1 : \left(y_{01} : A_0\right)\ \ \left(z_{01} : B_0\, y_{01}\right)\ \ \left(y_{10} : A_0\right)\ \ \left(z_{10} : B_0\, y_{10}\right)\ \ \left(\beta_{11} : A_1\, y_{01}\, y_{10}\right) \to \mathsf{Type}$$
$$B_1\, y_{01}\, z_{01}\, y_{10}\, z_{10}\, \beta_{11} \equiv \mathsf{Z}^{\mathsf{dd}}\ \left(\mathsf{S}^{\mathsf{d}}\, B\, y_{01}\, z_{01}\right)\, y_{10}\, z_{10}\, \beta_{11},$$

which mirrors the formula for A_1.

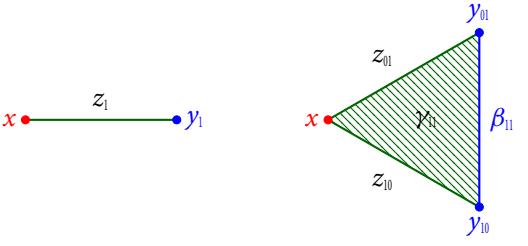
Then, putting all of this together again, if we have a 0-simplex $x_{001} : A_0$, then we take $B \equiv \mathsf{S}\, A\, x_{001}$. For $x_{010} : A_0$, we have that $B_0\, x_{010} \equiv \mathsf{Z}^{\mathsf{d}}\ \ (\mathsf{S}\, A\, x_{001})\ \ x_{010} \equiv A_1\, x_{001}\, x_{010}$. We thus get that:

$$A_2 : (x_{001} : A_0)\ \ (x_{010} : A_0)\ \ (\beta_{011} : A_1\, x_{001}\, x_{010})$$
$$\quad (x_{100} : A_0)\ \ (\beta_{101} : A_1\, x_{001}\, x_{100})\ \ (\beta_{110} : A_1\, x_{010}\, x_{100}) \to \mathsf{Type}$$
$$A_2\, x_{001}\, x_{010}\, \beta_{011}\, x_{100}\, \beta_{101}\, \beta_{110} \equiv \mathsf{Z}^{\mathsf{dd}}\ \left(\mathsf{S}^{\mathsf{d}}\ \ (\mathsf{S}\, A\, x_{001})\ \ x_{010}\, \beta_{011}\right)\, x_{100}\, \beta_{101}\, \beta_{110}.$$

In general, this pattern continues in higher dimensions and the process described lets us extract n-simplex types.

We can visualise what's going on in two different ways. The first visualisation shows how the n-simplices of the slice of A over x live dependently over the n-simplices of A:

$$z_1 : (\mathsf{S}\, A\, x)_0\ \ y_1 \qquad\qquad \gamma_{11} : (\mathsf{S}\, A\, x)_1\ \ y_{01}\, z_{01}\, y_{10}\, z_{10}\, \beta_{11}$$



For example, if z_1 is a 0-simplex of the slice of A over x displayed over the zero simplex y_1 of A, then z_1 is a 1-simplex of A joining x to y_1. Similarly, suppose z_{01} and z_{10} are 0-simplices of the slice of A over x displayed over y_{01} and y_{10}, respectively, and β_{11} is a 1-simplex of A joining y_{01} to y_{11}. Then if γ_{11} is a 1-simplex of the slice of A over x displayed over β_{11} and joining z_{01} to z_{10}, then γ_{11} is a 2-simplex of A with the specified boundary.

Geometrically, we are using the fact that the $(n+1)$-simplex is the *cone* of the n-simplex. Thus, with the above definition, assuming we know inductively that every semi-simplicial type has a type of n-simplices, then for every 0-simplex x, the displayed semi-simplicial type $\mathsf{S}\, A\, x$ has a type of '*displayed n-simplices*' over this. Such a displayed n-simplex depends on x (the cone vertex) as well as on an n-simplex of A (the base face, opposite the cone vertex) and thus can be viewed geometrically as an $(n+1)$-simplex.

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The second visualisation explains our formulas in terms of iterated slicing:

$$x_1 : \mathsf{Z}\ A \qquad \beta_{11} : \mathsf{Z}^{\mathsf{d}}\ (\mathsf{S}\ A\ x_{01})\ x_{10} \qquad \mathfrak{f}_{111} : \mathsf{Z}^{\mathsf{dd}}\ \left(\mathsf{S}^{\mathsf{d}}\ (\mathsf{S}\ A\ x_{001})\ x_{010}\ \beta_{011}\right)\ x_{100}\ \beta_{101}\ \beta_{110}$$



Note that, to form each successive slice, you have to provide 2^n simplex data points. The true dependent n-simplices of a slice may then be viewed as *matching objects*. Hopefully this is sufficiently convincing for now; later we will give a precise justification.

As simple and appealing as this *'definition'* is, it is not meaningful in ordinary dependent type theory. The intuitive claim is that it defines a type SST by coinduction, with Z and S as destructors. For Z this is unproblematic (it is not even corecursive). However, the output of S is not an element of the type SST being defined, as would be usual for a corecursive destructor of a coinductive type, but a *'dependent element'*, or *'displayed element'*, of SST over the input of S. If we write $\mathsf{SST}^{\mathsf{d}}$ for this putative family of *'displayed elements'*, the types of Z and S are:

$$\begin{aligned} &\mathsf{Z} : \mathsf{SST} \to \mathsf{Type} \\ &\mathsf{S} : (X : \mathsf{SST}) \to \mathsf{Z}\ X \to \mathsf{SST}^{\mathsf{d}}\ X. \end{aligned} \tag{3}$$

We would like to regard this as a sort of *'higher coinductive type'*. Just as a higher *inductive* type can have constructors involving not just elements of the type being defined but also *paths* therein, here we have a putative coinductive type whose destructors involve not just elements of the type being defined but also *'displayed elements'* thereof. Thus, to make sense of this we need a type theory with a primitive operation $(-)^{\mathsf{d}}$ associating to a type its family of *'displayed elements'*. As it turns out, the precise notion of $(-)^{\mathsf{d}}$ that we require is a variant of *unary internal parametricity*.

External and internal parametricity

In general, by *'parametricity'* we mean a statement that every type (perhaps subject to contextual restrictions; see below) is equipped with a relation (of some arity), and every function (subject to the same restrictions) preserves those relations. The original form of parametricity, such as in Wadler (1989), is a meta-theoretic statement *about* type theory, in which the relations are meta-theoretic, and the contextual restriction is that it applies only to *closed* types and terms (those defined in the empty context). That is, in this *'external'* parametricity, every closed type is given a relation on its closed terms, which is preserved by every closed function. For instance, in the unary case, given a closed function $() \vdash f : A \to B$, if the A-relation holds of a closed term $() \vdash a : A$, then the B-relation holds of the closed term $() \vdash f\ a : B$. Indeed, the fact that f satisfies this condition is exactly the statement that the $(A \to B)$-relation holds of it, and is thus a special case of the *'fundamental theorem of logical relations'* that *every* closed term satisfies the relation on its type. Semantically, external parametricity is obtained by interpreting type theory in a *'gluing'* or *'relational'* model.

By contrast, in type theories with fully *internal* parametricity such as Bernardy and Moulin (2012), Bernardy (2015), Moulin (2016), there is no contextual restriction, and the relations are internal (i.e. type families). In the unary case, this means for any type in any context, say $\Gamma \vdash A$ type, there is a type family that we will denote $\Gamma,\ x : A \vdash A^{\mathsf{d}}\ x$ type, and every function $\Gamma \vdash f : A \to B$ in any context lifts to a function $\Gamma \vdash f^{\mathsf{d}} : (x : A)(x' : A^{\mathsf{d}}\ x) \to B^{\mathsf{d}}\ (f\ x)$ between these type

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families. As in the external case, there is a formula for the relation of a function-type:

$$\big((a : A) \to B\ [\ a\ ] \big)^{\mathsf{d}}\ f \equiv \big(a : A\big)\ \big(a' : A^{\mathsf{d}}\ a\big) \to B^{\mathsf{d}}\ [\ a, a'\ ]\ \ \big(f\ a\big) \tag{4}$$

which says that so-called *'computability witnesses'* for f take computability witnesses of a to computability witnesses of $f\ a$. Thus the statement that f preserves computability witnesses is equivalent to saying it lifts to an element f^{d} of this type, and is a special case of a general rule that *every* term $\Gamma \vdash a : A$ lifts to a computability witness $\Gamma \vdash a^{\mathsf{d}} : A^{\mathsf{d}}\ a$. (The punning of notation is intentional, and indeed consistent, as we will see: the type $(-)^{\mathsf{d}}$ coincides with the term $(-)^{\mathsf{d}}$ applied to elements of the universe.)

Such an internalisation of parametricity introduces the new possibility of *iterating* it, leading to A^{dd}, A^{ddd}, and so on. Semantically, this means that each type must be interpreted by a *cubical type* (or set), where the arity of the relations determines the number of *'boundary points'* on each side of the cubes, and the dependencies of each iterated relation on the previous ones supplies the *faces* of a cube. Our primary interest is in unary parametricity, where A^{d} depends only on one copy of A; in this case the semantics involves *'unary cubes'* where each edge has only one vertex. (Geometrically, these can be thought of as powers of a half-open interval $[0, 1)^n$, or closed orthants $[0, \infty)^n$ in n-dimensional Euclidean space.)

The rule lifting any a to a^{d} then implies that these cubical types must have *degeneracies*, taking any cube to a higher-dimensional one with some boundaries trivial. More surprisingly, it seems that to have good computational behaviour and semantic models, the cubical types must also include *symmetries* (transpositions): from $d \equiv \lambda\ x.\ x^{\mathsf{d}} : (x : X) \to X^{\mathsf{d}}\ x$, we can either directly obtain the type of d^{d} as $\big(x : X\big)\ \big(x' : X^{\mathsf{d}}\ x\big) \to X^{\mathsf{dd}}\ x\ x'\ x^{\mathsf{d}}$ or first perform the computation $d^{\mathsf{d}} \equiv \lambda\ x\ x'.\ x'^{\mathsf{d}}$ and obtain its type as $\big(x : X\big)\ \big(x' : X^{\mathsf{d}}\ x\big) \to X^{\mathsf{dd}}\ x\ x^{\mathsf{d}}\ x'$, and it these must be related by a symmetry operation.

The general advantages of internal parametricity over external are clear: we can reason about computability witnesses while staying within a single type theory, using a single proof assistant. Moreover, internal parametricity allows us to at least try to make sense of a type SST with the destructors (3). However, fully internal parametricity is not conservative over ordinary type theory: it is highly nonclassical, incompatible with axioms such as the law of excluded middle; and its semantics is not as general as we would like. We would like our type theory to be interpretable in any $(\infty, 1)$-topos (generalising Shulman 2019), in such a way that our type SST is interpreted by a category-theoretic *'classifier'* of semi-simplicial objects; but as we have just observed, the semantics of internal parametricity seems to live only in a category of cubical objects. Now the category of cubical objects in an $(\infty, 1)$-topos is again an $(\infty, 1)$-topos, so (using Shulman 2019) we can expect to interpret an internally parametric type theory in the latter; but the connection of this interpretation to the *original* $(\infty, 1)$-topos is not clear.

Displayed type theory

Our solution is to use a *less internally* parametric type theory. We can think of internal parametricity as arising in three stages. The first stage is an *external* parametricity model, where types are interpreted by pairs $(A_0 : \mathsf{Type},\ A_1 : A_0 \to \mathsf{Type})$. In this case $(-)^{\mathsf{d}}$ maps one model to a *different* model, interpreting an ordinary type by a pair of types; this yields parametricity results as metatheorems. In the second stage, to make A^{d} live in the same model as A, we iterate this construction infinitely many times; now types are interpreted by *semi-cubical types*, with faces but not degeneracies. Finally, in the third stage we add a *'degeneracy'* operation $(x : A) \to A^{\mathsf{d}}\ x$ making every *term* parametric. This allows proving parametricity theorems *inside* the type theory, and semantically moves us from semi-cubical types to cubical ones.

The solution to our problem is to stop after the *second* stage. This is advantageous semantically because semi-cubical sets are presheaves on a *direct* category (i.e. covariant diagrams on an *inverse* category), and in this case the model construction is much more concrete. Specifically, in Shulman

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(2015) it was shown that from any model of univalent dependent type theory inside of a type theoretic fibration category, one may form a derived model of Reedy fibrant presheaves on any direct category, with the type formers in the presheaf model constructed inductively in terms of those in the original model. In particular, in degree 0 all the type-formers act exactly as they do in the original model. Thus, semantically we can be sure that all our constructions, including SST, specialise to something meaningful in an arbitrary $(\infty, 1)$-topos.

dTT is a syntax corresponding to this model, which is likewise intermediate between external parametricity and fully internal parametricity: its parametricity primitive $(-)^{\mathsf{d}}$ has a contextual restriction that is weaker than the *'only closed terms'* requirement of external parametricity, but stronger than the *'any context goes'* laxity of internal parametricity. We start by observing that in either cubical *or* semi-cubical sets, the semantically fundamental parametricity operation actually changes the context: given $\Gamma \vdash t : A$, one has $\Gamma^{\mathsf{D}} \vdash t^{\mathsf{d}} : A^{\mathsf{d}}\ t$, where Γ^{D} augments Γ by computability witnesses of all its variables. For cubical sets with degeneracies, we can deduce a version of $(-)^{\mathsf{d}}$ that doesn't change the context by substituting along a degeneracy map $\Gamma \to \Gamma^{\mathsf{D}}$ (e.g. this is the isomorphism between the *'global'* and *'local'* models of Altenkirch et al. (2024)). But for semi-cubical sets this is impossible, so we have to bite the bullet and deal with context-modifying operations.

The notion of a non-binding operation that changes contexts is familiar from the realm of modal logic, where, to first approximation, a proof of necessity of some proposition, i.e. of $\Box\ A$, may only use necessary assumptions. Modalities in dependent type theory have previously been used to internalise meta-theoretic operations that don't make sense in arbitrary contexts, such as the right adjoint to a Π-type in Licata et al. (2018), and we use them similarly here. Specifically, in dTT we have a modality $\triangle\Box$ that partially internalises the notion of *'closed term'* appearing in external parametricity, and which restricts the domain of $(-)^{\mathsf{d}}$. Thus the only analogue of the above $d \equiv \lambda\ x.\ x^{\mathsf{d}} : (x : X) \to X^{\mathsf{d}}\ x$ in dTT is $\widetilde{d} \equiv \lambda\ x.\ x^{\mathsf{d}} : \left(x :^{\triangle\Box} X\right) \to X^{\mathsf{d}}\ x$. In particular, modal variables are protected from alteration *by* $(-)^{\mathsf{d}}$, so that we have $\widetilde{d}^{\mathsf{d}} \equiv \lambda\ x.\ x^{\mathsf{dd}} : \left(x :^{\triangle\Box} X\right) \to X^{\mathsf{dd}}\ x\ x^{\mathsf{d}}\ x^{\mathsf{d}}$, thereby avoiding the need for symmetry.

In fact, to emphasise further that dTT retains general semantics over an arbitrary $(\infty, 1)$-topos, we will use a multimodal type theory Gratzer et al. (2022, 2021) with two modes, one for the original topos and the other for the topos of semi-cubical sets. These modes are related by modalities $\triangle$ (the constant semi-cubical type), $\Diamond$ (the 0-cubes of a cubical type), and $\Box$ (the limit of a cubical type), and $\triangle\Box = \triangle \circ \Box$ is a composite endo-modality. Only $\triangle\Box$ is necessary to formulate display, but the other modalities are also useful to have around: in particular, $\Diamond$ internalises the process of passing from the model in semi-cubical types to the original model in $\mathcal{C}$. For instance, $\Diamond$ SST is what corresponds semantically to the classifier of semi-simplicial objects in $\mathcal{C}$.

Furthermore, display itself may be thought of as a modality, albeit one that is indexed over the original type. Display falls into the new and yet underdeveloped framework of *indexed modalities*, such as the path types of cubical type theory (treated modally in Gratzer et al. 2022) and the identity types of the forthcoming *Higher Observational Type Theory (HOTT)* Altenkirch et al. (2024, 2022). Moreover, analogously to those cases, we could formulate display either as an inert type-former defined by abstraction over an *'interval'* (like path types in cubical type theory), or as an operation that computes on most other canonical type-formers (like identity types in HOTT). In this paper we make the latter choice, so that rules like eq. (4) are actually definitional equalities. Formulating such rules computationally is actually easier for dTT than for HOTT, due mainly to the lack of symmetry (although there is a tradeoff, since the presence of modalities is an extra complicating factor).

Until now we have been talking about semi-cubical types to make the connection with parametricity clear, but in the unary case there is an intriguing coincidence: the unary semi-cube category is isomorphic to the *augmented* semi-*simplex* category, with a dimension shift: the n-cube corresponds to the $(n-1)$-simplex. (To see this geometrically, note that the $(n+1)$-dimensional

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orthant $[0,\infty)^{n+1}$ contains a standard face-preserving embedding of the n-simplex, $\Delta^n = \{(x_0,\dots,x_n) \in [0,\infty)^{n+1} \mid x_0 + \cdots + x_n = 1\}$, including the augmentation case $\Delta^{-1} = \emptyset$.) For this reason we refer to the two modes in our theory as the *discrete mode* dm and the *simplicial mode* sm. Thus, dTT can actually internalise semi-simplicial types in *two* ways: as the coinductive type SST mentioned above, and as the *universe* of types at the simplicial mode. The latter suggests that dTT could also be used similarly to simplicial type theory, with types at the simplicial mode treated as synthetic (augmented semi-) simplicial types. (Herbelin and Ramachandara Herbelin and Ramachandra (2024) have also used iterated parametricity inside ordinary type theory to define augmented semi-simplicial *sets* and semi-cubical *sets*. The restriction to sets is because when not modifying the type theory, the higher coherences must be dealt with explicitly.)

(Note that an augmented semi-simplicial type can be viewed as a family of ordinary semi-simplicial sets indexed by the type of (–1)-simplices. Thus, our observation that augmented semi-simplicial types support a better internal language than ordinary ones is analogous to the observation of Riley et al. (2021) that *parametrised* spectra are likewise preferable to unparametrised ones.)

Displayed structures

Our terminology *display* for the operation ^{d} is inspired by the fact that when applied to record types whose elements are algebraic structures, it produces *displayed structures* of the corresponding sort. Here a '*displayed structure*' over a structure B is a structure A of the same kind with a structure map $A \to B$, but reformulated in terms of the corresponding family of fibres $B \to \mathsf{Type}$. Working with displayed structures rather than morphisms is a technique for enforcing definitional equalities on images in B.

The most common displayed structure is a *displayed category*; here the terminology was introduced by Ahrens and Lumsdaine (2019). This arises from the record type of categories, defined in the usual dependently typed way (where we omit the axioms for concision):

```
record Cat : Type where
  ob : Type
  hom : ob → ob → Type
  id : (x : ob) → hom x x
  comp : {x y z : ob} → hom y z → hom x y → hom x z
  ...
```

We do not discuss record types (including Σ-types) in this paper, but the extension of ^{d} to them produces another record type whose fields have ^{d} applied to them. For instance, from a Σ-type:

```
record Σ (A : Type) (B : A → Type) : Type where
  fst : A
  snd : B fst
```

We obtain:

```
record Σᵈ (A : Type) (A′ : Typeᵈ A) (B : A → Type)
         (B′ : (A → Type)ᵈ B) : Typeᵈ (Σ A B) where
  fstᵈ : Aᵈ fst
  sndᵈ : (B fst)ᵈ sndᵈ
```

Applying (4) and the similar rule $\mathsf{Type}^{\mathsf{d}}\, A \equiv A \to \mathsf{Type}$, this becomes:

```
record Σᵈ (A : Type) (A′ : A → Type) (B : A → Type)
         (B′ : (x : A) → A′ x → B x → Type) (s : Σ A B) : Type where
  fstᵈ : A′ (fst s)
  sndᵈ : B′ (fst s) fstᵈ (snd s)
```

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In a similar way, the above definition of the record type of categories yields:

```
record Catᵈ (𝒞 : Cat) : Type where
  obᵈ : ob 𝒞 → Type
  homᵈ : (x : ob 𝒞) (x′ : obᵈ x) (y : ob 𝒞) (y′ : obᵈ y) → hom 𝒞 x y → Type
  idᵈ : (x : ob 𝒞) (x′ : obᵈ x) → homᵈ x x′ x x′ (id 𝒞 x)
  compᵈ : {x : ob 𝒞} {x′ : obᵈ x} {y : ob 𝒞} {y′ : obᵈ y} {z : ob 𝒞}
    {z′ : obᵈ z} (α : hom 𝒞 y z) (α′ : homᵈ y y′ z z′ α) (β : hom 𝒞 x y)
    (β′ : homᵈ x x′ y y′ β) → homᵈ x x′ z z′ (comp 𝒞 α β)
  …
```

Thus a displayed category over $\mathcal{C}$ has a type of objects indexed by those of $\mathcal{C}$, types of morphisms indexed by pairs of objects-over-objects and by a morphism of $\mathcal{C}$, identity and composition operations on displayed objects and morphisms that lie strictly over those in $\mathcal{C}$, and similarly for the axioms.

As observed in Ahrens and Lumsdaine (2019), one use of displayed categories is to state definitions such as Grothendieck fibrations in terms of the existence of cartesian liftings strictly over any morphism in $\mathcal{C}$, without internalising definitional equality. Another is to construct categories and prove their properties in a modular way out of dependent pieces, just as we do for types using Σ-types and more general records. It is *'well-known'* by now that any sort of algebro-categorical structure has a *'displayed version'* – for instance, displayed bicategories were used in Ahrens et al. (2021) to prove univalence modularly – but to our knowledge this has not previously been formalised. Our *Displayed Type Theory (dTT)* automatically generates the displayed version of any notion definable in type theory; hence the name.

Outline of the paper

The rest of this paper has three parts. In section 2 we describe the general syntax of dTT, including the modalities, the operation of display (in various different forms), and how they compute. We do not prove any canonicity or normalisation results, but we conjecture that they hold.

In section 3 we extend the syntax of dTT to define a type SST of semi-simplicial types. In fact, we obtain this as a special case of a general notion of *'displayed coinductive type'*, which is easier to work with abstractly, and also includes other important examples such as the type of semi-simplicial morphisms between two semi-simplicial types. (One might hope that it would also include the displayed versions $\mathsf{SST}^{\mathsf{d}}$, $\mathsf{SST}^{\mathsf{dd}}$, etc., but this does not seem to be the case unless we add symmetry to our theory.) Then we explore a few applications, to make the point that this coinductive notion of semi-simplicial type is useful and practical.

Finally, in section 4 we consider the semantics of dTT. We will show that from any model $\mathcal{C}$ of ordinary dependent type theory with countable inverse limits (roughly as considered in Kraus 2015), we can construct a model of dTT whose discrete mode is $\mathcal{C}$ and whose simplicial mode is the category of Reedy fibrant augmented semi-simplicial diagrams in $\mathcal{C}$, and that this model supports displayed coinductive types including a type SST of semi-simplicial types. The underlying ordinary type theory of this model at the simplicial mode is an instance of the inverse diagram models of Shulman (2015), Kapulkin and Lumsdaine (2021), but we construct it more explicitly by hand so as to be able to verify the needed formulas for the additional operations of dTT.

Thus, although we don't expect dTT to be conservative over ordinary dependent type theory, we can isolate exactly a kind of extra infinitary structure (namely, countable inverse limits) that yields a well-behaved theory for working with semi-simplicial types, which precisely includes the original model at one mode. In particular, by Shulman (2019) any $(\infty, 1)$-topos can be presented by a type-theoretic model topos, which is a model of type theory with countable inverse limits, and thus also yields a model of dTT. However, an object with the *internal* universal property of SST expressible in dTT has the potential to exist even in models that lack such infinitary limits, which may have implications for a notion of elementary $(\infty, 1)$-topos.

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2. Syntax

dTT is based on a multimodal type theory following Gratzer et al. (2022, 2021), with one mode for discrete types and another for (augmented semi-)simplicial types. It then adds a notion of *'display'* at the simplicial mode that partially internalises unary parametricity. To state the computation rules for display, we need a generalised form of it that incorporates dependence on a telescope, so we also include a calculus of telescopes. (It would probably be possible to collapse this to dependence on a single type, using Σ-types instead of telescope extension, as in Altenkirch et al. (2024), but this would be unaesthetic and less practical for implementation.)

2.1 The mode theory

We begin with a modal type theory based on the following 2-category $\mathcal{M}$:

- there are two modes (objects), dm for *discrete* and sm for *simplicial*
- there are five nonidentity morphisms, forming hom-posets:

$$\mathcal{M}(\mathsf{dm},\mathsf{dm}) = \{1_{\mathsf{dm}}\} \qquad \mathcal{M}(\mathsf{dm},\mathsf{sm}) = \{\triangle\}$$
$$\mathcal{M}(\mathsf{sm},\mathsf{dm}) = \{\square \le \Diamond\} \qquad \mathcal{M}(\mathsf{sm},\mathsf{sm}) = \{\triangle\square \le 1_{\mathsf{sm}} \le \triangle\Diamond\}$$

- composition is defined by the following tables (plus identity laws):

$\nu \circ \rho$	$\triangle$	$\triangle\Diamond$	$\triangle\square$
$\Diamond$	1_{dm}	$\Diamond$	$\square$
$\square$	1_{dm}	$\Diamond$	$\square$
$\triangle\Diamond$	$\triangle$	$\triangle\Diamond$	$\triangle\square$
$\triangle\square$	$\triangle$	$\triangle\Diamond$	$\triangle\square$

$\nu \circ \rho$	$\Diamond$	$\square$
$\triangle$	$\triangle\Diamond$	$\triangle\square$

Intuitively, $\triangle$ takes a discrete type and forms the constant (augmented semi-)simplicial type, while $\Diamond$ takes the (-1)-simplices of an (augmented semi-)simplicial type and $\square$ takes the limit of an (augmented semi-)simplicial diagram.

The symbol $\triangle$ is chosen to suggest a *'diagonal'*, and is sometimes used in category theory for any constant diagram functor $\mathcal{C} \to \mathcal{C}^X$, since this includes the ordinary diagonal $\mathcal{C} \to \mathcal{C} \times \mathcal{C}$ as the case when $X = 2$. The symbols $\square$ and $\Diamond$ are taken from classical modal logic, where $\square$ denotes *'necessarily'* and $\Diamond$ denotes *'possibly'*, although there is little formal connection except that $\triangle\square$, like necessity, is a comonad, and $\triangle\Diamond$, like possibility, is a monad.

One verifies that the following adjunctions hold in $\mathcal{M}$:

$$\Diamond \dashv \triangle \dashv \square \qquad \triangle\Diamond \dashv \triangle\square \qquad 1_p \dashv 1_p$$

Thus every morphism in $\mathcal{M}$, except for $\Diamond$ and $\triangle\Diamond$, has a right adjoint. ◁

2.2 The modal type theory

The basic structure of dTT follows MTT Gratzer et al. (2021), with judgments shown in Figure 1, where p denotes a mode dm or sm. Following Coquand, we parametrise the type judgment by a universe level; the judgment 'ℓ level' generates these by lzero and lsuc with a join $\sqcup$. Flatness will be explained below.

The basic inference rules in Figure 2 formally define a Generalised Algebraic Theory (GAT) with these generating sorts. Accordingly, we present them using combinators for substitutions, including weakening; but in practice will generally use named variables and leave weakening implicit. Most rules are standard from MTT Gratzer et al. (2021) or MATT Shulman (2023); for emphasis we have drawn boxes around the novel ones, which we now describe.

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$$\Gamma\ \mathsf{ctx}_p \qquad \Gamma \vdash_p A\ \mathsf{type}_\ell \qquad \Gamma \vdash_p t : A \qquad \theta : \Gamma \Rightarrow_p \Theta \qquad \ell\ \mathsf{level} \qquad \boxed{\Gamma\ \mathsf{flat}}$$

Figure 1. Basic judgment forms of dTT.

$$\frac{p\ \mathsf{mode}}{()_p\ \mathsf{ctx}_p} \qquad \frac{\mu : p \to q \qquad \Gamma\ \mathsf{ctx}_q}{(\Gamma,\ \mathsf{🔒}_\mu)\ \mathsf{ctx}_p} \qquad \frac{\mu : p \to q \qquad \Gamma\ \mathsf{ctx}_q \qquad \Gamma,\ \mathsf{🔒}_\mu \vdash_p A\ \mathsf{type}_\ell}{(\Gamma,\ x :^\mu A)\ \mathsf{ctx}_q}$$

$$\left(\Gamma,\ \mathsf{🔒}_{1_p}\right) \equiv \Gamma \qquad (\Gamma,\ \mathsf{🔒}_\mu,\ \mathsf{🔒}_\nu) \equiv (\Gamma,\ \mathsf{🔒}_{\mu\circ\nu}) \qquad \boxed{((),\ \mathsf{🔒}_\square) \equiv ()_{\mathsf{sm}}} \qquad ((),\ \mathsf{🔒}_\triangle) \equiv ()_{\mathsf{dm}}$$

$$\boxed{\frac{\Gamma\ \mathsf{ctx}_{\mathsf{dm}}}{(\Gamma,\ \mathsf{🔒}_\Diamond)\ \mathsf{flat}}} \qquad \boxed{\frac{\Gamma\ \mathsf{ctx}_{\mathsf{sm}} \qquad \Gamma\ \mathsf{flat} \qquad \mu : p \to \mathsf{sm} \qquad \Gamma,\ \mathsf{🔒}_\mu \vdash_p A\ \mathsf{type}_\ell}{(\Gamma,\ x :^\mu A)\ \mathsf{flat}}}$$

$$\frac{\Gamma\ \mathsf{ctx}_p}{1_\Gamma : \Gamma \Rightarrow_p \Gamma} \qquad \frac{\Gamma\ \mathsf{ctx}_p}{[\,]_p : \Gamma \Rightarrow_p ()_p} \qquad \frac{\mu : p \to q \qquad \Gamma\ \mathsf{ctx}_q \qquad \Gamma,\ \mathsf{🔒}_\mu \vdash_p A\ \mathsf{type}_\ell}{\uparrow^{x:^\mu A} : (\Gamma,\ x :^\mu A) \Rightarrow_q \Gamma} \qquad \frac{\theta : \Gamma \Rightarrow_p \Theta \qquad \nu : \Theta \Rightarrow_p \Upsilon}{\nu \circ \theta : \Gamma \Rightarrow_p \Upsilon}$$

$$\frac{\theta : \Gamma \Rightarrow_q \Theta \qquad \Gamma,\ \mathsf{🔒}_\mu \vdash_p t : A\ [\ \theta,\ \mathsf{🔒}_\mu\ ]}{[\ \theta,\ t\ ] : \Gamma \Rightarrow_q (\Theta,\ x :^\mu A)} \qquad \frac{\theta : \Gamma \Rightarrow_q \Theta}{[\ \theta,\ \mathsf{🔒}_\mu\ ] : (\Gamma,\ \mathsf{🔒}_\mu) \Rightarrow_p (\Theta,\ \mathsf{🔒}_\mu)} \qquad \frac{\mu \le \nu}{\mathsf{🔑}^{\mu\le\nu} : (\Gamma,\ \mathsf{🔒}_\nu) \Rightarrow_p (\Gamma,\ \mathsf{🔒}_\mu)}$$

$$\boxed{\frac{\Gamma\ \mathsf{flat}}{\mathsf{🔑}^{\triangle\Diamond \ge 1_{\mathsf{sm}}} : \Gamma \Rightarrow_{\mathsf{sm}} (\Gamma,\ \mathsf{🔒}_{\triangle\Diamond})}}$$

$$\frac{\Gamma,\ \mathsf{🔒}_\mu \vdash_p A\ \mathsf{type}_{\ell_1} \qquad \Gamma,\ x :^\mu A \vdash_q B\ \mathsf{type}_{\ell_2}}{\Gamma \vdash_q (x :^\mu A) \to B\ \mathsf{type}_{\ell_1 \sqcup \ell_2}} \qquad \frac{\ell\ \mathsf{level}}{\Gamma \vdash_{\mathsf{dm}} \mathsf{Disc}_\ell\ \mathsf{type}_{\mathsf{lsuc}\,\ell}} \qquad \frac{\ell\ \mathsf{level}}{\Gamma \vdash_{\mathsf{sm}} \mathsf{Type}_\ell\ \mathsf{type}_{\mathsf{lsuc}\,\ell}}$$

$$\frac{\Gamma \vdash_{\mathsf{dm}} A : \mathsf{Disc}_\ell}{\Gamma \vdash_{\mathsf{dm}} \mathsf{El}\ A\ \mathsf{type}_\ell} \qquad \frac{\Gamma \vdash_{\mathsf{sm}} A : \mathsf{Type}_\ell}{\Gamma \vdash_{\mathsf{sm}} \mathsf{El}\ A\ \mathsf{type}_\ell} \qquad \frac{\theta : \Gamma \Rightarrow_p \Theta \qquad \Theta \vdash_p A\ \mathsf{type}_\ell}{\Gamma \vdash_p A\ [\ \theta\ ]\ \mathsf{type}_\ell}$$

$$\frac{\Gamma,\ \mathsf{🔒}_\square \vdash_{\mathsf{sm}} A\ \mathsf{type}_\ell}{\Gamma \vdash_{\mathsf{dm}} \square\ A\ \mathsf{type}_\ell} \qquad \frac{\Gamma,\ \mathsf{🔒}_\triangle \vdash_{\mathsf{dm}} A\ \mathsf{type}_\ell}{\Gamma \vdash_{\mathsf{sm}} \triangle\ A\ \mathsf{type}_\ell} \qquad \frac{\Gamma,\ \mathsf{🔒}_\Diamond \vdash_{\mathsf{sm}} A\ \mathsf{type}_\ell}{\Gamma \vdash_{\mathsf{dm}} \Diamond\ A\ \mathsf{type}_\ell} \qquad \frac{\mu \le \mathsf{locks}(\Theta)}{\Gamma,\ x :^\mu A,\ \Theta \vdash_q x : A\ [\ 1_\Gamma,\ \uparrow_\mu^{x:^\mu A,\,\Theta}\ ]}$$

$$\boxed{\frac{\Gamma\ \mathsf{flat} \qquad \mathsf{locks}(\Theta) = 1_{\mathsf{sm}}}{\Gamma,\ x :^{\triangle\Diamond} A,\ \Theta \vdash_{\mathsf{sm}} x : A\ [\ 1_\Gamma,\ \uparrow_{\triangle\Diamond}^{x:^{\triangle\Diamond} A,\,\Theta}\ ]\ [\ \mathsf{🔑}^{\triangle\Diamond\ge 1_{\mathsf{sm}}}\ ]}} \qquad \frac{\theta : \Gamma \Rightarrow_p \Theta \qquad \Theta \vdash_p t : A}{\Gamma \vdash_p t\ [\ \theta\ ] : A\ [\ \theta\ ]} \qquad \frac{\Gamma,\ x :^\mu A \vdash_q t : B}{\Gamma \vdash_q \lambda\, x.\, t : (x :^\mu A) \to B}$$

$$\frac{\Gamma \vdash_q f : (x :^\mu A) \to B \qquad \Gamma,\ \mathsf{🔒}_\mu \vdash_p a : A}{\Gamma \vdash_q f\ a : B\ [\ a/x\ ]}$$

$$\frac{\Gamma \vdash_{\mathsf{dm}} A\ \mathsf{type}_\ell}{\Gamma \vdash_{\mathsf{dm}} \mathsf{Code}\ A : \mathsf{Disc}_\ell} \qquad \frac{\Gamma \vdash_{\mathsf{sm}} A\ \mathsf{type}_\ell}{\Gamma \vdash_{\mathsf{sm}} \mathsf{Code}\ A : \mathsf{Type}_\ell}$$

$$\frac{\Gamma,\ \mathsf{🔒}_\square \vdash_{\mathsf{sm}} t : A}{\Gamma \vdash_{\mathsf{dm}} \square\ t : \square\ A} \qquad \frac{\Gamma,\ \mathsf{🔒}_\triangle \vdash_{\mathsf{dm}} t : A}{\Gamma \vdash_{\mathsf{sm}} \triangle\ t : \triangle\ A} \qquad \frac{\Gamma,\ \mathsf{🔒}_\Diamond \vdash_{\mathsf{sm}} t : A}{\Gamma \vdash_{\mathsf{dm}} \Diamond\ t : \Diamond\ A}$$

$$\frac{\Gamma,\ \mathsf{🔒}_\triangle \vdash_{\mathsf{dm}} t : \square\ A}{\Gamma \vdash_{\mathsf{sm}} \blacksquare^A\ t : A\ [\ \mathsf{🔑}^{\triangle\square\le 1_{\mathsf{sm}}}\ ]} \qquad \frac{\Gamma,\ \mathsf{🔒}_\Diamond \vdash_{\mathsf{sm}} t : \triangle\ A}{\Gamma \vdash_{\mathsf{dm}} \blacktriangle^A\ t : A} \qquad \boxed{\frac{\Gamma\ \mathsf{flat} \qquad \Gamma,\ \mathsf{🔒}_\triangle \vdash_{\mathsf{dm}} t : \Diamond\ A}{\Gamma \vdash_{\mathsf{sm}} \blacklozenge^A\ t : A\ [\ \mathsf{🔑}^{\triangle\Diamond\ge 1_{\mathsf{sm}}}\ ]}}$$

Figure 2. Basic rules of dTT.

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$$\Gamma \vdash_p \Theta\ \mathsf{tel}_\ell \qquad \Gamma \vdash_p \Theta\ \mathsf{stel}_\ell \qquad \Gamma \vdash \sigma : \Upsilon \qquad \Gamma \vdash_p A\ \ \mathsf{type}_{\ell_1} \big/ {}_{\upsilon : \Upsilon}$$

Figure 3. Telescope and meta-abstraction judgments.

In addition to functoriality of locks, we stipulate that $\mathbin{🔒}_\square$ and $\mathbin{🔒}_\triangle$ (but not $\mathbin{🔒}_\diamondsuit$) preserve empty contexts. (The rule for $\mathbin{🔒}_\triangle$ is redundant, since $(()_{\mathsf{sm}}, \mathbin{🔒}_\triangle) = (()_{\mathsf{dm}}, \mathbin{🔒}_\square, \mathbin{🔒}_\triangle) = (()_{\mathsf{dm}}, \mathbin{🔒}_{\square \circ \triangle}) = (()_{\mathsf{dm}}, \mathbin{🔒}_{1_{\mathsf{dm}}}) = ()_{\mathsf{dm}}$.) These equations mean that contexts do not have a unique presentation, but there are ways to select canonical presentations.

Semantically, each lock is left adjoint to its modality; thus $\mathbin{🔒}_\square \cong \triangle$ and $\mathbin{🔒}_\triangle \cong \diamondsuit$, so in particular $\mathbin{🔒}_\square$ and $\mathbin{🔒}_\triangle$ are also right adjoints. Intuitively, $\mathbin{🔒}_\diamondsuit$ is an equivalence from the discrete types to the simplicial types that are empty in dimensions $n > -1$. This subcategory is a sieve, so $\mathbin{🔒}_\diamondsuit$ is a *parametric* right adjoint, with a left adjoint defined on this subcategory, and which coincides with $\mathbin{🔒}_\triangle$ there. This subcategory is represented syntactically by the *flatness* judgment on sm-contexts, which are those containing a $\mathbin{🔒}_\diamondsuit$ which is not to the left of any $\mathbin{🔒}_\triangle$, as generated by the rules shown in Figure 2. The most important property of this subcategory is that $\mathbin{🔒}_{\triangle\diamondsuit}$ acts as the identity on it; we represent this by the *'exceptional'* key rule $\mathbin{🗝}^{\triangle\diamondsuit \geq 1_{\mathsf{sm}}}$ and its related variable rule.

We represent all three modalities in Fitch-style as in Gratzer et al. (2022), Shulman (2023), but with concrete parametric left adjoints rather than abstract ones. (We omit $\triangle\diamondsuit$ or $\triangle\square$, since their modalities can be obtained up to isomorphism by composing the others.) For $\square$ and $\triangle$ we use the actual left adjoints $\mathbin{🔒}_\triangle \dashv \mathbin{🔒}_\square$ and $\mathbin{🔒}_\diamondsuit \dashv \mathbin{🔒}_\triangle$ with rules as in Shulman (2023), while for $\diamondsuit$ the parametric right adjoint rules from Gratzer et al. (2022) become a flatness assumption on the context. ◁

2.3 Telescopes and meta-abstractions

Telescopes are suffixes of contexts not containing locks; they have a level that is no less than those of the types in them. Formally they are just another sort of the GAT, with judgments given in Figure 3 and rules given in Figure 4. We regard an operation on telescopes as *'defined'* when we specify how it computes on empty telescopes and extensions by a type, such as is done in Figure 4 for the extension of a context by a telescope, $(\Gamma \mid \Theta)$. This operation is left-associative with the comma, so for instance $\Gamma \mid \Theta, \mathbin{🔒}_\mu$ means $(\Gamma \mid \Theta), \mathbin{🔒}_\mu$.

A *strict telescope* (Θ stel) is one with no nontrivially modal variables. We treat the obvious map from strict telescopes to telescopes, and the operation of lifting a telescope to a higher level, as implicit coercions, using the same name for a telescope and its lifted and strict versions. We omit the obvious defining equations of these operations, and likewise for weakening by a telescope.

The judgement $\Gamma \vdash \sigma : \Upsilon$ describes an *'element'* of a telescope Υ, which we call a *'partial substitution'*, thinking of it as a substitution $\Gamma \Rightarrow (\Gamma \mid \Upsilon)$ that is the identity on Γ. They are built out of terms, can extend ordinary substitutions, and uniquely determined by their components; the latter is ensured by detecting their equality by induced substitutions.

The judgement form $\Gamma \vdash_p A\ \ \mathsf{type}_{\ell_1} \big/ {}_{\upsilon : \Upsilon}$ defined in Figure 5 is more novel. We call it a *meta-abstraction* and think of it as saying that A is a type depending on the variables υ in the telescope Υ, i.e. belonging to a *'framework-level* Π*-type'* $A : (\upsilon : \Upsilon) \to \mathsf{type}_{\ell_1}$. Accordingly, it is governed by binding and application, with a β and η-rule. We also regard $\mathcal{A}\ \ \mathsf{type}_{\ell_1} \big/ {}_{\upsilon : \Upsilon}$ as standing in for its own Π-type *'*$(\upsilon : \Upsilon) \to \mathcal{A}\ \upsilon$*'*, so such an $\mathcal{A}$ can have its own terms belonging to it, with binding and application and a β and η-rule. ◁

2.4 Décalage and displayed types

Semantically, the fundamental operation is shifting the dimensions of a simplicial type. In classical simplicial homotopy theory, this is called *décalage*:

$$\left(A^{\mathsf{D}}\right)_n = A_{n+1}$$

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$$\frac{\Gamma\ \mathsf{ctx}_p}{\Gamma \vdash_p ()_p\ \mathsf{tel}_\ell} \qquad \frac{\mu : p \to q \qquad \Gamma \vdash_q \Theta\ \mathsf{tel}_\ell \qquad \Gamma \mid \Theta,\ \clubsuit_\mu \vdash_p A\ \mathsf{type}_{\ell'} \qquad \ell' \le \ell}{\Gamma \vdash_q (\Theta,\ x :^\mu A)\ \mathsf{tel}_\ell}$$

$$\frac{\Gamma \vdash_p \Theta\ \mathsf{tel}_\ell}{(\Gamma \mid \Theta)\ \mathsf{ctx}} \qquad (\Gamma \mid ()_p) \equiv \Gamma \qquad (\Gamma \mid (\Theta,\ x :^\mu A)) \equiv ((\Gamma \mid \Theta),\ x :^\mu A) \qquad \frac{\Gamma \vdash_p \Theta\ \mathsf{tel}_\ell \qquad \ell \le \ell'}{\Gamma \vdash_p \Theta\ \mathsf{tel}_{\ell'}}$$

$$\frac{\Gamma\ \mathsf{ctx}_p}{\Gamma \vdash_p ()_p\ \mathsf{stel}_\ell} \qquad \frac{\Gamma \vdash_p \Theta\ \mathsf{stel}_\ell \qquad \Gamma \mid \Theta \vdash_p A\ \mathsf{type}_{\ell'} \qquad \ell' \le \ell}{\Gamma \vdash_p (\Theta,\ x :^{1_p} A)\ \mathsf{stel}_\ell} \qquad \frac{\Gamma \vdash_p \Theta\ \mathsf{tel}_\ell}{\Gamma \vdash_p \Theta\ \mathsf{stel}_\ell} \qquad \frac{\Gamma \vdash \Theta\ \mathsf{tel}_\ell}{\uparrow^\Theta : (\Gamma \mid \Theta) \Rightarrow \Gamma}$$

$$\frac{}{\Gamma \vdash_p [\,]_p : ()_p} \qquad \frac{\Gamma \vdash_q \sigma : \Upsilon \qquad \Gamma \mid \Upsilon,\ \clubsuit_\mu \vdash_p A\ \mathsf{type}_\ell \qquad \Gamma,\ \clubsuit_\mu \vdash_p t : A\,[\,1_\Gamma \mid \sigma,\ \clubsuit_\mu\,]}{\Gamma \vdash_q [\,\sigma,\ t\,] : (\Upsilon,\ x :^\mu A)}$$

$$\frac{\theta : \Gamma \Rightarrow_p \Delta \qquad \Delta \vdash_p \Upsilon\ \mathsf{tel}_\ell \qquad \Gamma \vdash_p \sigma : \Upsilon\,[\,\theta\,]}{[\,\theta \mid \sigma\,] : \Gamma \Rightarrow_p (\Delta \mid \Upsilon)} \qquad [\,\theta \mid [\,]_p\,] \equiv \theta \qquad [\,\theta \mid [\,\sigma,\ t\,]\,] \equiv [\,[\,\theta \mid \sigma\,],\ t\,]$$

$$\frac{\Gamma \vdash_p \sigma : \Upsilon \qquad \Gamma \vdash_p \tau : \Upsilon \qquad [\,1_\Gamma \mid \sigma\,] \equiv [\,1_\Gamma \mid \tau\,] : \Gamma \Rightarrow_p (\Gamma \mid \Upsilon)}{\Gamma \vdash_p \sigma \equiv \tau : \Upsilon}$$

Figure 4. Telescope rules.

$$\frac{\Gamma \mid (\upsilon : \Upsilon) \vdash_p A\ \mathsf{type}_{\ell_1}}{\Gamma \vdash_p (\!(\, A \,)\!)_{\upsilon:\Upsilon}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon}} \qquad \frac{\Gamma \vdash_p \mathcal{A}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \vdash_p \sigma : \Upsilon}{\Gamma \vdash_p \mathcal{A}\,\sigma\ \mathsf{type}_{\ell_1}} \qquad \frac{\Gamma \mid (\upsilon : \Upsilon) \vdash_p A\ \mathsf{type}_{\ell_1} \qquad \Gamma \vdash_p \sigma : \Upsilon}{\Gamma \vdash_p (\!(\, A \,)\!)_{\upsilon:\Upsilon}\,\sigma \equiv A\,[\,1_\Gamma \mid \sigma\,]}$$

$$\frac{\Gamma \vdash_p \mathcal{A}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \vdash_p \mathcal{B}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \mid (\upsilon : \Upsilon) \vdash_p \mathcal{A}\,\upsilon \equiv \mathcal{B}\,\upsilon}{\Gamma \vdash_p \mathcal{A} \equiv \mathcal{B}}$$

$$\frac{\Gamma \vdash_p \mathcal{A}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \mid (\upsilon : \Upsilon) \vdash_p t : \mathcal{A}\,\upsilon}{\Gamma \vdash_p [\![\, t \,]\!]_{\upsilon:\Upsilon} : \mathcal{A}} \qquad \frac{\Gamma \vdash_p \mathcal{A}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \vdash_p \mathsf{t} : \mathcal{A} \qquad \Gamma \vdash_p \sigma : \Upsilon}{\Gamma \vdash_p \mathsf{t}\,\sigma\ :\ \mathcal{A}\,\sigma\ \mathsf{type}_{\ell_1}}$$

$$\frac{\Gamma \vdash_p \mathcal{A}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \mid (\upsilon : \Upsilon) \vdash_p t : \mathcal{A}\,\upsilon \qquad \Gamma \vdash_p \sigma : \Upsilon}{\Gamma \vdash_p [\![\, t \,]\!]_{\upsilon:\Upsilon}\,\sigma \equiv t\,[\,1_\Gamma \mid \sigma\,]}$$

$$\frac{\Gamma \vdash_p \mathcal{A}\ \mathsf{type}_{\ell_1}/_{\upsilon:\Upsilon} \qquad \Gamma \vdash_p \mathsf{t} : \mathcal{A} \qquad \Gamma \vdash_p \mathsf{s} : \mathcal{A} \qquad \Gamma \mid (\upsilon : \Upsilon) \vdash_p \mathsf{t}\,\upsilon \equiv \mathsf{s}\,\upsilon}{\Gamma \vdash_p \mathsf{t} \equiv \mathsf{s}}$$

Figure 5. Meta-abstraction rules.

The simplicial structure maps of A^{D} are a subset of those of A, while the unused ones assemble into a simplicial map $A^{\mathsf{D}} \to A$. If A is Reedy fibrant, then this map $A^{\mathsf{D}} \to A$ is a Reedy fibration, and so we can regard A^{D} as the total space of a type dependent on A, which we denote A^{d} and call *display*. That is, A^{d} is the *fibre* of the map $A^{\mathsf{D}} \to A$; syntactically

$$A^{\mathsf{D}} = (x : A,\ x' : A^{\mathsf{d}}\ x).$$

Display is our version of the *'logical relation'* assigned to every type by internal parametricity.

However, in contrast to fully internal parametricity theories, because we don't have degeneracies in our cube category, décalage and display can only be applied in restricted contexts. In external parametricity, the logical relations apply only to types in the empty context; but our modalities allow us to say more generally that they apply to any *'boxed'* type. Here by *'box'* we mean not $\Box$ but the corresponding *endofunctor* of the simplicial mode, namely $\triangle\Box$. Thus, informally display should have the type $d : (A :^{\triangle\Box} \mathsf{Type}_\ell) \to A \to \mathsf{Type}_\ell$, with computability witnesses

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$$\frac{\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \Upsilon\ \mathsf{tel}_\ell}{\Gamma \vdash_{\mathsf{sm}} \Upsilon^{\mathsf{D}}\ \mathsf{tel}_\ell} \qquad \frac{\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \Upsilon\ \mathsf{tel}_\ell \qquad \Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \sigma : \Upsilon}{\Gamma \vdash_{\mathsf{sm}} \sigma^{\mathsf{D}} : \Upsilon^{\mathsf{D}}}$$

$$\frac{\Gamma \vdash_{\mathsf{sm}} \sigma^{+} : \Upsilon^{\mathsf{D}}}{\Gamma \vdash_{\mathsf{sm}} \sigma^{+\mathsf{ev}} : \Upsilon\ [\, \mathsf{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\,]} \qquad \sigma^{\mathsf{D}\,\mathsf{ev}} \equiv \sigma\ [\, \mathsf{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\,]$$

$$\frac{\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \mathcal{A}\ \ \mathsf{type}_{\ell_1}\big/_{\upsilon : \Upsilon}}{\Gamma \vdash_{\mathsf{sm}} \mathcal{A}^{\mathsf{d}}\ \ \mathsf{type}_{\ell_1}\big/_{\upsilon^{+} : \Upsilon^{\mathsf{D}},\, a : \mathcal{A}\,[\, \mathsf{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\,]\,\upsilon^{+\mathsf{ev}}}} \qquad \frac{\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \mathcal{A}\ \ \mathsf{type}_{\ell_1}\big/_{\upsilon : \Upsilon} \qquad \Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \mathsf{t} : \mathcal{A}}{\Gamma \vdash_{\mathsf{sm}} \mathsf{t}^{\mathsf{d}} : (\!(\, \mathcal{A}^{\mathsf{d}}\ \upsilon^{+}\ \ (\mathsf{t}\ \upsilon^{+\mathsf{ev}})\ )\!)_{\upsilon^{+} : \Upsilon^{\mathsf{D}}}}$$

Figure 6. Décalage, evens, and display.

being assigned by a function $d : (A :^{\triangle\square} \mathsf{Type}_\ell)(x :^{\triangle\square} A) \to A^{\mathsf{d}}\ x$. If we reformulate these without to Π-types, we obtain the following rules:

$$\frac{\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} A\ \mathsf{type}_\ell \qquad \Gamma \vdash_{\mathsf{sm}} t : A\ [\, \mathsf{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\,]}{\Gamma \vdash_{\mathsf{sm}} A^{\mathsf{d}}\ t\ \mathsf{type}_\ell} \qquad \frac{\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} t : A}{\Gamma \vdash t^{\mathsf{d}} : A^{\mathsf{d}}\ \big(t\ [\, \mathsf{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\,]\big)}$$

However, to compute with this, we need a version that incorporates dependence on a telescope to the right of the lock. The corresponding action on that telescope is décalage, which doubles the variables and groups each type with its displayed version, e.g.

$$(x : A,\ y : B)^{\mathsf{D}} \equiv (x : A,\ x' : A^{\mathsf{d}}\ x,\ y : B,\ y' : B^{\mathsf{d}}\ y).$$

The classical projection from décalage to the identity becomes an *'evens'* substitution from Υ^{D} to Υ that throws away the elements of the displayed types (the primed variables above).

Now, the more general version of display can informally be thought of as:

$$¿\quad \frac{\Gamma,\ \blacksquare_{\triangle\square} \mid \Upsilon \vdash_{\mathsf{sm}} A\ \mathsf{type}_\ell}{\Gamma \mid \Upsilon^{\mathsf{D}}, a : A \vdash_{\mathsf{sm}} A^{\mathsf{d}}\ a\ \mathsf{type}_\ell}\quad ?$$

However, this is not a well-behaved rule because the context of the conclusion is not fully general. We solve this by saying that general display acts on a meta-abstracted type, yielding the rules in Figure 6. Note that we put a superscript '+' on variables in décalaged telescopes, and a prime on variables in displayed types and telescopes; these symbols are just a (mnemonic) part of the variable name. Our first, simple, sort of display is just an abbreviation for the case when the telescope Υ is empty:

$$A^{\mathsf{d}}\ t \equiv (\!(\, A\ )\!)_{[\,] : ()_{\mathsf{sm}}}{}^{\mathsf{d}}\ [\, t\,].$$

The more general display enables the rules *'defining'* telescope décalage in Figure 7. When extending by a non-modal variable, we also extend by its displayed version, which depends on Θ^{D}. Extending by a nontrivially modal variable is simpler: the modality has the form $\triangle \circ \mu$, hence semantically is a constant simplicial type. So we should have informally $(\triangle\ A)^{\mathsf{D}} = \triangle\ A$, and therefore $(\triangle\ A)^{\mathsf{d}}$ is trivial, meaning we omit the displayed variables.

Finally, in Figure 8 we compute display on the basic type-formers, which as we discussed in section 1 acts like the identity types of HOTT. Note that the abstracting telescope changes as we compute, so these rules could not be stated for ordinary display alone. These rules represent the traditional behaviour of parametricity and logical relations on functions (a computability witness for a function says that it preserves computability witnesses) and universes (a computability witness for a type is a relation on it). ◁

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$$
\begin{aligned}
()^{\mathsf{D}}_{\mathsf{sm}} &\equiv ()_{\mathsf{sm}} \\
[\,]^{\mathsf{D}}_{\mathsf{sm}} &\equiv [\,]_{\mathsf{sm}} \\
[\,]^{\mathsf{ev}}_{\mathsf{sm}} &\equiv [\,]_{\mathsf{sm}} \\
(\theta : \Theta,\ x : A)^{\,\mathsf{D}} &\equiv \left(\theta^{+} : \Theta^{\mathsf{D}},\ x : A\,[\,\mathsf{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}} \mid \theta^{+\mathsf{ev}}\,],\ x' : (\!(\, A \,)\!)_{\theta : \Theta^{\mathsf{d}}\ \theta^{+}\ x}\right) \\
[\,\sigma,\ t\,]^{\mathsf{D}} &\equiv [\,\sigma^{\mathsf{D}},\ t,\ t^{\mathsf{d}}\,] \\
[\,\sigma^{+},\ t,\ t'\,]^{\mathsf{ev}} &\equiv [\,\sigma^{+\mathsf{ev}},\ t\,] \\
\left(\theta : \Theta,\ x :^{\triangle\circ\mu} A\right)^{\,\mathsf{D}} &\equiv \left(\theta^{+} : \Theta^{\mathsf{D}},\ x :^{\triangle\circ\mu} A\,[\,\mathsf{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}} \mid \theta^{+\mathsf{ev}},\ \mathsf{🔒}_{\triangle\circ\mu}\,]\right) \\
[\,\sigma,\ t\,]^{\mathsf{D}} &\equiv [\,\sigma^{\mathsf{D}},\ t\,] \\
[\,\sigma^{+},\ t\,]^{\mathsf{ev}} &\equiv [\,\sigma^{+\mathsf{ev}},\ t\,]
\end{aligned}
$$

Figure 7. Computation of décalage and evens.

$$
\begin{aligned}
(\!(\, (x : A) \to B \,)\!)_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv (\!(\, (x : A\,[\,\mathsf{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}} \mid \upsilon^{+\mathsf{ev}}\,])\ (x' : (\!(\, A \,)\!)_{\upsilon : \Upsilon^{\mathsf{d}}\ \upsilon^{+}\ x}) \to \\
&\qquad (\!(\, B \,)\!)_{\upsilon : \Upsilon, x : A^{\mathsf{d}}\ \upsilon^{+}\ x\ x'\ (f\ x)}\ )\!)_{\upsilon^{+} : \Upsilon^{\mathsf{D}}, f : (x : A) \to B} \\
[\![\, \lambda\, x.\ t \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv [\![\, \lambda\, x\, x'.\ [\![\, t \,]\!]_{\upsilon : \Upsilon, x : A^{\mathsf{d}}\ \upsilon^{+}\ x\ x'} \,]\!]_{\upsilon^{+} : \Upsilon^{\mathsf{D}}} \\
[\![\, f\ a \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv [\![\, ([\![\, f \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}\ \upsilon^{+}})\ (a\,[\,\upsilon^{+\mathsf{ev}}\,])\ ([\![\, a \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}\ \upsilon^{+}}) \,]\!]_{\upsilon^{+} : \Upsilon^{\mathsf{D}}} \\
(\!(\, (x :^{\triangle\circ\mu} A) \to B \,)\!)_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv (\!(\, (x :^{\triangle\circ\mu} A\,[\,\mathsf{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}} \mid \upsilon^{+\mathsf{ev}},\ \mathsf{🔒}_{\triangle\circ\mu}\,]) \to \\
&\qquad (\!(\, B \,)\!)_{\upsilon : \Upsilon, x :^{\triangle\circ\mu} A^{\mathsf{d}}\ \upsilon^{+}\ x\ (f\ x)}\ )\!)_{\upsilon^{+} : \Upsilon^{\mathsf{D}}, f : (x :^{\triangle\circ\mu} A) \to B} \\
[\![\, \lambda\, x.\ t \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv [\![\, \lambda\, x.\ [\![\, t \,]\!]_{\upsilon : \Upsilon, x :^{\triangle\circ\mu} A^{\mathsf{d}}\ \upsilon^{+}\ x} \,]\!]_{\upsilon^{+} : \Upsilon^{\mathsf{D}}} \\
[\![\, f\ a \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv [\![\, [\![\, f \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}\ \upsilon^{+}}\ (a\,[\,\upsilon^{+\mathsf{ev}}\,]) \,]\!]_{\upsilon^{+} : \Upsilon^{\mathsf{D}}} \\
(\!(\, \mathsf{Type}_{\ell} \,)\!)_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv (\!(\, \mathsf{El}\ A \to \mathsf{Type}_{\ell} \,)\!)_{\upsilon^{+} : \Upsilon^{\mathsf{D}}, A : \mathsf{Type}_{\ell}} \\
[\![\, \mathsf{Code}\ A \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv [\![\, \lambda\, a.\ \mathsf{Code}\ ((\!(\, A \,)\!)_{\upsilon : \Upsilon^{\mathsf{d}}\ \upsilon^{+}\ a}) \,]\!]_{\upsilon^{+} : \Upsilon^{\mathsf{D}}} \\
(\!(\, \mathsf{El}\ A \,)\!)_{\upsilon : \Upsilon^{\mathsf{d}}} &\equiv (\!(\, \mathsf{El}\ ([\![\, A \,]\!]_{\upsilon : \Upsilon^{\mathsf{d}}\ \upsilon^{+}\ a}) \,)\!)_{\upsilon^{+} : \Upsilon^{\mathsf{D}}, a : \mathsf{El}\ A}
\end{aligned}
$$

Figure 8. Computation of display.

2.5 Displayed telescopes

The rules given so far contain all the really important ideas of dTT, but in various contexts we find ourselves forced to introduce more general judgments and operations, involving telescopes dependent on other telescopes. In this section we briefly sketch these generalizations, which should be fairly intuitive; detailed rules governing them can be found in the extended version of the paper.

- The judgement $\Gamma \vdash_{p} \Phi\ \ \mathsf{tel}\big/_{\upsilon : \Upsilon}$ meta-abstracts a telescope over another telescope.
- The operation $\Gamma \vdash (\Upsilon \mid \Phi)\ \mathsf{tel}$ concatenates telescopes: it is a strictly associative Σ-type.
- The Π-telescope $(\upsilon : \Upsilon) \to \Theta$ of one telescope over another has the usual binding and application rules, and also computes on the structure of both Υ or Θ.
- A meta-abstracted telescope can be dependently décalaged: $\Phi^{\mathsf{D}}\ \ \mathsf{tel}\big/_{\upsilon^{+} : \Upsilon^{\mathsf{D}}}$. This reduces to ordinary décalage on constant meta-abstractions, commutes with telescope concatenation, and computes on telescopes made out of types.
- We also have *display* for telescopes: if $\sigma : \Upsilon\ [\,\mathsf{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}}\,]$ then $(\Upsilon^{\mathsf{d}}\ \sigma)\ \mathsf{stel}$. This always yields a *strict* telescope: its computation rules discard all nontrivially modal variables.

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- Compared to décalage, telescope display reorders the variables. For instance:

$$(x : A,\ y : B)^{\mathsf{D}} \equiv (x : A,\ x' : A^{\mathsf{d}}\ x,\ y : B,\ y' : B^{\mathsf{d}}\ y)$$
$$(x : A,\ y : B)^{\mathsf{d}} \equiv (\!(\ x' : A^{\mathsf{d}}\ x,\ y' : B^{\mathsf{d}}\ y\ )\!)_{x:A,\,y:B}$$
$$(x : A,\ y : B) \mid (x : A,\ y : B)^{\mathsf{d}} \equiv (x : A,\ y : B,\ x' : A^{\mathsf{d}}\ x,\ y' : B^{\mathsf{d}}\ y)$$
$$\not\equiv (x : A,\ y : B)^{\mathsf{D}}.$$

  This difference is mediated by an *'odds'* operation that picks out the elements of displayed types, and a *'pairing'* operation that interleaves them together, such that evens and odds together form an isomorphism with pairing as inverse:

$$\frac{\Gamma \vdash_{\mathsf{sm}} \sigma^{+} : \Upsilon^{\mathsf{D}}}{\Gamma \vdash_{\mathsf{sm}} \sigma^{+\mathsf{od}} : \Upsilon^{\mathsf{d}}\ \sigma^{\mathsf{ev}}} \qquad \frac{\Gamma \vdash_{\mathsf{sm}} \sigma : \Upsilon\ [\ \mathsf{key}^{\triangle\Box \le 1_{\mathsf{sm}}}\ ] \qquad \Gamma \vdash_{\mathsf{sm}} \sigma' : \Upsilon^{\mathsf{d}}\ \sigma}{\Gamma \vdash_{\mathsf{sm}} \langle\, \sigma\, ,\, \sigma'\, \rangle : \Upsilon^{\mathsf{D}}}$$

$$\sigma^{+} \equiv \langle\, \sigma^{+\mathsf{ev}}\, ,\, \sigma^{+\mathsf{od}}\, \rangle \qquad \langle\, \sigma\, ,\, \sigma'\, \rangle^{\mathsf{ev}} \equiv \sigma \qquad \langle\, \sigma\, ,\, \sigma'\, \rangle^{\mathsf{od}} \equiv \sigma'$$

- As with types, we have display for meta-abstracted telescopes, which décalages the domain telescope. This has a more general odds/evens isomorphism, and gives a way to compute display of a concatenated telescope.
- Finally, either kind of display of a Π-telescope computes by décalaging the domain and displaying the codomain. (It does not seem possible to consistently compute décalage of Π-telescopes.)

Remark 5. We can now give a comparison to the *'local theory'* from Altenkirch et al. (2024), with our telescopes and telescope concatenation replacing their types and Σ-types. Our décalaged telescope Υ^{D} from section 2.4 corresponds to their $\forall A$, while the meta-abstracted analogue corresponds to their $\forall d(x.B)$. Our décalaged partial substitution σ^{D} from section 2.4 corresponds to their R (although with an added modal lock), while its meta-abstracted analogue corresponds to their apd (with modal lock). We don't have their ap represented explicitly, but one of their rules says it is equivalent to apd in a constant family. Our ^{ev} is their k (in the unary case, so $k = 0$), and our modal guards make their S unnecessary. ◁

3. Semi-Simplicial and Displayed Coinductive Types

Recall from the introduction that our primary goal in formulating dTT (at the moment) is to have a type theory in which we can make precise our coinductive definition of the type SST of semi-simplicial types. In this section we give that definition, along with a more general notion of *'displayed coinductive type'*, and explore some applications.

3.1 Semi-simplicial types

In a proof assistant implementing displayed type theory, one would expect to define semi-simplicial types as an instance of a general **codata** declaration. In addition to the display primitive, this requires an enhancement of the usual syntaxes for coinductive definitions that allows the coinductive input of each destructor to be named and referred to in its type. For instance, using an Agda-like syntax for records we might write the following:

```
codata SST : Type where
  Z : SST → Type
  S : (A : SST) → Z A → SSTᵈ A
```

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In fact, this is already possible: the second author has implemented an experimental proof assistant called `Narya` for *'observational'* higher type theories like dTT and HOTT. `Narya` doesn't yet include the modalities of dTT, but it does have a general notion of coinductive definition that includes SST, with the following syntax:

```
def SST : Type ≔ codata [
| A .Z : Type
| A .S : A .Z → SST(d) A ]
```

Note that `Narya` regards Z and S as *fields*, written with postfix syntax, rather than *functions* written applicatively. Also it requires parentheses around ^{d}, for reasons we'll mention later.

A sufficiently broad framework encompassing all such *'displayed coinductive'* definitions (let alone a full specification of `Narya`) is beyond the scope of this paper, but we will describe one general paradigmatic class of them, analogous to W-types as paradigmatic inductive types and M-types as paradigmatic coinductive types.

We begin by discussing the motivating example of SST in more detail, starting with its type formation law and destructors. Of course, since SST is a sort of *'universe'*, its elements consisting of types, it must also be parametrised by a level.

$$\frac{}{\Gamma \vdash_{\mathsf{sm}} \mathsf{SST}_\ell\ \mathsf{type}_{\mathsf{lsuc}\,\ell}}$$

$$\Gamma \vdash_{\mathsf{sm}} \mathsf{Z} : (\!(\ \mathsf{Type}_\ell\ )\!)_{A\,:\,\mathsf{SST}_\ell}$$

$$\Gamma \vdash_{\mathsf{sm}} \mathsf{S} : (\!(\ \mathsf{SST}^{\mathsf{d}}_\ell\ A\ )\!)_{A\,:\,\mathsf{SST}_\ell,\,a\,:\,\mathsf{El}\,(\mathsf{Z}\,A)}$$

Note that the destructors are defined as terms belonging to *'meta-abstractions'* as introduced in section 2.3. We have chosen this over the more common method of supplying the arguments in premises, for example:

$$\frac{\Gamma \vdash_{\mathsf{sm}} A : \mathsf{SST}_\ell}{\Gamma \vdash_{\mathsf{sm}} \mathsf{Z}\,A : \mathsf{Type}_\ell},$$

because it makes it easier to compute ^{d} of them:

$$\Gamma \vdash_{\mathsf{sm}} \mathsf{SST}^{\mathsf{d}}_\ell\ \ \mathsf{type}_{\mathsf{lsuc}\,\ell}\,/_{A_3\,:\,\mathsf{SST}_\ell}$$

$$\Gamma \vdash_{\mathsf{sm}} \mathsf{Z}^{\mathsf{d}} : (\!(\ \mathsf{El}\ \ (\mathsf{Z}\,A_3) \to \mathsf{Type}_\ell\ )\!)_{\{A_3\,:\,\mathsf{SST}_\ell\},\,A_x\,:\,\mathsf{SST}^{\mathsf{d}}_\ell\,A_3}$$

$$\Gamma \vdash_{\mathsf{sm}} \mathsf{S}^{\mathsf{d}} : (\!(\ \mathsf{SST}^{\mathsf{dd}}_\ell\,A_3\,A_x\ \ (\mathsf{S}\,A_3\,a_3)\ )\!)_{\{A_3\,:\,\mathsf{SST}_\ell\},\,A_x\,:\,\mathsf{SST}^{\mathsf{d}}_\ell\,A_3,\,a_3\,:\,\mathsf{El}\,(\mathsf{Z}\,A_3),\,a_x\,:\,\mathsf{El}\,(\mathsf{Z}^{\mathsf{d}}\,A_x\,a_3)}$$

This suggests that the family $\mathsf{SST}^{\mathsf{d}}$ should behave as though defined by computing ^{d} on all of the destructors of SST, regarded as functions:

```
codata SSTᵈ (A₃ : SST) : Type where
  Zᵈ : SSTᵈ A → Z A → Type
  Sᵈ : (Aₓ : SSTᵈ A) (a₃ : Z A₃) → Zᵈ Aₓ a₃ → SSTᵈᵈ A₃ Aₓ (S A₃ a₃)
```

Unfortunately, as we will see this is not actually possible in the version of dTT we have described in this paper, but it is the correct intuition.

Now suppose we want to construct a function mapping into SST from some telescope. We first think purely in terms of code, written in the style of `Agda`-esque copattern matching, with the goal of writing down something that can conceivably be justified:

```
f : X → SST
Z (f t) = (?Z₀ : Type)
S (f t) a = fᵈ t (?S₀ : Xᵈ t)
```

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Here, suppose that $\Gamma,\ \mathbb{\triangle\square} \vdash_{\mathsf{sm}} \Upsilon\ \mathsf{tel}_{\ell'}$. If we think of Υ as a *state space* and $[\ \sigma : \Upsilon\ ]$ as a *state*, then the above definition suggests that we are able to define $f : (\upsilon : \Upsilon) \to \mathsf{SST}_\ell$ provided that we are able to provide two ingredients. First, we need a way of extracting $[\ \widetilde{Z}\ \sigma : \mathsf{Type}_\ell\ ]$, a type of 0-simplices, from a state σ. Second, we need a way of extracting $[\ \widetilde{S}\ \sigma\ a : \Upsilon^{\mathsf{d}}\ \sigma\ ]$, a dependent section of Υ over σ, from a state σ and a 0-simplex $[\ a : \widetilde{Z}\ \sigma\ ]$. This suggests that a reasonable coinduction principle for SST_ℓ is the following:

$$\frac{\Gamma,\ \mathbb{\triangle\square} \vdash_{\mathsf{sm}} \Upsilon\ \mathsf{ctx}_{\ell'} \qquad \Gamma,\ \mathbb{\triangle\square} \vdash_{\mathsf{sm}} \widetilde{Z} : (\!|\ \mathsf{Type}_\ell\ |\!)_{\delta\,:\,\Upsilon} \qquad \Gamma,\ \mathbb{\triangle\square} \vdash_{\mathsf{sm}} \widetilde{S} : (\!|\ \Upsilon^{\mathsf{d}}\ \delta\ |\!)_{\delta\,:\,\Upsilon,\, a\,:\,\mathsf{El}\,(\widetilde{Z}\ \delta)}}{\Gamma \vdash_{\mathsf{sm}} \mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S} : (\!|\ \mathsf{SST}_\ell\ |\!)_{\delta\,:\,\Upsilon}}$$

and that its computation rules should be:

$$\mathsf{Z}\ (\mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S}\ \sigma) \equiv \widetilde{Z}\ \sigma$$
$$\mathsf{S}\ (\mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S}\ \sigma)\ a_{\mathfrak{Z}} \equiv (\mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S})^{\mathsf{d}}\ \langle\ \sigma\ ,\widetilde{S}\ \sigma\ a_{\mathfrak{Z}}\ \rangle.$$

Now $(\mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S})^{\mathsf{d}}$ belongs to $(\!|\ \mathsf{SST}^{\mathsf{d}}_\ell\ (\mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S}\ \upsilon^{+^{\mathsf{ev}}})\ |\!)_{\upsilon^+\,:\,\Upsilon^{\mathsf{D}}}$, and one might hope to compute this using a corecursor for $\mathsf{SST}^{\mathsf{d}}_\ell$. More generally, let us try to work out the coinduction principle for defining $f : (x : X) \to \mathsf{SST}^{\mathsf{d}}_\ell\ (A\ x)$, where $\Gamma,\ \mathbb{\triangle\square} \vdash_{\mathsf{sm}} A : X \to \mathsf{SST}^{\mathsf{d}}_\ell$. Using the previous methodology, we start with reasonable-looking code:

```
f : (t : X) → SST^d (A t)
Z^d (f t) a = (?Z₂ : Type)
S^d (f t) a b = f^d t (?S₃ : X^d t)
```

However, this doesn't actually typecheck, because we have

$$\Gamma,\ \mathbb{\triangle\square},\ t : X,\ a : \mathsf{El}\ (\mathsf{Z}\ (A\ t))\,,\ b : \mathsf{El}\ ?_{Z_2} \vdash_{\mathsf{sm}} \mathsf{S}^{\mathsf{d}}\ (f\ t)\ a\ b : \mathsf{SST}^{\mathsf{dd}}_\ell\ (A\ t)\ (f\ t)\ (\mathsf{S}\ (A\ t)\ a)$$

but

$$\Gamma,\ \mathbb{\triangle\square},\ t : X,\ a : \mathsf{El}\ (\mathsf{Z}\ (A\ t))\,,\ b : \mathsf{El}\ ?_{Z_2} \vdash_{\mathsf{sm}} \qquad f^{\mathsf{d}}\ t\ ?_{S_3} : \mathsf{SST}^{\mathsf{dd}}_\ell\ (A\ t)\ \left(A^{\mathsf{d}}\ t\ ?_{S_3}\right)\ (f\ t)\,.$$

Thus, there is an index ordering mismatch that seems to prevent us from writing down a coinduction principle for $\mathsf{SST}^{\mathsf{d}}_\ell$. The solution is to extend dTT with *symmetries*; this would allow us to line up the $f\ t$ indices and impose the definitional equality $\mathsf{S}\ (A\ t)\ a \equiv A^{\mathsf{d}}\ t\ ?_{S_3}$ as a corecursor premise. In fact, `Narya` implements symmetries, so this is possible there:

```
def f (t : X) : SST^(d) (A t) ≔ [
| .Z ↦ a ↦ (?Z₂ : Type)
| .S ↦ a b ↦ (f^(d) t (?S₃ : X^(d) t))^(21) ]
```

`Narya` doesn't require the $^{(\mathsf{d})}$ on Z and S, since that is implicit in the type $\mathsf{SST}^{(\mathsf{d})}$ of f. Symmetries are also the reason for `Narya`'s parentheses: its $^{(\mathsf{d})}$ is an instance of a general operation that also includes all higher-dimensional symmetries represented by numerical permutation notations, such as the $^{(21)}$ used here for a transposition.

In this paper we have chosen to develop a theory without symmetries, so we must abandon this approach. Instead we can leave $(\mathsf{R}_\Upsilon\ \widetilde{Z}\ \widetilde{S})^{\mathsf{d}}$ as a stuck form, but specify how to compute Z^{d} and S^{d} on it. The main idea is that if we define:

```
f : X → SST
Z (f t) = ʒ t
S (f t) a = f^d t (ꜱ t a)
```

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then we can compute display on each line of this definition to obtain:

```
f^d : (t : X) → X^d t → SST^d (f t)
Z^d (f^d t s) = λ a → ȝ^d t s a
S^d (f^d t s) a₀ a₁ = f^dd t s (ƨ t a₀) (ƨ^d t s a₀ a₁)
```

Thus we obtain the computation laws:

$$\mathsf{Z}^{\mathsf{d}}\left(\left(\mathsf{R}_{\Upsilon}\ \widetilde{\mathsf{Z}}\ \widetilde{\mathsf{S}}\right)^{\mathsf{d}}\ \sigma^{+}\right)\ a \equiv \widetilde{\mathsf{Z}}^{\mathsf{d}}\ \sigma^{+}\ a$$

$$\mathsf{S}^{\mathsf{d}}\left(\left(\mathsf{R}_{\Upsilon}\ \widetilde{\mathsf{Z}}\ \widetilde{\mathsf{S}}\right)^{\mathsf{d}}\ \sigma^{+}\right)\ a_0\ a_1 \equiv \left(\mathsf{R}_{\Upsilon}\ \widetilde{\mathsf{Z}}\ \widetilde{\mathsf{S}}\right)^{\mathsf{dd}}\ \left(\langle\ -\ ,\ -\ \rangle^{\mathsf{D}}\ \left(\sigma^{+}\right)\ \left(\widetilde{\mathsf{S}}^{\mathsf{D}}\ \sigma^{+}\ a_0\ a_1\right)\right)$$

If we think of pairing as a Π-telescope, $\langle\ -\ ,\ -\ \rangle_{\Upsilon} : (\upsilon : \Upsilon) \to \Upsilon^{\mathsf{d}}\ \upsilon \to \Upsilon^{\mathsf{D}}$, then specialising this to Υ^{D} gives $\langle\ -\ ,\ -\ \rangle_{\Upsilon^{\mathsf{D}}} : \left(\upsilon^{+} : \Upsilon^{\mathsf{D}}\right) \to \Upsilon^{\mathsf{D}\mathsf{d}}\ \upsilon^{+} \to \Upsilon^{\mathsf{D}\mathsf{D}}$. On the other hand, taking Π-telescope décalage gives $\langle\ -\ ,\ -\ \rangle^{\mathsf{D}}_{\Upsilon} : \left(\upsilon^{+} : \Upsilon^{\mathsf{D}}\right) \to \Upsilon^{\mathsf{d}\mathsf{D}}\ \upsilon^{+} \to \Upsilon^{\mathsf{D}\mathsf{D}}$. As we shall note in Section 3.3, display and décalage do not commute, so we must be careful to distinguish these operations. In the case of a single type, $\langle\ -\ ,\ -\ \rangle^{\mathsf{D}}_{(x:A)}$ reduces to concatenation, and more generally, $\langle\ -\ ,\ -\ \rangle^{\mathsf{D}}_{\Upsilon}$ reduces to interwoven concatenation for each variable in Υ.

We can iterate this to obtain:

$$\mathsf{Z}^{\mathsf{d}^n}\left(\left(\mathsf{R}_{\Upsilon}\ \widetilde{\mathsf{Z}}\ \widetilde{\mathsf{S}}\right)^{\mathsf{d}^n}\ \sigma^{n}\right)\ \partial a \equiv \widetilde{\mathsf{Z}}^{\mathsf{d}^n}\ \sigma^{n}\ \partial a$$

$$\mathsf{S}^{\mathsf{d}^n}\left(\left(\mathsf{R}_{\Upsilon}\ \widetilde{\mathsf{Z}}\ \widetilde{\mathsf{S}}\right)^{\mathsf{d}^n}\ \sigma^{n}\right)\ \partial a\ a \equiv \left(\mathsf{R}_{\Upsilon}\ \widetilde{\mathsf{Z}}\ \widetilde{\mathsf{S}}\right)^{\mathsf{d}^{n+1}}\ \left(\langle\ -\ ,\ -\ \rangle^{\mathsf{D}^n}\ \left(\sigma^{n}\right)\ \left(\widetilde{\mathsf{S}}^{\mathsf{D}^n}\ \sigma^{n}\ \partial a\ a\right)\right)$$

The situation is not unlike (co)pattern matches in `Agda` (and `Narya`), which define new normal forms rather than expanding to a first-class introduction or elimination form; such names only reduce when their defining patterns occur. ◁

3.2 Examples of semi-simplicial types

Of course, simply *defining* a type of semi-simplicial types is only the first step: we also want to be able to work with such things conveniently. Developing a full theory of semi-simplicial types is beyond the scope of this paper, but in this section we give a few examples to suggest that this may be possible with our definition of SST and its corecursion principle. We will use `Agda`-esque copattern-matching, and assume that our type theory has plenty of other structure rather than the bare-bones version of dTT that we have studied formally in this paper. Many of our examples have been formalised in `Narya`. (However, in attempting to go beyond the examples here, we have found that symmetries are generally unavoidable.)

3.2.1 The singular semi-simplicial types

Display is independent from any notion of propositional equality, whether that be Martin-Löf, cubical, or observational. However, we now want to define a semi-simplicial type that arises from the ∞-groupoid structure of a type in HoTT, and for concreteness we will use a cubical notion of equality. In `Cubical Agda` notation, we have:

$$A : \mathsf{Type}_{\ell},\ x : A,\ y : A \vdash_{\mathsf{sm}} \mathsf{Path}\ A\ x\ y\ \mathsf{type}_{\ell}$$

$$A : \mathsf{Type}_{\ell},\ P : A \to \mathsf{Type}_{\ell},\ x : A,\ x' : P\ x,\ y : A,\ y' : P\ y,\ p : \mathsf{Path}\ A\ x\ y \vdash_{\mathsf{sm}} \mathsf{PathP}\ \left(\lambda\ i.\ P\ \left(p\ i\right)\right)\ x'\ y'\ \mathsf{type}_{\ell}.$$

Thus, it is well-typed to declare the latter to be the display of the former:

$$\left(\!\left(\ \mathsf{Path}\ A\ x\ y\ \right)\!\right)_{A : \mathsf{Type}_{\ell}, x : A, y : A}{}^{\mathsf{d}} \equiv \left(\!\left(\ \mathsf{PathP}\ \left(\lambda\ i.\ P\ \left(p\ i\right)\right)\ x'\ y'\ \right)\!\right)_{A : \mathsf{Type}_{\ell}, P : A \to \mathsf{Type}_{\ell}, x : A, x' : P\ x, y : A, y' : P\ y, p : \mathsf{Path}\ A\ x\ y}.$$

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With this given, the singular semi-simplicial types are defined by corecursion. Rather than write this explicitly using the corecursor from section 3.1, we use a copattern-matching syntax, including a *'displayed corecursive call'* $\mathsf{Sing}^{\mathsf{d}}$:

```
Sing : Type → SST
Z (Sing A) = A
S (Sing A) x = Sing^d A (λ y → Path A x y)
```

A calculation then yields:

```
Z (Sing A) = A
Z^d (S (Sing A) x₀₁) x₁₀ = Path A x₀₁ x₁₀
Z^dd (S^d (S (Sing A) x₀₀₁) x₀₁₀ β₀₁₁) x₁₀₀ β₁₀₁ β₁₁₀
     = PathP (λ i → Path A x₀₀₁ (β₁₁₀ i)) β₀₁₁ β₁₀₁
Z^ddd (S^dd (S^d (S (Sing A) x₀₀₀₁) x₀₀₁₀ β₀₀₁₁) x₀₁₀₀ β₀₁₀₁ β₀₁₁₀ f₀₁₁₁) x₁₀₀₀ β₁₀₀₁ β₁₀₁₀ f₁₀₁₁ β₁₁₀₀ f₁₁₀₁ f₁₁₁₀
     = PathP (λ i → PathP (λ j → Path A x₀₀₀₁ (f₁₁₁₀ i j)) β₀₀₁₁ (f₁₁₀₁ i)) f₀₁₁₁ f₁₀₁₁
```
◁

3.2.2 Nerves of categories

Let Cat denote the type of 1-categories, defined as a record inside dTT (extended by record types), and recall that in section 1 we computed $\mathsf{Cat}^{\mathsf{d}}$ to consist of *'displayed categories'* in the usual sense Ahrens and Lumsdaine (2019). Thus we can define:

```
Nerve : Cat → SST
Z (Nerve 𝒞) = ob 𝒞
S (Nerve 𝒞) x = Nerve^d 𝒞 (coslice 𝒞 x)
```

Here by coslice $\mathcal{C}$ x we mean the coslice category $x/\mathcal{C}$, regarded as a displayed category over $\mathcal{C}$ via the forgetful functor. Note that defining coslice : $(\mathcal{C} : \mathsf{Cat}) \to \mathsf{ob}\ \mathcal{C} \to \mathsf{Cat}^{\mathsf{d}}\ \mathcal{C}$ globally in dTT automatically induces the definition of $\mathsf{coslice}^{\mathsf{d}}$. A similar idea works for bicategories, and any other kind of category that has a displayed (co)slice. ◁

3.2.3 Topological singular complexes

For any type Top of *'topological space'* definable inside of dTT as a record, we have a displayed version $\mathsf{Top}^{\mathsf{d}}$. In some cases an element of the resulting $\mathsf{Top}^{\mathsf{d}}\ X$ is more general than an $Y : \mathsf{Top}$ with a map $Y \to X$; but at least *from* such a Y we can construct its fibers as a displayed space. Thus, as long as we can construct, for any $x : X$, a space of *'continuous paths in X starting at x'* with an endpoint projection down to X, we can make it a displayed space paths X x over X, and use this to construct the singular semi-simplicial types:

```
Sing : Top → SST
Z (Sing X) = pt X
S (Sing X) x = Sing^d X (paths X x)
```
◁

3.2.4 Fibers and higher spans

As we will see in section 4, semantically *each* type at mode sm is already an augmented semi-simplicial type. Thus, we expect that if we fix a (-1)-simplex in such an augmented semi-simplicial type, we should obtain an (unaugmented) semi-simplicial type as its *'fibre'*. And indeed, we can define this operation:

```
Fib : (X :^□ Type) (ʒ : X) → SST
Z (Fib X ʒ) = X^d ʒ
S (Fib X ʒ) x = Fib^d X ʒ x
```

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Note that X is required to be modal so that we can take display of it. Then we have, for instance, if $\mathfrak{Z}_{00} : X$ and $x_{01}, x_{10} : X^{\mathsf{d}}\ \mathfrak{Z}_{00}$, that:

$$(\mathsf{Fib}\ X\ \mathfrak{Z}_{00})_0 \equiv \mathsf{Z}\ (\mathsf{Fib}\ X\ \mathfrak{Z}_{00}) \equiv X^{\mathsf{d}}\ \mathfrak{Z}_{00}$$
$$(\mathsf{Fib}\ X\ \mathfrak{Z}_{00})_1\ x_{01}\ x_{10} \equiv \mathsf{Z}^{\mathsf{d}}\ (\mathsf{S}\ (\mathsf{Fib}\ X\ \mathfrak{Z}_{00})\ x_{01})\ x_{10} \equiv \mathsf{Z}^{\mathsf{d}}\ (\mathsf{Fib}^{\mathsf{d}}\ X\ \mathfrak{Z}_{00}\ x_{01})\ x_{10} \equiv X^{\mathsf{dd}}\ \mathfrak{Z}_{00}\ x_{01}\ x_{10}$$

and as a last example:

$$\begin{aligned}(\mathsf{Fib}\ X\ \mathfrak{Z}_{000})_2\ x_{001}\ x_{010}\ \beta_{011}\ x_{100}\ \beta_{101}\ \beta_{110} &\equiv \mathsf{Z}^{\mathsf{dd}}\ (\mathsf{S}^{\mathsf{d}}\ (\mathsf{S}\ (\mathsf{Fib}\ X\ \mathfrak{Z}_{000})\ x_{001})\ x_{010}\ \beta_{011})\ x_{100}\ \beta_{101}\ \beta_{110}\\ &\equiv \mathsf{Z}^{\mathsf{dd}}\ (\mathsf{S}^{\mathsf{d}}\ (\mathsf{Fib}^{\mathsf{d}}\ X\ \mathfrak{Z}_{000}\ x_{001})\ x_{010}\ \beta_{011})\ x_{100}\ \beta_{101}\ \beta_{110}\\ &\equiv \mathsf{Z}^{\mathsf{dd}}\ (\mathsf{Fib}^{\mathsf{dd}}\ X\ \mathfrak{Z}_{000}\ x_{001}\ x_{010}\ \beta_{011})\ x_{100}\ \beta_{101}\ \beta_{110}\\ &\equiv X^{\mathsf{ddd}}\ \mathfrak{Z}_{000}\ x_{001}\ x_{010}\ \beta_{011}\ x_{100}\ \beta_{101}\ \beta_{110}.\end{aligned}$$

In particular, if we let $X = \mathsf{Type}_\ell$ be a universe and $\mathfrak{Z} = \top$ be a unit type, we have:

$$(\mathsf{Fib}\ \mathsf{Type}_\ell\ \top)_0 \equiv \mathsf{Type}_\ell^{\mathsf{d}}\ \top \equiv (\top \to \mathsf{Type}_\ell) \cong \mathsf{Type}_\ell$$
$$(\mathsf{Fib}\ \mathsf{Type}_\ell\ \top)_1\ X_{01}\ X_{10} \equiv \mathsf{Type}_\ell^{\mathsf{dd}}\ \top\ X_{01}\ X_{10} \cong X_{01} \to X_{10} \to \mathsf{Type}_\ell$$
$$\begin{aligned}(\mathsf{Fib}\ \mathsf{Type}_\ell\ \top)_2\ X_{001}\ X_{010}\ B_{011}\ X_{100}\ B_{101}\ B_{110} &\equiv \mathsf{Type}_\ell^{\mathsf{ddd}}\ \top\ X_{001}\ X_{010}\ B_{011}\ X_{100}\ B_{101}\ B_{110}\\ &\cong (x_{001} : X_{001})(x_{010} : X_{010})(\beta_{011} : B_{011}\ x_{001}\ x_{010})\\ &\quad (x_{100} : X_{100})(\beta_{101} : B_{101}\ x_{001}\ x_{100})(\beta_{110} : B_{110}\ x_{010}\ x_{100}) \to \mathsf{Type}_\ell\end{aligned}$$

Thus $\mathsf{Fib}\ \mathsf{Type}_\ell\ \top$ is the semi-simplicial type of types, spans, and a sort of simplicial *'higher spans'* that could also be called *'heterogeneous simplices'*. More generally, $\mathsf{Fib}\ \mathsf{Type}_\ell\ A$ for any type A consists of types, spans, and simplicial higher spans indexed by A. ◁

3.2.5 Operations on semi-simplicial types

We can also use corecursion to define operations on semi-simplicial types that are essentially levelwise. For instance, any two semi-simplicial types have a product:

```
_×_ : SST → SST → SST
Z (X × Y) = Z X × Z Y
S (X × Y) ⟨ x , y ⟩ = (S X x) ×ᵈ (S Y y)
```

Here we treat the non-displayed arguments of $\times^{\mathsf{d}}$ as implicit – its full type is:

$$_\times^{\mathsf{d}}_ : \{X : \mathsf{SST}\}\ (X' : \mathsf{SST}^{\mathsf{d}}\ X)\ \{Y : \mathsf{SST}\}\ (Y' : \mathsf{SST}^{\mathsf{d}}\ Y) \to \mathsf{SST}^{\mathsf{d}}\ (X \times Y)$$

There is an empty semi-simplicial type. Note that the S case can be omitted, since one of its arguments would belong to the empty type $\bot$.

```
∅ : SST
Z ∅ = ⊥
```

Similarly, there is a trivial one:

```
⊤ : SST
Z ⊤ = ⊤
S ⊤ u = ⊤ᵈ
```

We can also take the product of any family of semi-simplicial types indexed by a *discrete* type. The discreteness of A means that it doesn't need a displayed version in $\bigtimes^{\mathsf{d}}$.

```
⨯ : (A :^△ Disc) → ((a :^△ A)) → SST) → SST
Z (⨯ A X) = ((a :^△ A) → Z (X a))
S (⨯ A X) p = ⨯ᵈ A X (λ a → S (X a) (p a))
```

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However, some things do not seem possible with our current theory. For example, the disjoint union of semi-simplicial types should certainly have the disjoint union of 0-simplices, but the slice over a 0-simplex should come only from one side: $\mathsf{S}\ (X + Y)\ (\mathsf{inl}\ x)$ should morally be $\mathsf{S}\ X\ x$. But $\mathsf{S}\ X\ x$ belongs to $\mathsf{SST}^{\mathsf{d}}\ X$, whereas $\mathsf{S}\ (X + Y)\ (\mathsf{inl}\ x)$ must belong to $\mathsf{SST}^{\mathsf{d}}\ (X + Y)$. We need to take its disjoint union with an *'empty semi-simplicial type displayed over Y'*; but what is that? We defined a *'global'* empty semi-simplicial type above, and it seems intuitively that we should be able to define a *'constant'* version of this displayed over Y. But as noted in section 3.1, without symmetry it does not seem possible to formulate a useful corecursor for $\mathsf{SST}^{\mathsf{d}}$, so it is unclear how to define *'constantly displayed'* semi-simplicial types. There are many other examples of this sort that suggest that further work in this direction will require the inclusion of symmetries. ◁

3.3 Displayed coinductive types

Generalising the discussion of SST, we now formulate a fairly general notion of *'indexed displayed coinductive type'*. It depends on a telescope Φ of *'non-uniform parameters'*, and every element of it has a *'head'* belonging to some specified type family A and a *'tail'*, depending on a telescope of parameters $\mathcal{B}$, and belonging to the displayed version of the coinductive type itself. The parameters of this displayed version of the very type being defined are Φ^{D}, which we can assemble provided that the data of the old parameters $\varphi : \Phi$, the head $x : A\ \varphi$, and the new dependencies $\mathfrak{b} : \mathcal{B}\ \varphi\ a$, are sufficient to extract a section $\sigma\ \varphi\ x\ \mathfrak{b} : \Phi^{\mathsf{d}}\ \varphi$. The idea is analogous to an *'indexed M-type'*, but with the output of the tail being displayed, and with $\mathcal{B}$ being a telescope rather than a simple type. The pseudo-Agda corresponding to this would be:

```
module _ (Φ :^△□ Tel) (A :^△□ Φ → Type) (B :^△□ (φ : Φ) (a : A φ) → Tel)
         (σ :^△□ (φ : Φ) (a : A φ) (b : B φ a) → Φ^d φ) where
  codata dCoind (φ : Φ) : Type where
    head : dCoind φ → A φ
    tail : (x : dCoind φ) (b : B φ (head x)) → dCoind^d ⟨ φ , σ φ (head x) b ⟩ x
```

We can thus write down the formation and introduction rules for dCoind as follows:

$$\frac{\begin{array}{c}\Gamma,\ \text{🔒}_{\triangle\square} \vdash_{\mathsf{sm}} \Phi\ \mathsf{tel}_{\ell_0} \qquad \Gamma,\ \text{🔒}_{\triangle\square} \vdash_{\mathsf{sm}} A\ \ \mathsf{type}_{\ell_1}\big/_{\varphi:\Phi} \\ \Gamma,\ \text{🔒}_{\triangle\square} \vdash_{\mathsf{sm}} \mathcal{B}\ \ \mathsf{tel}_{\ell_2}\big/_{\varphi:\Phi,\, a:A\,\varphi} \qquad \Gamma,\ \text{🔒}_{\triangle\square} \vdash_{\mathsf{sm}} \sigma : \big(\!\big(\ \Phi^{\mathsf{d}}\ \varphi\ \big)\!\big)_{\varphi:\Phi,\, a:A\,\varphi,\, \mathfrak{b}:\mathcal{B}\,\varphi\,x}\end{array}}{\begin{array}{c}\Gamma \vdash_{\mathsf{sm}} \mathsf{dCoind}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ \ \mathsf{type}_{\ell_1 \sqcup \ell_2}\big/_{\varphi:\Phi\ [\ \text{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\ ]} \\ \Gamma \vdash_{\mathsf{sm}} \mathsf{head}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]} : \big(\!\big(\ A\ [\ \text{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\ ]\ \varphi\ \big)\!\big)_{\varphi,\, x:\mathsf{dCoind}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\,\varphi} \\ \Gamma \vdash_{\mathsf{sm}} \mathsf{tail}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]} : \big(\!\big(\ \mathsf{dCoind}^{\mathsf{d}}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ \langle\ \varphi\ ,\sigma\ \varphi\ \ (\mathsf{head}\ x)\ \ \mathfrak{b}\ \rangle\ x\ \big)\!\big)_{\varphi,\, x,\, \mathfrak{b}:\mathcal{B}[\ \text{🔑}^{\triangle\square \le 1_{\mathsf{sm}}}\ ]\,\varphi\ (\mathsf{head}\,\varphi\,x)}\end{array}}$$

Note that the universe level of dCoind is governed by those of A and $\mathcal{B}$, but does not depend on the level of the telescope of non-uniform parameters Φ.

Following the example of SST, we will begin by attempting to write down a reasonable template for a coinduction principle. In the same module context, we can attempt to map into a dCoind type from a length two context as follows:

```
f : (t : X) (s : Y t) → dCoind (ϕ t s)
head (f t s) = (?h1 : A (ϕ t s))
tail (f t s) b = f^d t (?t1 : X^d t) s (?t2 : Y^d t ?t1 s)
```

The types that we have are then:

$$\mathsf{tail}\ \ (f\ t\ s)\ \ \mathfrak{b} : \mathsf{dCoind}^{\mathsf{d}}\ \langle\ \phi\ t\ s\ ,\sigma\ \ (\phi\ t\ s)\ \ ?_{\mathsf{h}_1}\ \ \mathfrak{b}\ \rangle\ \ (f\ t\ s)$$

$$f^{\mathsf{d}}\ t\ ?_{\mathsf{t}_1}\ s\ ?_{\mathsf{t}_2} : \mathsf{dCoind}^{\mathsf{d}}\ \langle\ \phi\ t\ s\ ,\phi^{\mathsf{d}}\ t\ ?_{\mathsf{t}_1}\ s\ ?_{\mathsf{t}_2}\ \rangle\ \ (f\ t\ s)$$

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Thus there is a non-trivial condition that needs to be imposed for this definition template to be well typed. Fortunately, unlike in the case of $\mathsf{SST}^{\mathsf{d}}$, we generally have terms lining up in the sense that the terminal $(f\ t\ s)$ terms align. We get the following rule:

$$\frac{\begin{array}{c}\Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \Upsilon\ \mathsf{tel}_{\ell'} \qquad \Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \phi : (\!|\ \Phi\ |\!)_{\upsilon\,:\,\Upsilon} \\ \Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \widetilde{h} : (\!|\ A\ \ (\phi\ \upsilon)\ |\!)_{\upsilon\,:\,\Upsilon} \qquad \Gamma,\ \blacksquare_{\triangle\square} \vdash_{\mathsf{sm}} \widetilde{\tau} : (\!|\ \Upsilon^{\mathsf{d}}\ \upsilon\ |\!)_{\upsilon\,:\,\Upsilon,\ \mathfrak{b}\,:\,\mathcal{B}\ (\phi\ \upsilon)\ (\widetilde{h}\ \upsilon)} \\ \Gamma,\ \blacksquare_{\triangle\square} \mid \upsilon : \Upsilon,\ \mathfrak{b} : \mathcal{B}\ (\phi\ \upsilon)\ (\widetilde{h}\ \upsilon) \vdash \phi^{\mathsf{d}}\ \langle\ \upsilon\ ,\widetilde{\tau}\ \upsilon\ \mathfrak{b}\ \rangle \equiv \sigma\ (\phi\ \upsilon)\ (\widetilde{h}\ \upsilon)\ \mathfrak{b}\end{array}}{\Gamma \vdash_{\mathsf{sm}} \mathsf{corec}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ [\ \Upsilon,\ \phi,\ \widetilde{h},\ \widetilde{\tau}\ ] : (\!|\ \mathsf{dCoind}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ (\phi\ \upsilon)\ |\!)_{\upsilon\,:\,\Upsilon}}$$

This comes with the following computation rules:

$$\mathsf{head}\ (\phi\ \upsilon)\ \left(\mathsf{corec}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ [\ \Upsilon,\ \phi,\ \widetilde{h},\ \widetilde{\tau}\ ]\ \upsilon\right) \equiv \widetilde{h}\ \upsilon$$

$$\mathsf{tail}\ (\phi\ \upsilon)\ \left(\mathsf{corec}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ [\ \Upsilon,\ \phi,\ \widetilde{h},\ \widetilde{\tau}\ ]\ \upsilon\right)\ \mathfrak{b} \equiv \left(\mathsf{corec}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ [\ \Upsilon,\ \phi,\ \widetilde{h},\ \widetilde{\tau}\ ]\right)^{\mathsf{d}}\ \langle\ \upsilon\ ,\widetilde{\tau}\ \upsilon\ \mathfrak{b}\ \rangle$$

Now we can define SST as a particular indexed displayed coinductive type (which happens to have trivial indexing). In fact, it is the *'universal'* such instance, where the family $\mathcal{B}$ indexed by A is the universal family El indexed by Type_{ℓ}. This may be compared with the fact that the W-type of El is the type of *'presentations of well-founded sets'* Aczel (1978), while its M-type is the type of *'presentations of ill-founded sets'* Lindström (1989).

$$\mathsf{SST}_{\ell} \equiv \mathsf{dCoind}\ \left[\ ()_{\mathsf{sm}},\ (\!|\ \mathsf{Type}_{\ell}\ |\!)_{\varphi\,:\,()_{\mathsf{sm}}},\ (\!|\ \mathsf{El}\ X\ |\!)_{X\,:\,\mathsf{Type}_{\ell}},\ [\![\ ]\!]_{\mathsf{sm}}\ ]\!]_{X\,:\,\mathsf{Type}_{\ell}, x\,:\,\mathsf{El}\ X}\ \right]$$

Then we can *deduce* the rules for SST_{ℓ}, defining $\mathsf{Z} \equiv \mathsf{head}$ and $\mathsf{S} \equiv \mathsf{tail}$ and $\mathsf{R} \equiv \mathsf{corec}$.

The problem of giving a corecursion rule for $\mathsf{SST}^{\mathsf{d}}$ carries over to the general case. Specifically, just as ^{d} of a Π-type is another Π-type and so on for records and ordinary coinductive types, we'd like to compute ^{d} of a dCoind to be another dCoind, with something like this:

$$\begin{array}{rc}\mathsf{dCoind}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}{}^{\mathsf{d}} \equiv \mathsf{dCoind}\ \Big[& \left(\varphi^{+} : \Phi^{\mathsf{D}},\ c : \mathsf{dCoind}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ \varphi^{+\mathsf{ev}}\right), \\ \text{¿} & (\!|\ A^{\mathsf{d}}\ \varphi^{+}\ \ (\mathsf{head}\ c)\ |\!)_{\varphi^{+}, c},\ (\!|\ \mathcal{B}^{\mathsf{D}}\ \varphi^{+}\ \ (\mathsf{head}\ c)\ \ a'\ |\!)_{\varphi^{+}, x, a'}, \qquad ? \\ & [\![\ \sigma^{\mathsf{D}}\ \varphi^{+}\ [\ \mathsf{head}\ c,\ a'\ ]\ \mathfrak{b}^{+},\ \mathsf{tail}\ c\ \mathfrak{b}^{+\mathsf{ev}}\ ]\!]_{\varphi^{+}, c, a', \mathfrak{b}^{+}}\ \Big]\end{array}$$

To see whether this is well-typed, observe that we have:

$$\varphi : \Phi,\ a : A\ \varphi,\ \mathfrak{b} : \mathcal{B}\ \varphi\ a \vdash_{\mathsf{sm}} \sigma\ \varphi\ a\ \mathfrak{b} : \Phi^{\mathsf{d}}\ \varphi$$

$$\varphi^{+} : \Phi^{\mathsf{D}},\ a : A,\ a' : A^{\mathsf{d}}\ a,\ \mathfrak{b}^{+} : \mathcal{B}^{\mathsf{D}}\ \varphi^{+}\ [\ a,\ a'\ ] \vdash_{\mathsf{sm}} \sigma^{\mathsf{D}}\ \varphi^{+}\ [\ a,\ a'\ ]\ \mathfrak{b}^{+} : \Phi^{\mathsf{dD}}\ \varphi^{+}$$

whereas the σ of the resulting dCoind must lie in $\left(\varphi^{+} : \Phi^{\mathsf{D}},\ c : \mathsf{dCoind}_{[\ \Phi, A, \mathcal{B}, \sigma\ ]}\ \varphi^{+\mathsf{ev}}\right)^{\mathsf{d}}$. Thus, in particular, we need to compare $\Phi^{\mathsf{dD}}\ \varphi^{+}$ to $\Phi^{\mathsf{Dd}}\ \varphi^{+}$, where $\varphi^{+} : \Phi^{\mathsf{D}}$. In the case of a one-type telescope $\Phi = (a : A)$, this becomes:

$$\Phi^{\mathsf{D}} \equiv \left(a : A,\ a' : A^{\mathsf{d}}\ a\right)$$

$$\Phi^{\mathsf{d}}\ a \equiv \left(a' : A^{\mathsf{d}}\ a\right)$$

$$\Phi^{\mathsf{dD}}\ [\ a, a'\ ] \equiv \left(a'' : A^{\mathsf{d}}\ a,\ a''' : A^{\mathsf{dd}}\ a\ a'\ a''\right)$$

$$\Phi^{\mathsf{Dd}}\ [\ a, a'\ ] \equiv \left(a'' : A^{\mathsf{d}}\ a,\ a''' : A^{\mathsf{dd}}\ a\ a''\ a'\right)$$

Unfortunately the last two are *not* the same! This is not just about ordering the variables in a telescope; although the second and third arguments of A^{dd} both lie in $A^{\mathsf{d}}\ a$, it need not be symmetrical with respect to those arguments. So again we see that without adding symmetry to the

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theory, it seems we can't give a general corecursor for $\mathsf{dCoind}^{\mathsf{d}}$, and hence we can't compute $\mathsf{corec}^{\mathsf{d}}$ to something more primitive.

In `Narya`, which does have symmetries, the analogues of $\mathsf{dCoind}^{\mathsf{d}}$ are again displayed coinductive types, which therefore permit copattern-matching with displayed corecursive calls; although specific constructions like $\mathsf{corec}^{\mathsf{d}}$ do not actually compute to anything until their copatterns apply, as with ordinary corecursive definitions. However, since `Narya` presently lacks modalities, we cannot formulate in it a *generic* displayed coinductive such as dCoind with arbitrary parameters such as Φ, A, $\mathcal{B}$, only specific examples. ◁

3.4 Examples of displayed coinductive types

We continue our exploration of the theory of semi-simplicial types from section 3.2, now using the general notion of displayed coinductive type. As in section 3.2, we will use Agda-esque codata and copattern-matching definitions, and assume that our type theory has plenty of other structure.

We have already noted that SST is in some sense the *'universal'* (unparametrised) displayed coinductive type, whose determining family $x : A \vdash \mathcal{B}\ x\ \mathsf{type}_{\ell_2}$ is the universal one $X : \mathsf{Type}_\ell \vdash \mathsf{El}\ X\ \mathsf{type}_\ell$. Moreover, it seems likely that in order for an unparametrised displayed coinductive type to be interesting, the types A and $\mathcal{B}$ must have nontrivial display structure, i.e. they must not be discrete. But the simplicial universe Type_ℓ is the primary source of types with nontrivial display, just as the universe in homotopy type theory is a primary source of types with higher homotopy structure. (In section 5.9 we will speculate about a notion of *'display inductive type'* analogous to higher inductive types, which are the other source of higher homotopy structure in homotopy type theory.) For these reasons, we do not have a lot of interesting examples of other *unparametrised* displayed coinductive types, but there is at least one: augmented semi-simplicial types.

3.4.1 Augmented semi-simplicial types

If we simply omit the Z input of S in the definition of SST, we obtain a definition of *augmented* semi-simplicial types:

```
codata ASST : Type where
  Z⁺ : ASST → Type
  S⁺ : (A : ASST) → ASSTᵈ A
```

We can extract the types of low-dimensional simplices from an $X : \mathsf{ASST}$:

$$\vdash X_{-1} \equiv \mathsf{Z}^+\ X$$

$$\mathfrak{z}_0 : X_{-1} \vdash X_0\ \mathfrak{z}_0 \equiv \mathsf{Z}^{+\mathsf{d}}\ (\mathsf{S}^+\ X)$$

$$\mathfrak{z}_{00} : X_{-1},\ x_{01} : X_0\ \mathfrak{z}_{00},\ x_{10} : X_0\ \mathfrak{z}_{00} \vdash X_1\ \mathfrak{z}_{00}\ x_{01}\ x_{10} \equiv \mathsf{Z}^{+\mathsf{dd}}\ (\mathsf{S}^{+\mathsf{d}}\ (\mathsf{S}^+\ X))\ \mathfrak{z}_{00}\ x_{01}\ x_{10}$$

and so on. Now we can observe that the construction Fib of section 3.2.4 factors through ASST via a pair of maps, both defined by copattern-matching:

```
Int : (X :^□ Type) → ASST
Z⁺ (Int X) = X
S⁺ (Int X) = Intᵈ X

Fib′ : (X : ASST) (x : Z⁺ X) → SST
Z (Fib′ X x) = Z^{+d} (S⁺ X) x
S (Fib′ X x) y = Fib′ᵈ X (S⁺ X) x y
```
◁

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3.4.2 Pointed semi-simplicial types

More interesting examples of displayed coinductive types have nontrivial parametrisations, often involving more semi-simplicial types. For instance, we can define the structure of a *pointing* on a semi-simplicial type displayed-coinductively:

```
codata Pt (X : SST) : Type where
  zp : Pt X → Z X
  sp : (p : Pt X) → Pt^d X (S X (zp p)) p
```

We then have, for $p : \mathsf{Pt}\ X$, that:

$$\mathsf{zp}\ p : \mathsf{Z}\ X \equiv X_0$$
$$\mathsf{zp}^{\mathsf{d}}\ (\mathsf{sp}\ p) : \mathsf{Z}^{\mathsf{d}}\ (\mathsf{S}\ X\ (\mathsf{zp}\ p))\ (\mathsf{zp}\ p) \equiv X_1\ (\mathsf{zp}\ p)\ (\mathsf{zp}\ p)$$
$$\mathsf{zp}^{\mathsf{dd}}\ (\mathsf{sp}^{\mathsf{d}}\ (\mathsf{sp}\ p)) : X_2\ (\mathsf{zp}\ p)\ (\mathsf{zp}\ p)\ (\mathsf{zp}^{\mathsf{d}}\ (\mathsf{sp}\ p))\ (\mathsf{zp}\ p)\ (\mathsf{zp}^{\mathsf{d}}\ (\mathsf{sp}\ p))\ (\mathsf{zp}^{\mathsf{d}}\ (\mathsf{sp}\ p))$$

and so on. That is, an element of $\mathsf{Pt}\ X$ equips X with a *'fat point'*, i.e. a chosen 0-simplex zp that comes with all of the higher *'degenerate simplices'* that one would expect to be associated to zp if it were in a *simplicial* set rather than a semi-simplicial one. ◁

3.4.3 Morphisms of semi-simplicial types

With a double parametrisation, we can define a type of morphisms of semi-simplicial types:

```
codata Hom (X Y : SST) : Type where
  zhom : Hom X Y → Z X → Z Y
  shom : (f : Hom X Y) (x : Z X) → Hom^d X (S X x) Y (S Y (zhom f x)) f
```

As usual, we can unravel this a few steps to see what it looks like. $\mathsf{zhom}\ f$ is a function between types of 0-simplices, which we may denote f_0. At the next dimension we have:

$$\mathsf{zhom}^{\mathsf{d}}\ (\mathsf{shom}\ f\ x_{01})\ x_{01}\ \beta_{11} : \mathsf{Z}^{\mathsf{d}}\ (\mathsf{S}\ Y\ (\mathsf{zhom}\ f\ x_{01}))\ (\mathsf{zhom}\ f\ x_{01})$$

which is to say that:

$$\mathsf{zhom}^{\mathsf{d}}\ (\mathsf{shom}\ f\ x_{01})\ x_{01} : X_1\ x_{01}\ x_{10} \to Y_1\ (f_0\ x_{01})\ (f_0\ x_{10}).$$

We may denote this function by f_1, and go on to extract a function f_2 between types of 2-simplices and so on. We expect other basic operations on semi-simplicial types to be internalisable in a similar way. ◁

4. Semantics

We now construct, from any model $\mathcal{C}$ of ordinary dependent type theory with limits of countably infinite towers of types, a model of dTT in which $\mathcal{C}$ sits as the discrete mode.

Remark 6. For simplicity and space, we will omit the type-former $\triangle$. It should be possible to model $\triangle$ as well, as long as $\mathcal{C}$ has a unit type, but we leave the details for the future. We will still model $\triangle$-modal variables and function types such as $(x :^{\triangle} A) \to B\ x$.

In section 4.1 we extend the semantics of dependent type theory to telescopes and meta-abstractions, plus countable infinite limits. Then in section 4.2 we use these limits to construct a model in augmented semi-simplicial Reedy diagrams. This is essentially an instance of Shulman (2015), Kapulkin and Lumsdaine (2021), but our explicit version builds in display and décalage. In section 4.3 we add modalities, and in section 4.4 we discuss the *general* notion of model of dTT. Finally, in section 4.5 we construct displayed coinductive types in these models, including SST.

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4.1 The semantics of dependent type theory

4.1.1 CwFs and natural models

As our notion of model for type theory we will use *categories with families* (CwF) Castellan et al. (2021), in their equivalent presentation as *natural models* Awodey (2018), with universe levels:

Definition 7. A *natural model with levels* is a category $\mathcal{C}$ with a terminal object and a family of algebraically representable natural transformations $\mathsf{pr}_\ell : \mathsf{Tm}_\ell \to \mathsf{Ty}_\ell$ of presheaves on $\mathcal{C}$.

We view a presheaf X on $\mathcal{C}$ as a *hypothetical judgment form*, and elements of $X(\Gamma)$ as X-judgments in context Γ. For instance, $A \in \mathsf{Ty}_\ell(\Gamma)$ is the judgment $\Gamma \vdash A\ \mathsf{type}_\ell$, which we will also write as $\gamma : \Gamma \vdash A\ \gamma\ \mathsf{type}_\ell$ to emphasize the dependence of A on Γ. We denote the functorial action of $\sigma : \Delta \to \Gamma$ on types A and terms t by A^σ and t^σ:

$$\delta : \Delta \vdash A^\sigma\ \delta\ \mathsf{type}_\ell \qquad\qquad \delta : \Delta \vdash t^\sigma\ \delta : A^\sigma\ \delta$$

or equivalently by taking the formal variables γ, δ more seriously and *'applying'* σ to them:

$$\delta : \Delta \vdash A\ (\sigma\ \delta)\ \mathsf{type}_\ell \qquad\qquad \delta : \Delta \vdash t\ (\sigma\ \delta) : A\ (\sigma\ \delta)$$

This can be formalized using the internal logic of presheaves. Similarly, plain sets such as the objects and morphisms of $\mathcal{C}$ can be viewed as *'absolute'* or *'context-less' judgments*.

The representability of pr gives, for each type $\gamma : \Gamma \vdash A\ \ \gamma\ \mathsf{type}_\ell$, a *context extension* $(\gamma : \Gamma,\ a : A\ \gamma)$, with a family of bijections, natural in Δ:

$$(\Delta \to (\gamma : \Gamma,\ a : A\ \gamma)) \simeq ((\sigma\ :\ \Delta \to \Gamma) \times (\delta : \Delta \vdash t\ \delta : A\ \ (\sigma\ \delta)))$$

By the Yoneda lemma, the forwards direction of this bijection is determined by a *fundamental context projection* or *parent map*, and a *zero variable* term:

$$\mathsf{pt}^A : (\gamma : \Gamma,\ a : A\ \gamma) \twoheadrightarrow \Gamma \qquad\qquad \delta : (\gamma : \Gamma,\ a : A\ \gamma) \vdash \mathsf{zv}^A\ \delta : A\ \ \left(\mathsf{pt}^A\ \delta\right).$$

Then the forwards direction of the bijection sends a substitution $\tau : \Delta \to (\gamma : \Gamma,\ a : A\ \gamma)$ to the pair $\left(\mathsf{pt}^A \circ \tau,\ \ \left(\mathsf{zv}^A\right)^\tau\right)$, and its inverse is a substitution extension operation:

$$\frac{\sigma : \Delta \to \Gamma \qquad \delta : \Delta \vdash t\ \delta : A\ \ (\sigma\ \delta)}{[\ \sigma,\ t\ ] : \Delta \to (\gamma : \Gamma,\ a : A\ \gamma)}$$

such that for $\sigma : \Delta \to \Gamma$ and $\delta : \Delta \vdash t\ \delta : A\ \ (\sigma\ \delta)$ and $\tau : \Delta \to (\gamma : \Gamma,\ a : A\ \gamma)$, we have:

$$\mathsf{pt}^A \circ [\ \sigma,\ t\ ] \equiv \sigma \qquad \left(\mathsf{zv}^A\right)^{[\ \sigma,\ t\ ]} \equiv t \qquad [\ \mathsf{pt}^A \circ \tau,\ \ \left(\mathsf{zv}^A\right)^\tau\ ] \equiv \tau. \tag{8}$$

As a corollary of this, the following diagram is a pullback:

$$\begin{array}{ccc} (\delta : \Delta,\ a : A\ \ (\sigma\ \delta)) & \xrightarrow{[\ \sigma \circ \mathsf{pt}^{A^\sigma},\ \mathsf{zv}^{A^\sigma}\ ]} & (\gamma : \Gamma,\ a : A\ \gamma) \\ \downarrow{\scriptstyle \mathsf{pt}^{A^\sigma}} & \lrcorner & \downarrow{\scriptstyle \mathsf{pt}^A} \\ \Delta & \xrightarrow[\sigma]{} & \Gamma \end{array}$$

We call the map in the top row above the *weakening two* of σ by A, and write it $\mathsf{W}_2^A\ \sigma$. Note that in Altenkirch et al. (1995), they define $\mathsf{W}_1^A\ \sigma \equiv \sigma\ \circ\ \mathsf{pt}^A : (\delta : \Delta,\ a : A) \to \Gamma$ for $\delta : \Delta \vdash A\ \delta\ \mathsf{type}_\ell$.

Finally, instead of $\delta : (\gamma : \Gamma,\ a : A\ \gamma) \vdash B\ \delta\ \mathsf{type}_{\ell_1}$, we may write $\gamma : \Gamma,\ a : A\ \gamma \vdash B\ \gamma\ a\ \mathsf{type}_{\ell_1}$, and so on. Thus, in particular, zv is just $\gamma : \Gamma,\ a : A\ \gamma \vdash a : A\ \gamma$. As before, this can be justified formally as an interpretation in the internal type theory of $\mathsf{Set}^{\mathcal{C}^{\mathrm{op}}}$. ◁

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4.1.2 Polynomial endofunctors

Each representable map pr induces a polynomial endofunctor P_{pr} of $\mathsf{Set}^{\mathcal{C}^{\mathrm{op}}}$, where for any presheaf (i.e. judgment) X, elements of $P_{\mathsf{pr}}(X)(\Gamma)$ (i.e. $P_{\mathsf{pr}}(X)$-judgments in context Γ) are bijective to pairs of a type $A \in \mathsf{Ty}(\Gamma)$ in context Γ and an element of $X(\gamma : \Gamma, a : A\ \gamma)$.

Similarly, $(P_{\mathsf{pr}} \circ P_{\mathsf{pr}})(X)$ contains X-judgments in a context $(\gamma : \Gamma,\ a : A\ \gamma,\ b : B\ \gamma\ a)$. Thus, it is the polynomial functor associated to the map $\mathsf{pr}^2 : \mathsf{Tm}^2 \to P_{\mathsf{pr}}(\mathsf{Ty})$ where the fiber of $\mathsf{Tm}^2(\Gamma)$ over (A, B) consists of a pair of terms $\gamma : \Gamma \vdash a : A\ \gamma$ and $\gamma : \Gamma \vdash b : B\ \gamma\ a$.

For example, $P_{\mathsf{pr}_{\ell_0}}(\mathsf{Ty}_{\ell_1})$ represents families of types of level ℓ_1 indexed by a type of level ℓ_0. Thus the formation rule of Π-types:

$$\frac{\gamma : \Gamma \vdash A\ \gamma\ \mathsf{type}_{\ell_0} \qquad \gamma : \Gamma,\ a : A\ \gamma \vdash B\ \gamma\ a\ \mathsf{type}_{\ell_1}}{\gamma : \Gamma \vdash (\Pi\ A\ B)\ \ \gamma\ \mathsf{type}_{\ell_0 \sqcup \ell_1}}$$

is represented by a morphism $\Pi : P_{\mathsf{pr}_{\ell_0}}(\mathsf{Ty}_{\ell_1}) \to \mathsf{Ty}_{\ell_0 \sqcup \ell_1}$. Other types and term rules can also be represented this way; see Awodey (2018). We will assume Π-types and universes. ◁

4.1.3 Telescopes

Definition 9. A natural model with levels $\mathcal{C}$ **has telescopes** if it is equipped with:

- Another family of representable transformations $\mathsf{tpr}_\ell : \mathsf{PSub}_\ell \to \mathsf{Tel}_\ell$.
- Morphisms of polynomial functors $1_{\mathcal{C}} \to P_{\mathsf{tpr}_\ell}$, yielding empty telescopes.
- Morphisms of polynomial functors $P_{\mathsf{tpr}_\ell} \circ P_{\mathsf{pr}_{\ell'}} \to P_{\mathsf{tpr}_\ell}$ whenever $\ell' \leq \ell$, yielding the extension of a telescope by a type. Note $P_{\mathsf{tpr}} \circ P_{\mathsf{pr}}$ is the polynomial functor of a map with codomain $P_{\mathsf{tpr}}(\mathsf{Ty})$, the presheaf of types in a context extended by a telescope.
- The rules $(\Gamma \mid ()_p) \equiv \Gamma$ and $(\Gamma \mid (\Theta,\ x : A)) \equiv ((\Gamma \mid \Theta),\ x : A)$ from section 2.3 hold, along with their analogues for partial substitutions. These are equalities of objects of $\mathcal{C}$, and only make sense if pr and tpr are *algebraically* representable.
- A morphism of polynomial functors $P_{\mathsf{tpr}_{\ell_0}} \circ P_{\mathsf{tpr}_{\ell_1}} \to P_{\mathsf{tpr}_{\ell_0 \sqcup \ell_1}}$, yielding the concatenation of telescopes mentioned in section 2.5, such that the relevant rules hold.

Additionally, it has Π-**telescopes** if there is a pullback square:

$$\begin{array}{ccc} P_{\mathsf{tpr}_{\ell_0}}(\mathsf{PSub}_{\ell_1}) & \longrightarrow & \mathsf{PSub}_{\ell_0 \sqcup \ell_1} \\ {\scriptstyle P_{\mathsf{tpr}_{\ell_0}}(\mathsf{tpr}_{\ell_1})}\downarrow & \lrcorner & \downarrow{\scriptstyle \mathsf{tpr}_{\ell_0 \sqcup \ell_1}} \\ P_{\mathsf{tpr}_{\ell_0}}(\mathsf{Tel}_{\ell_1}) & \xrightarrow[\Pi]{} & \mathsf{Tel}_{\ell_0 \sqcup \ell_1}, \end{array}$$

satisfying the computation rules alluded to in section 2.5.

This represents the rules from section 2.3 in the non-modal case. By using presheaves, it implicitly includes substitution into telescopes that commute with the other operations.

As for contexts, the rules do not require that every telescope is obtained by successive extension of the empty telescope by types. Indeed, there is no way to assert such a thing in a Generalised Algebraic Theory. But it holds *'admissibly'* in the initial syntactic model, and any CwF can be extended with telescopes in this way.

Theorem 10. Any natural model with levels can be equipped with telescopes. If it has Π-types, it also has Π-telescopes.

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Proof. We define tpr_ℓ to be the map such that $P_{\mathsf{tpr}_\ell} = \sum_{\substack{n\in\mathbb{N} \\ \forall i\le n, \ell_i \le \ell}} P_{\mathsf{pr}_{\ell_0}} \circ \cdots \circ P_{\mathsf{pr}_{\ell_n}}$. Thus an element of $\mathsf{Tel}_\ell(\Gamma)$ is a tower of n types over Γ of level $\le \ell$, and similarly for terms. We define context extension in the obvious way by iterating context extension by types, and the operations and equations are immediate. Similarly, we define Π-telescopes by using the rules for computing them on extended telescopes. □ ◁

4.1.4 Meta-abstractions

Because meta-abstractions are not *'reified'* as types, they do not require any additional structure. Specifically, the rules for the judgment $\Gamma \vdash A \ \mathsf{type}\big/_{\delta : \Upsilon}$ simply say that it is the object $P_{\mathsf{tpr}}(\mathsf{Ty})$ that classifies types indexed by a telescope, and similarly elements are classified by $P_{\mathsf{tpr}}(\mathsf{Tm})$. Thus meta-abstractions of types are classified by the map $P_{\mathsf{tpr}}(\mathsf{pr})$, and similarly meta-abstractions of telescopes are classified by $P_{\mathsf{tpr}}(\mathsf{tpr})$; thus any natural model with telescopes also has meta-abstractions of types and telescopes. Semantically, we don't need to discuss meta-abstractions explicitly, since $\Gamma \vdash A \ \mathsf{type}_\ell\big/_{\delta : \Upsilon}$ is equivalent to $\Gamma \mid \Upsilon \vdash A \ \mathsf{type}_\ell$, and so on. Thus we will generally talk only about types and telescopes in contexts. ◁

4.1.5 Infinite telescopes

We now define a new judgement $\gamma : \Gamma \vdash \widetilde{\Upsilon}\ \gamma \ \mathsf{stel}^\infty_\ell$ whose elements are *'infinite telescopes'*. As with meta-abstractions, this is not (yet) introducing new structure on a CwF, rather it is a definition that can be made in the presheaf category of any CwF. The idea is that an infinite telescope consists of an infinite sequence of types each dependent on all the previous ones:

$$\begin{aligned}
\gamma : \Gamma &\vdash \widetilde{\Upsilon}^0\ \gamma \ \mathsf{type}_{\ell_0} \\
\gamma : \Gamma,\ \upsilon^0 : \widetilde{\Upsilon}^0\ \gamma &\vdash \widetilde{\Upsilon}^1\ \gamma\ \upsilon^0 \ \mathsf{type}_{\ell_1} \\
\gamma : \Gamma,\ \upsilon^0 : \widetilde{\Upsilon}^0\ \gamma,\ \upsilon^1 : \widetilde{\Upsilon}^1\ \gamma\ \upsilon^0 &\vdash \widetilde{\Upsilon}^1\ \gamma\ \upsilon^0\ \upsilon^1 \ \mathsf{type}_{\ell_2} \\
&\vdots
\end{aligned}$$

where $\ell_n \le \ell$ for all n. Formally, we define this along with its finite approximations:

$$\begin{aligned}
\widetilde{\Upsilon}^{\partial 0}\ \gamma &\equiv () \\
\widetilde{\Upsilon}^{\partial 1}\ \gamma &\equiv \left(\upsilon^0 : \widetilde{\Upsilon}^0\ \gamma\right) \\
\widetilde{\Upsilon}^{\partial 2}\ \gamma &\equiv \left(\upsilon^0 : \widetilde{\Upsilon}^0\ \gamma,\ \upsilon^1 : \widetilde{\Upsilon}^1\ \gamma\ \upsilon^0\right) \\
&\vdots
\end{aligned}$$

so that we can say that in general $\widetilde{\Upsilon}^n$ is a type in context $(\gamma : \Gamma \mid \upsilon : \widetilde{\Upsilon}^{\partial n})$. In syntax, this means we give the following bidirectional rule with infinitely many premises:

$$\frac{\left(\gamma : \Gamma \vdash \widetilde{\Upsilon}^{\partial n}\ \gamma \ \mathsf{tel}_\ell\right)_{n\in\mathbb{N}} \qquad \left(\gamma : \Gamma \mid \partial\upsilon : \widetilde{\Upsilon}^{\partial n}\ \gamma \vdash \widetilde{\Upsilon}^n\ \gamma\ \partial\upsilon \ \mathsf{type}_{\ell_n}\right)_{n\in\mathbb{N}} \qquad (\ell_n \le \ell)_{n\in\mathbb{N}} \qquad \widetilde{\Upsilon}^{\partial 0}\ \gamma \equiv () \qquad \left(\gamma : \Gamma \vdash \widetilde{\Upsilon}^{\partial(n+1)}\ \gamma \equiv (\partial\upsilon : \widetilde{\Upsilon}^{\partial n}\ \gamma,\ \upsilon : \widetilde{\Upsilon}^n\ \gamma\ \partial\upsilon)\right)_{n\in\mathbb{N}}}{\gamma : \Gamma \vdash \widetilde{\Upsilon}\ \gamma \ \mathsf{stel}^\infty_\ell}$$

Similarly, the judgement $\gamma : \Gamma \vdash \widetilde{\upsilon}\ \gamma : \widetilde{\Upsilon}\ \gamma$ of infinite partial substitutions is characterised by a bidirectional rule:

$$\frac{\left(\gamma : \Gamma \vdash \widetilde{\upsilon}^{\partial n}\ \gamma : \widetilde{\Upsilon}^{\partial n}\ \gamma\right)_{n\in\mathbb{N}} \qquad \left(\gamma : \Gamma \vdash \widetilde{\upsilon}^n\ \gamma : \widetilde{\Upsilon}^n\ \gamma\ (\widetilde{\upsilon}^{\partial n}\ \gamma)\right)_{n\in\mathbb{N}} \qquad \widetilde{\upsilon}^{\partial 0}\ \gamma \equiv [\,] \qquad \left(\gamma : \Gamma \vdash \widetilde{\upsilon}^{\partial(n+1)}\ \gamma \equiv [\,\widetilde{\upsilon}^{\partial n}\ \gamma,\ \widetilde{\upsilon}^n\ \gamma\,]\right)_{n\in\mathbb{N}}}{\gamma : \Gamma \vdash \widetilde{\upsilon}\ \gamma : \widetilde{\Upsilon}\ \gamma}$$

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Substitution extends level-wise to infinite telescopes and their partial substitutions. Categorically, these rules mean the map $\mathsf{pr}^\infty{}_\ell : \mathsf{PSub}^\infty{}_\ell \to \mathsf{Tel}^\infty{}_\ell$ is the limit of the sequence:

$$\cdots \to \mathsf{pr}_\ell^n \to \cdots \to \mathsf{pr}_\ell^3 \to \mathsf{pr}_\ell^2 \to \mathsf{pr}_\ell \to \mathbb{1}$$

where pr_ℓ^n is such that $P_{\mathsf{pr}_\ell^n} = \sum_{\forall i \le n, \ell_i \le \ell} P_{\mathsf{pr}_{\ell_0}} \circ \cdots \circ P_{\mathsf{pr}_{\ell_n}}$. Thus $P_{\mathbb{1}}$ is the identity functor, $\mathbb{1}$ is the identity map of the terminal object, and $\mathsf{pr}_\ell{}^{n+1} \to \mathsf{pr}_\ell{}^n$ discards the last type in a telescope of length $n+1$ (there being no other way to get a telescope of length n). ◁

4.1.6 ω-limits

Finally, we define the structure of infinite (sequential, Reedy) limits on a CwF. These are an *'infinitary rule'* (i.e. a non-elementary structure) that is not part of dTT or any implementable type theory, but we will use them to build our intended models of dTT. Syntactically, they are essentially just a kind of Σ-type of an infinite telescope.

Definition 11. A CwF with levels has ω-**limits** if it is equipped with pullback squares:

$$\begin{array}{ccc} \mathsf{PSub}^\infty_\ell & \xrightarrow{\lim} & \mathsf{Tm}_\ell \\ {\scriptstyle \mathsf{pr}^\infty{}_\ell}\downarrow & \lrcorner & \downarrow{\scriptstyle \mathsf{pr}_\ell} \\ \mathsf{Tel}^\infty_\ell & \xrightarrow[\lim]{} & \mathsf{Ty}_\ell, \end{array}$$

In syntax, this means we have the following structure and properties. First, having a merely *commutative* square as above gives the following rules:

$$\frac{\gamma : \Gamma \vdash \widetilde{\Upsilon}\ \gamma\ \mathsf{stel}^\infty_\ell}{\gamma : \Gamma \vdash \lim\left(\widetilde{\Upsilon}\ \gamma\right) \mathsf{type}_\ell} \qquad \frac{\gamma : \Gamma \vdash \widetilde{v}\ \gamma : \widetilde{\Upsilon}\ \gamma}{\gamma : \Gamma \vdash \lim\left(\widetilde{v}\ \gamma\right) : \lim\left(\widetilde{\Upsilon}\ \gamma\right)}$$

Second, we have restriction operations:

$$\frac{\gamma : \Gamma \vdash u : \lim\left(\widetilde{\Upsilon}\ \gamma\right)}{\gamma : \Gamma \vdash \mathsf{res}^{\partial n}\ \gamma\ u : \widetilde{\Upsilon}^{\partial n}\ \gamma} \qquad \frac{\gamma : \Gamma \vdash u : \lim\left(\widetilde{\Upsilon}\ \gamma\right)}{\gamma : \Gamma \vdash \mathsf{res}^{n}\ \gamma\ u : \widetilde{\Upsilon}^{n}\ \gamma\ \left(\mathsf{res}^{\partial n}\ \gamma\ u\right)}$$

We require that $\mathsf{res}^{\partial n}$ is derived from res^n via:

$$\mathsf{res}^{\partial 0}\ \gamma\ u \equiv [\,]$$
$$\mathsf{res}^{\partial (n+1)}\ \gamma\ u \equiv [\,\mathsf{res}^{\partial n}\ \gamma\ u,\ \mathsf{res}^{n}\ \gamma\ u\,]$$

and that the following computation and uniqueness rules hold:

$$\mathsf{res}^{\partial n}\ \gamma\ \ (\lim\left(\widetilde{a}\ \gamma\right)) \equiv \widetilde{a}^{\partial n}\ \gamma$$
$$\mathsf{res}^{n}\ \gamma\ \ (\lim\left(\widetilde{a}\ \gamma\right)) \equiv \widetilde{a}^{n}\ \gamma$$
$$u \equiv \lim\left(\mathsf{res}^n\ \gamma\ u\right)_n$$

Of course, all these constructions must also be stable under substitution. ◁

4.2 The simplicial model

Fix a natural model with all of the structure described above, which we call the *discrete model* (dm). We will construct a new model called the *simplicial model* (sm), by first constructing the *truncated simplicial models* (sm^n) for $n \ge -2$.

Let $\mathbb{B}$ be the type of binary digits, $\mathbb{0}$, $\mathbb{1} : \mathbb{B}$. For $n \ge m \ge -1$, let $\mathbb{B}^{\langle n \rangle, \langle m \rangle}$ be the type of length $n+1$ binary sequences of which exactly $m+1$ are $\mathbb{1}$. The *augmented semi-simplex category* Δ^+ has

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objects $\langle n \rangle$ and morphisms $\Delta^+(\langle m \rangle, \langle n \rangle) = \mathbb{B}^{\langle n \rangle, \langle m \rangle}$. The identity $1_{\langle n \rangle}$ is all $\mathbb{1}$s, and the composite of $b_1 : \mathbb{B}^{\langle n \rangle, \langle m \rangle}$ and $b_0 : \mathbb{B}^{\langle m \rangle, \langle k \rangle}$ replaces the $\mathbb{1}$s in b_1 with the digits of b_0; e.g. $\mathbb{1}\mathbb{0}\mathbb{1}\mathbb{0}\mathbb{0}\mathbb{1}\mathbb{1} \circ \mathbb{0}\mathbb{1}\mathbb{1}\mathbb{0} = \mathbb{0}\mathbb{0}\mathbb{1}\mathbb{0}\mathbb{0}\mathbb{1}\mathbb{0}$. Note that each representable $\mathbb{B}^{\langle n \rangle, -}$ is finite.

We write $\mathbb{0}b : \mathbb{B}^{\langle n+1 \rangle, \langle k \rangle}$ and $\mathbb{1}b : \mathbb{B}^{\langle n+1 \rangle, \langle k+1 \rangle}$ for left appending digits. Then:

$$\mathbb{0}b_1 \circ b_0 \equiv \mathbb{0}\left(b_1 \circ b_0\right) \qquad \mathbb{1}b_1 \circ \mathbb{1}b_0 \equiv \mathbb{1}\left(b_1 \circ b_0\right) \qquad \mathbb{1}b_1 \circ \mathbb{0}b_0 \equiv \mathbb{0}\left(b_1 \circ b_0\right)$$

Note that by the second rule, along with the fact that $\mathbb{1}1_{\langle n \rangle} \equiv 1_{\langle n+1 \rangle}$, the assignments $\langle n \rangle \mapsto \langle n+1 \rangle$ and $b \mapsto \mathbb{1}b$ define an endofunctor of Δ^+.

4.2.1 The truncated simplicial models

For every $n \geq -2$, let Δ^+_n be the full subcategory of Δ^+ on the $\langle k \rangle$ with $k \leq n$. Thus Δ^+_{-2} is empty, while Δ^+_{-1} is terminal. The *n-truncated simplicial model* sm^n has underlying category $\mathcal{C}^{\Delta^{+\mathrm{op}}_n}$. For an object of this category, written $\Gamma\ \mathsf{ob}_{\mathsf{sm}^n}$, we have $\Gamma_m\ \mathsf{ob}_{\mathsf{dm}}$ whenever $n \geq m \geq -2$, where $\Gamma_{-2} \equiv ()_{\mathsf{dm}}$ is terminal; and if $b : \mathbb{B}^{\langle n \rangle, \langle m \rangle}$ then $\Gamma^b : \Gamma_n \to \Gamma_m$, contravariantly functorially. We sometimes write γ^b for $\Gamma^b\ \gamma$. A morphism $\sigma : \Delta \to \Gamma$ consists of $\sigma_n : \Delta_n \to \Gamma_n$ for each n such that $\Delta^b \circ \sigma_n \equiv \sigma_m \circ \Gamma^b$ for any $b : \mathbb{B}^{\langle n \rangle, \langle m \rangle}$.

The sm^n are related by *truncation* and *décalage* functors and a natural transformation:

$$\pi : \mathcal{C}^{\Delta^{+\mathrm{op}}_{n+1}} \to \mathcal{C}^{\Delta^{+\mathrm{op}}_n} \qquad (-)^{\mathsf{D}} : \mathcal{C}^{\Delta^{+\mathrm{op}}_{n+1}} \to \mathcal{C}^{\Delta^{+\mathrm{op}}_n} \qquad \rho : (-)^{\mathsf{D}} \Rightarrow \pi$$

$$(\pi\Gamma)_{m+1} \equiv \Gamma_{m+1} \qquad \left(\Gamma^{\mathsf{D}}\right)_{m+1} \equiv \Gamma_{m+2} \qquad (\rho_\Gamma)_{m+1} \equiv \Gamma^{\mathbb{0}1_{\langle m+1 \rangle}}$$

$$(\pi\Gamma)^b \equiv \Gamma^b \qquad \left(\Gamma^{\mathsf{D}}\right)^b \equiv \Gamma^{\mathbb{1}b}$$

$$(\pi\sigma)_{m+1} \equiv \sigma_{m+1} \qquad \left(\sigma^{\mathsf{D}}\right)_{m+1} \equiv \sigma_{m+2}$$

We will now construct the type-theoretical/fibrant structure of the models sm^n with a large set of mutually inductive definitions. Our first goal is to define the judgement:

$$\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} A\ \gamma\ \mathsf{type}_\ell$$

Intuitively, such a simplicial type A consists of discrete m-simplex types for $m \leq n+1$:

$$\gamma_{-1} : \Gamma_{-1} \vdash_{\mathsf{dm}} A_{-1}\ \gamma_{-1}\ \mathsf{type}_\ell$$

$$\gamma_0 : \Gamma_0,\ \mathbb{3}_{\mathbb{0}} : A_{-1}\ \gamma_0{}^{\mathbb{0}} \vdash_{\mathsf{dm}} A_0\ \gamma_0\ \mathbb{3}_{\mathbb{0}}\ \mathsf{type}_\ell$$

$$\gamma_1 : \Gamma_1,\ \mathbb{3}_{\mathbb{0}\mathbb{0}} : A_{-1}\ \gamma_1{}^{\mathbb{0}\mathbb{0}},\ x_{\mathbb{0}\mathbb{1}} : A_0\ \gamma_1{}^{\mathbb{0}\mathbb{1}}\ \mathbb{3}_{\mathbb{0}\mathbb{0}},\ x_{\mathbb{1}\mathbb{0}} : A_0\ \gamma_1{}^{\mathbb{1}\mathbb{0}}\ \mathbb{3}_{\mathbb{0}\mathbb{0}} \vdash_{\mathsf{dm}} A_1\ \gamma_1\ \mathbb{3}_{\mathbb{0}\mathbb{0}}\ x_{\mathbb{0}\mathbb{1}}\ x_{\mathbb{1}\mathbb{0}}\ \mathsf{type}_\ell$$

$$\vdots$$

Note that Γ_0 denotes the 0-component of the presheaf Γ (a dm-object), while γ_0 is an atomic variable name in this object. And the type of $x_{\mathbb{0}\mathbb{1}}$ is well-typed because the outer rectangle of the following diagram is a distinguished pullback:

$$\begin{array}{ccccc}
\left(\gamma_1 : \Gamma_1,\ \mathbb{3}_{\mathbb{0}\mathbb{0}} : A_{-1}\ \left(\gamma_1{}^{\mathbb{0}\mathbb{1}}\right)^{\mathbb{0}}\right) & \longrightarrow & \left(\gamma_0 : \Gamma_0,\ \mathbb{3}_{\mathbb{0}} : A_{-1}\ \gamma_0{}^{\mathbb{0}}\right) & \longrightarrow & \left(\gamma_{-1} : \Gamma_{-1},\ \mathbb{3}_{\mathbb{0}} : A_{-1}\ \gamma_{-1}\right) \\
\downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\
\left(\gamma_1 : \Gamma_1\right) & \xrightarrow{\Gamma^{\mathbb{0}\mathbb{1}}} & \left(\gamma_0 : \Gamma_0\right) & \xrightarrow{\Gamma^{\mathbb{0}}} & \left(\gamma_{-1} : \Gamma_{-1}\right)
\end{array}$$

$$\Gamma^{\mathbb{0}} \circ \Gamma^{\mathbb{0}\mathbb{1}} \equiv \Gamma^{\mathbb{0}\mathbb{0}}$$

We will write the type declarations of A_n generically as:

$$\gamma_{n+1} : \Gamma_{n+1},\ \partial a : \pi A_{\partial(n+1)}\ \gamma_{n+1} \vdash_{\mathsf{dm}} A_{n+1}\ \gamma_{n+1}\ \partial a\ \mathsf{type}_{\ell_{n+1}}.$$

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Here $\pi A_{\partial(n+1)}$ is a telescope consisting of the *'boundary'* of an $(n+1)$-simplex, also known as the Reedy *'matching object'* of an augmented semi-simplicial type. For example:

$$\begin{aligned} A_{\partial(-1)}\ \gamma_{-1} &\equiv () \\ A_{\partial 0}\ \gamma_0 &\equiv \left(z_0 : A_{-1}\ \gamma_0{}^{\mathbb{0}}\right) \\ A_{\partial 1}\ \gamma_1 &\equiv \left(z_{00} : A_{-1}\ \gamma_1{}^{\mathbb{00}},\ x_{01} : A_0\ \gamma_1{}^{\mathbb{01}}\ z_{00},\ x_{10} : A_0\ \gamma_1{}^{\mathbb{10}}\ z_{00}\right). \end{aligned}$$

Similarly, the term judgement

$$\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} t\ \gamma : A\ \gamma$$

will be defined to consist of the data:

$$\begin{aligned} &\gamma_{-1} : \Gamma_{-1} \vdash_{\mathsf{dm}} t_{-1}\ \gamma_{-1} : A_{-1}\ \gamma_{-1} \\ &\gamma_0 : \Gamma_0 \vdash_{\mathsf{dm}} t_0\ \gamma_0 : A_0\ \gamma_0 \quad \left(t_{-1}\ \gamma_0{}^{\mathbb{0}}\right) \\ &\gamma_1 : \Gamma_1 \vdash_{\mathsf{dm}} t_1\ \gamma_1 : A_1\ \gamma_1 \quad \left(t_{-1}\ \gamma_0{}^{\mathbb{00}}\right) \quad \left(t_0\ \gamma_1{}^{\mathbb{01}}\right) \quad \left(t_0\ \gamma_1{}^{\mathbb{10}}\right) \\ &\qquad\vdots \end{aligned}$$

Similarly to before, we will write this generically as:

$$\gamma_{n+1} : \Gamma_{n+1} \vdash_{\mathsf{dm}} t_{n+1}\ \gamma_{n+1} : A_{n+1}\ \gamma_{n+1} \quad \left(\pi t_{\partial(n+1)}\ \gamma_{n+1}\right)$$

where $\pi t_{\partial(n+1)}\ \gamma_{n+1}$ acts by the lower-dimensional parts of t on the boundary of γ_{n+1}.

Therefore, we need to mutually define these *matching telescopes* and *matching substitutions*. To define these, in turn, we need a truncated form of *display*, which takes an $(n+1)$-truncated semi-simplicial diagram A to an n-truncated one dependent on πA. Since we have no modal locks available yet, this version of display must be in a totally décalaged context. Finally, to make the context extension a presheaf, we need the actions of morphisms in Δ^+_{n+1} on matching telescopes and on types. The complete list of judgments and operations being defined in our first mutual induction is shown in Figure 9.

The inductive definitions of these judgments and operations are shown in Figure 10. The judgments are defined by bidirectional rules, meaning they are presheaves of ordered pairs, with the premises giving notation for the two components. We also have to define the substitution action of these presheaves in terms of these components. Similarly, after defining context extension, we have to specify the functorial action to make the extended context an augmented semi-simplicial object. The matching telescopes and matching substitutions are defined inductively on n; display is defined in terms of components; and the simplicial action on matching objects is defined based on the simplicial morphisms.

There are also many equations to verify as part of the induction: everything has to be well-typed and stable under substitution, and we need computation rules for display:

$$\left(\gamma : \Gamma,\ a : A\ \gamma\right)^{\mathsf{D}} \equiv \left(\gamma^+ : \Gamma^{\mathsf{D}},\ a : \pi A^{\rho_\Gamma}\ \gamma^+,\ a' : A^{\mathsf{d}}\ \gamma^+\ a\right) \tag{12}$$

$$\left[\ \sigma,\ t\ \right]^{\mathsf{D}} \equiv \left[\ \sigma^{\mathsf{D}},\ \pi t^{\rho_\Delta},\ t^{\mathsf{d}}\ \right] \tag{13}$$

and functoriality of the act operations:

$$\mathsf{act}^{\pi^{n-m}A}_{\partial b_0}\ \left(\Gamma^{b_1}\ \gamma_{n+1}\right)\ \left(\mathsf{act}^{A}_{\partial b_1}\ \gamma_{n+1}\ \partial a\right) \equiv \mathsf{act}^{A}_{\partial(b_1 \circ b_0)}\ \gamma_{n+1}\ \partial a \tag{14}$$

$$\mathsf{act}^{\pi^{n-m}A}_{b_0}\ \left(\Gamma^{b_1}\ \gamma_{n+1}\right)\ \left(\mathsf{act}^{\pi A}_{\partial b_1}\ \gamma_{n+1}\ \partial a\right)\left(\mathsf{act}^{A}_{b_1}\ \gamma_{n+1}\ \partial a\ a\right) \equiv \mathsf{act}^{A}_{b_1 \circ b_0}\ \gamma_{n+1}\ \partial a\ a. \tag{15}$$

Some of these proofs can be found in the extended version of this paper.

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$$\gamma : \Gamma \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell \qquad \gamma : \Gamma \vdash_{\mathsf{sm}^n} t\ \gamma : A\ \gamma$$

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{(\gamma : \Gamma,\ a : A\ \gamma)\ \mathsf{ob}_{\mathsf{sm}^n}} \qquad \frac{\sigma : \Delta \to_{\mathsf{sm}^n} \Gamma \qquad \delta : \Delta \vdash_{\mathsf{sm}^n} t\ \delta : A\ (\sigma\ \delta)}{[\ \sigma,\ t\ ] : \Delta \to_{\mathsf{sm}^n} (\gamma : \Gamma,\ a : A\ \gamma)}$$

$$\frac{\gamma^- : \pi\Gamma \vdash_{\mathsf{sm}^n} A\ \gamma^-\ \mathsf{type}_\ell}{\gamma_{n+1} : \Gamma_{n+1} \vdash_{\mathsf{dm}} A_{\partial(n+1)}\ \gamma_{n+1}\ \mathsf{tel}_\ell} \qquad \frac{\gamma^- : \pi\Gamma \vdash_{\mathsf{sm}^n} t\ \gamma^- : A\ \gamma^-}{\gamma_{n+1} : \Gamma_{n+1} \vdash_{\mathsf{dm}} t_{\partial(n+1)}\ \gamma_{n+1} : A_{\partial(n+1)}\ \gamma_{n+1}}$$

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} A\ \gamma\ \mathsf{type}_\ell}{\gamma^+ : \Gamma^{\mathsf{D}},\ a : \pi A^{\rho\Gamma}\ \gamma^+ \vdash_{\mathsf{sm}^n} A^{\mathsf{d}}\ \gamma^+\ a\ \mathsf{type}_\ell} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} t\ \gamma : A\ \gamma}{\gamma^+ : \Gamma^{\mathsf{D}} \vdash_{\mathsf{sm}^n} t^{\mathsf{d}}\ \gamma^+ : A^{\mathsf{d}}\ \gamma^+\ \pi t^{\rho\Gamma}}$$

$$\frac{\gamma^- : \pi\Gamma \vdash_{\mathsf{sm}^n} A\ \gamma^-\ \mathsf{type}_\ell \qquad b : \mathbb{B}^{\langle n+1\rangle,\langle m+1\rangle}}{\gamma_{n+1} : \Gamma_{n+1},\ \partial a : A_{\partial(n+1)}\ \gamma_{n+1} \vdash_{\mathsf{dm}} \mathsf{act}^A_{\partial b}\ \gamma_{n+1}\ \partial a : \pi^{n-m} A_{\partial(m+1)}\ (\Gamma^b\ \gamma_{n+1})}$$

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} A\ \gamma\ \mathsf{type}_\ell \qquad b : \mathbb{B}^{\langle n+1\rangle,\langle m+1\rangle}}{\begin{array}{l}\gamma_{n+1} : \Gamma_{n+1},\ \partial a : \pi A_{\partial(n+1)}\ \gamma_{n+1},\ a : A_{n+1}\ \gamma_{n+1}\ \partial a \\ \qquad \vdash_{\mathsf{dm}} \mathsf{act}^A_b\ \gamma_{n+1}\ \partial a\ a : \pi^{n-m} A_{m+1}\ (\Gamma^b\ \gamma_{n+1})\ (\mathsf{act}_{\partial b}\ \gamma_{n+1}\ \partial a)\end{array}}$$

Figure 9. Structures defined mutually for truncated simplicial types and terms.

Now to make sm^n a natural model, we need fundamental projections and variables:

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{\mathsf{pt}^A_{\mathsf{sm}^n} : (\gamma : \Gamma,\ a : A\ \gamma) \to \Gamma} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma,\ a : A\ \gamma \vdash_{\mathsf{sm}^n} \mathsf{zv}^A_{\mathsf{sm}^n}\ \gamma\ a : A^{\mathsf{pt}}\ \gamma\ a}.$$

We define these inductively:

$$\begin{array}{ll} \left(\mathsf{pt}^A_{\mathsf{sm}^{-1}}\right)_{-1} \equiv \mathsf{pt}^{A_{-1}}_{\mathsf{dm}} & \left(\mathsf{zv}^A_{\mathsf{sm}^{-1}}\right)_{-1} \equiv \mathsf{zv}^{A_{-1}}_{\mathsf{dm}} \\ \left(\mathsf{pt}^A_{\mathsf{sm}^{n+1}}\right)_{m+1} \equiv \left(\mathsf{pt}^{\pi A}_{\mathsf{sm}^n}\right)_{m+1} \quad \text{for } m < n & \pi\ \left(\mathsf{zv}^A_{\mathsf{sm}^{n+2}}\right) \equiv \mathsf{zv}^{\pi A}_{\mathsf{sm}^n} \\ \left(\mathsf{pt}^A_{\mathsf{sm}^{n+2}}\right)_{n+2} \equiv \left(\mathsf{pt}^{\pi A^{\rho\Gamma}}_{\mathsf{sm}^{n+1}}\right)_{n+1} \circ \left(\mathsf{pt}^{A^{\mathsf{d}}}_{\mathsf{sm}^{n+1}}\right)_{n+1} & \left(\mathsf{zv}^A_{\mathsf{sm}^{n+2}}\right)_{n+2} \equiv \left(\mathsf{zv}^{A^{\mathsf{d}}}_{\mathsf{sm}^{n+1}}\right)_{n+1}. \end{array}$$

To make this well-typed, we have to simultaneously prove eq. (8) as well as:

$$\left(\mathsf{pt}^A_{\mathsf{sm}^{n+1}}\right)^{\mathsf{D}} \equiv \mathsf{pt}^{\pi A^{\rho\Gamma}}_{\mathsf{sm}^n} \circ \mathsf{pt}^{A^{\mathsf{d}}}_{\mathsf{sm}^n} \qquad \left(\mathsf{zv}^A_{\mathsf{sm}^{n+1}}\right)^{\mathsf{d}} \equiv \mathsf{zv}^{A^{\mathsf{d}}}_{\mathsf{sm}^n} \qquad \left(\mathsf{zv}^{\pi A}_{\mathsf{sm}^n}\right)^{\rho(\Gamma,A)} \equiv \left(\mathsf{zv}^{\pi A^{\rho\Gamma}}_{\mathsf{sm}^n}\right)^{\mathsf{pt}^{A^{\mathsf{d}}}_{\mathsf{sm}^n}}.$$

These proofs can also be found in the extended version. ◁

4.2.2 Π-types

Assuming the discrete model has Π-types, we construct them in sm^n inductively. Assuming Π-types at prior levels, we have the following two types in the same context:

$$\gamma^+ : \Gamma^{\mathsf{D}},\ f : \left(\Pi^{\mathsf{sm}^n}\ \pi A\ \pi B\right)^{\rho\Gamma}\ \gamma^+ \vdash_{\mathsf{sm}^n} \left(\Pi^{\mathsf{sm}^{n+1}}\ A\ B\right)^{\mathsf{d}}\ \gamma^+\ f\ \mathsf{type}_\ell$$

$$\gamma^+ : \Gamma^{\mathsf{D}},\ f : \left(\Pi^{\mathsf{sm}^n}\ \pi A\ \pi B\right)^{\rho\Gamma}\ \gamma^+ \vdash_{\mathsf{sm}^n} \left(a : \pi A^{\rho\Gamma}\ \gamma^+\right)\left(a' : A^{\mathsf{d}}\ \gamma^+\ a\right) \to B^{\mathsf{d}}\ [\ \gamma^+,\ a,\ a'\ ]\ \left(\mathsf{app}\ [\ \gamma^+,\ f,\ a,\ a'\ ]\ f\ a\right)\ \mathsf{type}_\ell.$$

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$$\frac{\gamma^- : \pi\Gamma \vdash_{\mathsf{sm}^n} \pi A\ \gamma^-\ \mathsf{type}_\ell \qquad \gamma_{n+1} : \Gamma_{n+1},\ \partial a : \pi A_{\partial(n+1)}\ \gamma_{n+1} \vdash_{\mathsf{dm}} A_{n+1}\ \gamma_{n+1}\ \partial a\ \mathsf{type}_\ell}{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} A\ \gamma\ \mathsf{type}_\ell}$$

$$\frac{\begin{array}{c}\gamma^- : \pi\Gamma \vdash_{\mathsf{sm}^n} \pi t\ \gamma^- : \pi A\ \gamma^- \\ \gamma_{n+1} : \Gamma_{n+1} \vdash_{\mathsf{dm}} t_{n+1}\ \gamma_{n+1} : A_{n+1}\ \gamma_{n+1}\ \left(\pi t_{\partial(n+1)}\ \gamma_{n+1}\right)\end{array}}{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} t\ \gamma : A\ \gamma}$$

$$\pi\left(A^\sigma\right) \equiv \pi A^{\pi\sigma} \qquad \left(A^\sigma\right)_{n+1} \equiv A_{n+1}^{\mathsf{W}_2^{\pi A_{\partial(n+1)}}\ \sigma_{n+1}}$$
$$\pi\left(t^\sigma\right) \equiv \pi t^{\pi\sigma} \qquad \left(t^\sigma\right)_{n+1} \equiv t_{n+1}^{\sigma_{n+1}}$$

$$\left(\gamma : \Gamma,\ a : A\ \gamma\right)_{m+1} \equiv \left(\gamma^- : \pi\Gamma,\ a^- : \pi A\ \gamma^-\right)_{m+1} \quad \text{for} \quad m < n$$
$$\left(\gamma : \Gamma,\ a : A\ \gamma\right)_{n+1} \equiv \left(\gamma_{n+1} : \Gamma_{n+1},\ \partial a : \pi A_{\partial(n+1)}\ \gamma_{n+1},\ a : A_{n+1}\ \gamma_{n+1}\ \partial a\right)$$
$$[\ \sigma,\ t\ ]_{m+1} \equiv [\ \pi\sigma,\ \pi t\ ]_{m+1} \quad \text{for} \quad m < n$$
$$[\ \sigma,\ t\ ]_{n+1} \equiv [\ \sigma_{n+1},\ \pi t_{\partial(n+1)},\ t_{n+1}\ ]$$

$$(\gamma : \Gamma,\ a : A\ \gamma)^b\ \ \gamma_{n+1}\ \partial a\ a \equiv [\ \Gamma^b\ \gamma_{n+1},\ \mathsf{act}^{\pi A}_{\partial b}\ \gamma_{n+1}\ \partial a,\ \mathsf{act}^{A}_{b}\ \gamma_{n+1}\ \partial a\ a\ ]$$

$$A_{\partial(-1)} \equiv ()_{\mathsf{dm}} \qquad t_{\partial(-1)} \equiv [\ ]_{\mathsf{dm}}$$

$$A_{\partial(n+2)}\ \gamma_{n+2} \equiv \left(\partial a : \left(\pi A^{\rho_{\pi\Gamma}}\right)_{\partial(n+1)}\ \gamma_{n+2},\ a : \left(A^{\rho_\Gamma}\right)_{n+1}\ \gamma_{n+2}\ \partial a, \partial a' : (A^{\mathsf{d}})_{\partial(n+1)}[\gamma_{n+2},\ \partial a,\ a]\right)$$
$$t_{\partial(n+2)}\ \gamma_{n+2} \equiv [\ (\pi t^{\rho_{\pi\Gamma}})_{\partial(n+1)},\ (t^{\rho_\Gamma})_{n+1},\ (t^{\mathsf{d}})_{\partial(n+1)}\ ]$$
$$\pi\left(A^{\mathsf{d}}\right) \equiv \pi A^{\mathsf{d}} \qquad \left(A^{\mathsf{d}}\right)_{n+1} \equiv A_{n+2}$$
$$\pi\left(t^{\mathsf{d}}\right) \equiv \pi t^{\mathsf{d}} \qquad \left(t^{\mathsf{d}}\right)_{n+1} \equiv t_{n+2}$$
$$\mathsf{act}^{A}_{\partial(\mathbb{0}b)}\ \gamma_{n+2}\ [\ \partial a,\ a,\ \partial a'\ ] \equiv \mathsf{act}^{\pi A^{\rho_{\pi\Gamma}}}_{\partial b}\ \gamma_{n+2}\ \partial a$$
$$\mathsf{act}^{A}_{\mathbb{0}b}\ \gamma_{n+2}\ [\ \partial a,\ a,\ \partial a'\ ]\ a' \equiv \mathsf{act}^{\pi A^{\rho_\Gamma}}_{b}\ \gamma_{n+2}\ \partial a\ a$$
$$\mathsf{act}^{A}_{\partial(\mathbb{1}b)}\ \gamma_{n+2}\ [\ \partial a,\ a,\ \partial a'\ ] \equiv [\ \mathsf{act}^{\pi A^{\rho_{\pi\Gamma}}}_{\partial b}\ \gamma_{n+2}\ \partial a,\ \mathsf{act}^{A^{\rho_\Gamma}}_{b}\ \gamma_{n+2}\ \partial a\ a,\ \mathsf{act}^{A^{\mathsf{d}}}_{\partial b}\ [\ \gamma_{n+2},\ \partial a,\ a\ ]\ \partial a'\ ]$$
$$\mathsf{act}^{A}_{\mathbb{1}b}\ \gamma_{n+2}\ [\ \partial a,\ a,\ \partial a'\ ]\ a' \equiv \mathsf{act}^{A^{\mathsf{d}}}_{b}\ [\ \gamma_{n+2},\ \partial a,\ a\ ]\ \partial a'\ a'$$

Figure 10. Mutual inductive definitions for truncated simplicial types and terms.

In addition to the definition of Π-types, we include in the induction the statement that these two types are equal. Using this assumption, we can give the base definitions at sm^{-1} as

$$\left(\Pi^{\mathsf{sm}^{-1}}\ A\ B\right)_{-1} \equiv \Pi^{\mathsf{dm}}\ A_{-1}\ B_{-1} \qquad \left(\lambda^{\mathsf{sm}^{-1}}\ t\right)_{-1} \equiv \lambda^{\mathsf{dm}}\ t_{-1} \qquad \left(\mathsf{app}^{\mathsf{sm}^{-1}}\ f\ s\right)_{-1} \equiv \mathsf{app}^{\mathsf{dm}}\ f_{-1}\ s_{-1}$$

and continue inductively to make the whole construction level-wise:

$$\pi\left(\Pi^{\mathsf{sm}^{n+2}}\ A\ B\right) \equiv \Pi^{\mathsf{sm}^{n+1}}\ \pi A\ \pi B$$
$$\left(\Pi^{\mathsf{sm}^{n+2}}\ A\ B\right)_{n+2} \equiv$$
$$\left(\Pi^{\mathsf{sm}^{n+1}}\ \left(\pi A^{\rho_\Gamma}\right)^{\mathsf{pt}}\ \Pi^{\mathsf{sm}^{n+1}}\ \left(A^{\mathsf{d}}\right)^{\mathsf{W}_2^{\pi A^{\rho_\Gamma}}\ \mathsf{pt}}\ \left(B^{\mathsf{d}}\right)^{[\ \mathsf{W}_2^{A^{\mathsf{d}}}\mathsf{W}_2^{\pi A^{\rho_\Gamma}}\ \mathsf{pt},\ \mathsf{app}\ \mathsf{zv}^{\mathsf{pt}\circ\mathsf{pt}}\ \mathsf{zv}^{\mathsf{pt}}\ ]}\right)_{n+1}$$

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$$\pi\left(\lambda^{\mathsf{sm}^{n+2}}\ t\right) \equiv \lambda^{\mathsf{sm}^{n+1}}\ \pi t$$
$$\left(\lambda^{\mathsf{sm}^{n+2}}\ t\right)_{n+2} \equiv \left(\lambda^{\mathsf{sm}^{n+1}}\ \left(\lambda^{\mathsf{sm}^{n+1}}\ t^{\mathsf{d}}\right)\right)_{n+1}$$
$$\pi\left(\mathsf{app}^{\mathsf{sm}^{n+2}}\ f\ s\right) \equiv \mathsf{app}^{\mathsf{sm}^{n+1}}\ \pi f\ \pi s$$
$$\left(\mathsf{app}^{\mathsf{sm}^{n+2}}\ f\ s\right)_{n+2} \equiv \left(\mathsf{app}^{\mathsf{sm}^{n+1}}\ \left(\mathsf{app}^{\mathsf{sm}^{n+1}}\ f^{\mathsf{d}}\ \pi s^{\rho_\Gamma}\right)\ s^{\mathsf{d}}\right)_{n+1}.$$

The mutual proofs that with these definitions the above two types are equal, and that the β and η laws hold, can be found in the extended version. ◁

4.2.3 Universes

Denoting universes of the discrete model by Disc_ℓ, we construct universes in sm^n inductively, assuming them at all prior levels. We include in the induction that:

$$\left(\mathsf{Type}_\ell^{\mathsf{sm}^{n+1}}\right)^{\mathsf{d}} \equiv \Pi^{\mathsf{sm}^n}\ (\mathsf{El}\ \mathsf{zv})\ \mathsf{Type}_\ell^{\mathsf{sm}^n} \tag{16}$$
$$\left(\mathsf{Code}^{\mathsf{sm}^{n+1}}\ A\right)^{\mathsf{d}} \equiv \lambda^{\mathsf{sm}^n}\ \left(\mathsf{Code}^{\mathsf{sm}^n}\ A^{\mathsf{d}}\right) \tag{17}$$
$$\left(\mathsf{El}^{\mathsf{sm}^{n+1}}\ A\right)^{\mathsf{d}} \equiv \mathsf{El}^{\mathsf{sm}^n}\ \left(\mathsf{app}^{\mathsf{sm}^n}\ \left(A^{\mathsf{d}}\right)^{\mathsf{pt}}\ \mathsf{zv}\right). \tag{18}$$

Assuming this inductively, for sm^{-1} we define:

$$\left(\mathsf{Type}_\ell^{\mathsf{sm}^{-1}}\right)_{-1} \equiv \mathsf{Disc}_\ell \qquad \left(\mathsf{Code}^{\mathsf{sm}^{-1}}\ A\right)_{-1} \equiv \mathsf{Code}^{\mathsf{dm}}\ A_{-1} \qquad \left(\mathsf{El}^{\mathsf{sm}^{-1}}\ A\right)_{-1} \equiv \mathsf{El}^{\mathsf{dm}}\ A_{-1}$$

and at the inductive step we make things level-wise:

$$\pi\left(\mathsf{Type}_\ell^{\mathsf{sm}^{n+2}}\right) \equiv \mathsf{Type}_\ell^{\mathsf{sm}^{n+1}}$$
$$\left(\mathsf{Type}_\ell^{\mathsf{sm}^{n+2}}\right)_{n+2} \equiv \left(\Pi^{\mathsf{sm}^{n+1}}\ (\mathsf{El}\ \mathsf{zv})\ \mathsf{Type}_\ell^{\mathsf{sm}^{n+1}}\right)_{n+1}$$
$$\pi\left(\mathsf{Code}^{\mathsf{sm}^{n+2}}\ A\right) \equiv \mathsf{Code}^{\mathsf{sm}^{n+1}}\ \pi A$$
$$\left(\mathsf{Code}^{\mathsf{sm}^{n+2}}\ A\right)_{n+2} \equiv \left(\lambda^{\mathsf{sm}^{n+1}}\ \left(\mathsf{Code}^{\mathsf{sm}^{n+1}}\ A^{\mathsf{d}}\right)\right)_{n+1}$$
$$\pi\left(\mathsf{El}^{\mathsf{sm}^{n+2}}\ A\right) \equiv \mathsf{El}^{\mathsf{sm}^{n+1}}\ \pi A$$
$$\left(\mathsf{El}^{\mathsf{sm}^{n+2}}\ A\right)_{n+2} \equiv \left(\mathsf{El}^{\mathsf{sm}^{n+1}}\ \left(\mathsf{app}^{\mathsf{sm}^{n+1}}\ \left(A^{\mathsf{d}}\right)^{\mathsf{pt}}\ \mathsf{zv}\right)\right)_{n+1}.$$

The proofs of eqs. (16)–(18), and that Code and El are mutual inverses, can be found in the extended version. ◁

4.2.4 ω-limits

Assuming ω-limits exist in dm, and inductively in sm^n, we first construct the display of infinite telescopes and partial substitutions in sm^{n+1}. These operations have types:

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} \widetilde{\Upsilon}\ \gamma\ \mathsf{stel}_\ell^\infty}{\gamma^+ : \Gamma^{\mathsf{D}},\ \mathfrak{u} : \mathsf{lim}^{\mathsf{sm}^n}\ \pi\widetilde{\Upsilon}^{\rho_\Gamma}\ \gamma^+ \vdash_{\mathsf{sm}^n} \widetilde{\Upsilon}^{\mathsf{d}}\ \gamma^+\ \mathfrak{u}\ \mathsf{stel}_\ell^\infty} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} \widetilde{\upsilon}\ \gamma : \widetilde{\Upsilon}\ \gamma\ \mathsf{stel}_\ell^\infty}{\gamma^+ : \Gamma^{\mathsf{D}} \vdash_{\mathsf{sm}^n} \widetilde{\upsilon}^{\mathsf{d}}\ \gamma^+ : \widetilde{\Upsilon}^{\mathsf{d}}\ \gamma^+\ \left(\mathsf{lim}^{\mathsf{sm}^n}\ \pi\widetilde{\upsilon}^{\rho_\Gamma}\right)}$$

and we define them by:

$$\left(\widetilde{\Upsilon}^{\mathsf{d}}\ \gamma^+\ \mathfrak{u}\right)^{\partial m} \equiv \left(\widetilde{\Upsilon}^{\partial m}\right)^{\mathsf{d}}\ \gamma^+\ \left(\mathsf{res}_{\mathsf{sm}^n}^{\partial m}\ \gamma^+\ \mathfrak{u}\right) \qquad \left(\widetilde{\upsilon}^{\mathsf{d}}\right)^{\partial m} \equiv \left(\widetilde{\upsilon}^{\partial m}\right)^{\mathsf{d}}$$
$$\left(\widetilde{\Upsilon}^{\mathsf{d}}\ \gamma^+\ \mathfrak{u}\right)^{m} \equiv \left(\widetilde{\Upsilon}^{m}\right)^{\mathsf{d}}\ \gamma^+\ \left(\mathsf{res}_{\mathsf{sm}^n}^{m}\ \gamma^+\ \mathfrak{u}\right) \qquad \left(\widetilde{\upsilon}^{\mathsf{d}}\right)^{m} \equiv \left(\widetilde{\upsilon}^{m}\right)^{\mathsf{d}}.$$

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We include in the induction the following theorems:

$$\left(\lim^{\mathsf{sm}^{n+1}} \widetilde{\Upsilon}\right)^{\mathsf{d}} \equiv \lim^{\mathsf{sm}^{n}} \widetilde{\Upsilon}^{\mathsf{d}} \qquad \left(\mathrm{res}^{\partial m}_{\mathsf{sm}^{n+1}}\, \mathfrak{u}\right)^{\mathsf{d}} \equiv \mathrm{res}^{\partial m}_{\mathsf{sm}^{n}}\, \mathfrak{u}^{\mathsf{d}}$$
$$\left(\lim^{\mathsf{sm}^{n+1}} \widetilde{\upsilon}\right)^{\mathsf{d}} \equiv \lim^{\mathsf{sm}^{n}} \widetilde{\upsilon}^{\mathsf{d}} \qquad \left(\mathrm{res}^{m}_{\mathsf{sm}^{n+1}}\, \mathfrak{u}\right)^{\mathsf{d}} \equiv \mathrm{res}^{m}_{\mathsf{sm}^{n}}\, \mathfrak{u}^{\mathsf{d}}.$$

Assuming this inductively, the definitions are:

$$\left(\lim^{\mathsf{sm}^{-1}} \widetilde{\Upsilon}\right)_{-1} \equiv \lim^{\mathsf{dm}} \widetilde{\Upsilon}_{-1} \qquad \left(\lim^{\mathsf{sm}^{-1}} \widetilde{\upsilon}\right)_{-1} \equiv \lim^{\mathsf{dm}} \widetilde{\upsilon}_{-1}$$
$$\left(\mathrm{res}^{\partial m}_{\mathsf{sm}^{-1}}\, \mathfrak{u}\right)_{-1} \equiv \mathrm{res}^{\partial m}_{\mathsf{dm}}\, \mathfrak{u}_{-1} \qquad \left(\mathrm{res}^{m}_{\mathsf{sm}^{-1}}\, \mathfrak{u}\right)_{-1} \equiv \mathrm{res}^{m}_{\mathsf{dm}}\, \mathfrak{u}_{-1}$$
$$\pi\left(\lim^{\mathsf{sm}^{n+2}} \widetilde{\Upsilon}\right) \equiv \lim^{\mathsf{sm}^{n+1}} \pi\widetilde{\Upsilon} \qquad \left(\lim^{\mathsf{sm}^{n+2}} \widetilde{\Upsilon}\right)_{n+2} \equiv \left(\lim^{\mathsf{sm}^{n}} \widetilde{\Upsilon}^{\mathsf{d}}\right)_{n+1}$$
$$\pi\left(\lim^{\mathsf{sm}^{n+2}} \widetilde{\upsilon}\right) \equiv \lim^{\mathsf{sm}^{n+1}} \pi\widetilde{\upsilon} \qquad \left(\lim^{\mathsf{sm}^{n+2}} \widetilde{\upsilon}\right)_{n+2} \equiv \left(\lim^{\mathsf{sm}^{n}} \widetilde{\upsilon}^{\mathsf{d}}\right)_{n+1}$$
$$\pi\left(\mathrm{res}^{\partial m}_{\mathsf{sm}^{n+2}}\, \mathfrak{u}\right) \equiv \mathrm{res}^{\partial m}_{\mathsf{sm}^{n+1}}\, \pi\mathfrak{u} \qquad \left(\mathrm{res}^{\partial m}_{\mathsf{sm}^{n+2}}\, \mathfrak{u}\right)_{n+2} \equiv \left(\mathrm{res}^{\partial m}_{\mathsf{sm}^{n}}\, \mathfrak{u}^{\mathsf{d}}\right)_{n+1}$$
$$\pi\left(\mathrm{res}^{m}_{\mathsf{sm}^{n+2}}\, \mathfrak{u}\right) \equiv \mathrm{res}^{m}_{\mathsf{sm}^{n+1}}\, \pi\mathfrak{u} \qquad \left(\mathrm{res}^{m}_{\mathsf{sm}^{n+2}}\, \mathfrak{u}\right)_{n+2} \equiv \left(\mathrm{res}^{m}_{\mathsf{sm}^{n}}\, \mathfrak{u}^{\mathsf{d}}\right)_{n+1}.$$

The proofs of all the necessary equations can be found in the extended version. ◁

4.2.5 The simplicial model

Finally, we obtain the untruncated *simplicial model* sm by taking a limit. We first define a *tail-cutting truncation functor*, extend *décalage* to an endofunctor, and ρ without truncation:

$$\pi_n : \mathcal{C}^{\Delta^{+\mathrm{op}}} \to \mathcal{C}^{\Delta^{+\mathrm{op}}_n} \qquad (-)^{\mathsf{D}} : \mathcal{C}^{\Delta^{+\mathrm{op}}} \to \mathcal{C}^{\Delta^{+\mathrm{op}}} \qquad \rho : (-)^{\mathsf{D}} \Rightarrow 1_{\mathcal{C}^{\Delta^{+\mathrm{op}}}}$$
$$(\pi_n \Gamma)_{m+1} \equiv \Gamma_{m+1} \qquad \left(\Gamma^{\mathsf{D}}\right)_{m+1} \equiv \Gamma_{m+2} \qquad (\rho_\Gamma)_{m+1} \equiv \Gamma^{\mathbb{01}\langle m+1\rangle}$$
$$(\pi_n \Gamma)^b \equiv \Gamma^b \qquad \left(\Gamma^{\mathsf{D}}\right)^b \equiv \Gamma^{\mathbb{1}b}$$
$$(\pi_n \sigma)_{m+1} \equiv \sigma_{m+1} \qquad \left(\sigma^{\mathsf{D}}\right)_{m+1} \equiv \sigma_{m+2}$$

Now we define the types and terms in sm to be compatible towers of types and terms in the sm^n. In syntax this can be expressed by the following infinitary bidirectional rules:

$$\frac{\left(\gamma : \pi_n \Gamma \vdash_{\mathsf{sm}_n} \pi_n A\ \gamma\ \mathsf{type}_\ell\right)_{n\geq -2} \qquad \left(\pi\ (\pi_{n+1} A) \equiv \pi_n A\right)_{n\geq -2}}{\gamma : \Gamma \vdash_{\mathsf{sm}} A\ \gamma\ \mathsf{type}_\ell}$$

$$\frac{\left(\gamma : \pi_n \Gamma \vdash_{\mathsf{sm}_n} \pi_n t\ \gamma : \pi_n A\ \gamma\right)_{n\geq -2} \qquad \left(\pi\ (\pi_{n+1} t) \equiv \pi_n t\right)_{n\geq -2}}{\gamma : \Gamma \vdash_{\mathsf{sm}} t\ \gamma : A\ \gamma}$$

We also define:

$$A_{n+1} \equiv (\pi_{n+1} A)_{n+1} \qquad t_{n+1} \equiv (\pi_{n+1} t)_{n+1}.$$

All the above constructions in sm^n extend to sm levelwise, since they are preserved strictly by the π_n. In particular, *display* is modified in the absence of truncation:

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}} A\ \gamma\ \mathsf{type}_\ell}{\gamma^+ : \Gamma^{\mathsf{D}},\ a : A^{\rho_\Gamma}\ \gamma^+ \vdash_{\mathsf{sm}} A^{\mathsf{d}}\ \gamma^+\ a\ \mathsf{type}_\ell} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{sm}} t\ \gamma : A\ \gamma}{\gamma^+ : \Gamma^{\mathsf{D}} \vdash_{\mathsf{sm}} t^{\mathsf{d}}\ \gamma^+ : A^{\mathsf{d}}\ \gamma^+\ t^{\rho_\Gamma}}.$$

The computation rules for display similarly hold in sm when modified to exclude π. ◁

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4.3 Modalities

4.3.1 Pieces of the triangle modality

The modality $\triangle$ is supposed to construct a constant (augmented semi-)simplicial diagram, while its left adjoint $\blacksquare_{\triangle}$ picks out the object of (-1)-simplices. Both are determined levelwise by their behaviour on truncated diagrams. (Recall that we will not be modelling the modality $\triangle$ on *types* itself.) We begin by defining a functor $\left(-,\ \blacksquare_{\triangle_{n+1}}\right) : \mathcal{C}^{\triangle^{+}_{n+1}{}^{\mathsf{op}}} \to \mathcal{C}$ via:

$$\left(\gamma : \Gamma,\ \blacksquare_{\triangle_{n+1}}\right) \equiv \Gamma_{-1} \qquad\qquad [\ \sigma,\ \blacksquare_{\triangle_{n+1}}\ ] \equiv \sigma_{-1}$$

Then we construct modal extension for $\triangle_{n+1}$ in sm^{n+1}:

$$\frac{\Gamma\ \mathsf{ob}_{\mathsf{sm}^{n+1}} \qquad \gamma : \Gamma,\ \blacksquare_{\triangle_{n+1}} \vdash_{\mathsf{dm}} A\ \gamma\ \mathsf{type}_{\ell}}{\left(\gamma : \Gamma,\ a :^{\triangle_{n+1}} A\ \gamma\right)\ \mathsf{ob}_{\mathsf{sm}^{n+1}}} \qquad\qquad \frac{\sigma : \Delta \to_{\mathsf{sm}^{n+1}} \Gamma \qquad \gamma : \Gamma,\ \blacksquare_{\triangle_{n+1}} \vdash_{\mathsf{dm}} t\ \gamma : A\ \gamma}{[\ \sigma,\ t\ ]_{\triangle_{n+1}} : \Delta \to_{\mathsf{sm}^{n+1}} \left(\gamma : \Gamma,\ a :^{\triangle_{n+1}} A\ \gamma\right)}$$

The construction is inductive, as before, and the induction includes the following equations:

$$\left(\gamma : \Gamma,\ a :^{\triangle_{n+2}} A\ \gamma\right)^{\mathsf{D}} \equiv \left(\gamma^{+} : \Gamma^{\mathsf{D}},\ a :^{\triangle_{n+1}} A^{[\rho_{\Gamma}, \blacksquare_{\triangle_{n+1}}]}\ \gamma^{+}\right)$$

$$\left([\ \sigma,\ t\ ]_{\triangle_{n+2}}\right)^{\mathsf{D}} \equiv [\ \sigma^{\mathsf{D}},\ t^{[\rho_{\Gamma}, \blacksquare_{\triangle_{n+1}}]}\ ]_{\triangle_{n+1}}$$

Assuming this, the definitions are:

$$\left(\gamma : \Gamma,\ a :^{\triangle_{-1}} A\ \gamma\right)_{-1} \equiv \left(\gamma_{-1} : \Gamma_{-1},\ a : A\ \gamma_{-1}\right)$$

$$\left([\ \sigma,\ t\ ]_{\triangle_{-1}}\right)_{-1} \equiv [\ \sigma_{-1},\ t\ ]$$

$$\left(\gamma : \Gamma,\ a :^{\triangle_{n+2}} A\ \gamma\right)_{m+1} \equiv \left(\gamma^{-} : \pi\Gamma,\ a :^{\triangle_{n+1}} A\ \gamma^{-}\right)_{m+1} \quad \text{for} \quad m \le n$$

$$\left(\gamma : \Gamma,\ a :^{\triangle_{n+2}} A\ \gamma\right)_{n+2} \equiv \left(\gamma^{+} : \Gamma^{\mathsf{D}},\ a :^{\triangle_{n+1}} A^{[\rho_{\Gamma}, \blacksquare_{\triangle_{n+1}}]}\ \gamma^{+}\right)_{n+1}$$

$$\left([\ \sigma,\ t\ ]_{\triangle_{n+2}}\right)_{m+1} \equiv \left([\ \pi\sigma,\ t\ ]_{\triangle_{n+1}}\right)_{m+1} \quad \text{for} \quad m \le n$$

$$\left([\ \sigma,\ t\ ]_{\triangle_{n+2}}\right)_{n+2} \equiv \left([\ \sigma^{\mathsf{D}},\ t^{[\rho_{\Gamma}, \blacksquare_{\triangle_{n+1}}]}\ ]_{\triangle_{n+1}}\right)_{n+1}$$

We next have fundamental context projections and zero variables:

$$\frac{\gamma : \Gamma,\ \blacksquare_{\triangle_{n+1}} \vdash_{\mathsf{dm}} A\ \gamma\ \mathsf{type}_{\ell}}{\mathsf{pt}^{A}_{\triangle_{n+1}} : \left(\gamma : \Gamma,\ a :^{\triangle_{n+1}} A\right) \to_{\mathsf{sm}^{n+1}} \Gamma} \qquad\qquad \frac{\gamma : \Gamma,\ \blacksquare_{\triangle_{n+1}} \vdash_{\mathsf{dm}} A\ \gamma\ \mathsf{type}_{\ell}}{\gamma : \Gamma,\ a :^{\triangle_{n+1}} A\ \gamma,\ \blacksquare_{\triangle_{n+1}} \vdash_{\mathsf{dm}} \mathsf{zv}^{A}_{\triangle_{n+1}}\ \gamma\ a : A^{[\mathsf{pt}, \blacksquare_{\triangle_{n+1}}]}\ \gamma\ a}$$

The definition is again inductive, and the induction includes that:

$$\left(\mathsf{pt}^{A}_{\triangle_{n+2}}\right)^{\mathsf{D}} \equiv \mathsf{pt}^{A^{[\rho_{\Gamma}, \blacksquare_{\triangle_{n+1}}]}}_{\triangle_{n+1}}.$$

Assuming this, the definitions are:

$$\left(\mathsf{pt}^{A}_{\triangle_{-1}}\right)_{-1} \equiv \mathsf{pt}^{A}_{\mathsf{dm}} \qquad\qquad \mathsf{zv}^{A}_{\triangle_{-1}} \equiv \mathsf{zv}^{A}_{\mathsf{dm}}$$

$$\pi\left(\mathsf{pt}^{A}_{\triangle_{n+2}}\right) \equiv \mathsf{pt}^{A}_{\triangle_{n+1}} \qquad \left(\mathsf{pt}^{A}_{\triangle_{n+2}}\right)_{n+2} \equiv \left(\mathsf{pt}^{A^{[\rho_{\Gamma}, \blacksquare_{\triangle_{n+1}}]}}_{\triangle_{n+1}}\right)_{n+1} \qquad \mathsf{zv}^{A}_{\triangle_{n+2}} \equiv \mathsf{zv}^{A}_{\triangle_{n+1}}$$

Finally, we construct modal Π-types:

$$\frac{\gamma : \Gamma,\ \blacksquare_{\triangle} \vdash_{\mathsf{dm}} A\ \gamma\ \mathsf{type}_{\ell_0} \qquad \gamma : \Gamma,\ a :^{\triangle_{n+1}} A\ \gamma \vdash_{\mathsf{sm}^{n+1}} B\ \gamma\ a\ \mathsf{type}_{\ell_1}}{\gamma : \Gamma \vdash_{\mathsf{sm}^{n+1}} \Pi^{\mathsf{sm}^{n+1}}_{\triangle}\ A\ B\ \gamma\ \mathsf{type}_{\ell_0 \sqcup \ell_1}} \qquad\qquad \frac{\gamma : \Gamma,\ a :^{\triangle_{n+1}} A\ \gamma \vdash_{\mathsf{sm}^{n}} t\ \gamma\ a : B\ \gamma\ a}{\gamma : \Gamma \vdash_{\mathsf{sm}} \lambda^{\mathsf{sm}^{n+1}}_{\triangle}\ t\ \gamma : \Pi^{\mathsf{sm}^{n+1}}_{\triangle}\ A\ B\ \gamma}$$

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}^{n}} f\ \gamma : \Pi^{\mathsf{sm}^{n}}_{\triangle}\ A\ B\ \gamma \qquad \gamma : \Gamma,\ \blacksquare_{\triangle} \vdash_{\mathsf{dm}} s\ \gamma : A\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{sm}} \mathsf{app}^{\mathsf{sm}^{n}}_{\triangle}\ f\ s\ \gamma : B^{[\ 1_{\Gamma}, s\ ]_{\triangle}}\ \gamma}$$

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This construction follows the pattern of the non-modal truncated case and is performed levelwise. We will inductively assert the following formulas for display:

$$\left(\Pi_\triangle^{\mathsf{sm}^{n+2}}\ A\ B\right)^{\mathsf{d}} \equiv \Pi_\triangle^{\mathsf{sm}^{n+1}}\left(A^{[\rho_\Gamma, \blacksquare_\triangle]}\right)^{\mathsf{pt}}\ \left(B^{\mathsf{d}}\right)^{[\ \mathsf{W}_2^{A^{[\rho_\Gamma,\blacksquare_\triangle]}}\ \mathsf{pt},\ \mathsf{app}_\triangle^{\mathsf{sm}^{n+1}}\ \mathsf{zv}^{\mathsf{pt}\circ\mathsf{pt}}\ \mathsf{zv}_\triangle^{\mathsf{pt}}\ ]}$$

$$\left(\lambda_\triangle^{\mathsf{sm}^{n+2}}\ t\right)^{\mathsf{d}} \equiv \lambda_\triangle^{\mathsf{sm}^{n+1}}\ t^{\mathsf{d}}$$

$$\left(\mathsf{app}_\triangle^{\mathsf{sm}^{n+2}}\ f\ s\right)^{\mathsf{d}} \equiv \mathsf{app}_\triangle^{\mathsf{sm}^{n+1}}\ f^{\mathsf{d}}\ s.$$

In dimension -1 we set:

$$\left(\Pi_\triangle^{\mathsf{sm}^{-1}}\ A\ B\right)_{-1} \equiv \Pi^{\mathsf{dm}}\ A\ B_{-1} \qquad \left(\lambda_\triangle^{\mathsf{sm}^{-1}}\ t\right)_{-1} \equiv \lambda^{\mathsf{dm}}\ t_{-1} \qquad \left(\mathsf{app}_\triangle^{\mathsf{sm}^{-1}}\ f\ s\right)_{-1} \equiv \mathsf{app}^{\mathsf{dm}}\ f_{-1}\ s.$$

Then we inductively define:

$$\pi\left(\Pi_\triangle^{\mathsf{sm}^{n+2}}\ A\ B\right) \equiv \Pi_\triangle^{\mathsf{sm}^{n+1}}\ A\ \pi B$$

$$\left(\Pi_\triangle^{\mathsf{sm}^{n+2}}\ A\ B\right)_{n+2} \equiv \left(\Pi_\triangle^{\mathsf{sm}^{n+1}}\left(A^{[\rho_\Gamma,\blacksquare_\triangle]}\right)^{\mathsf{pt}}\ \left(B^{\mathsf{d}}\right)^{[\ \mathsf{W}_2^{A^{[\rho_\Gamma,\blacksquare_\triangle]}}\ \mathsf{pt},\ \mathsf{app}_\triangle^{\mathsf{sm}^{n+1}}\ \mathsf{zv}^{\mathsf{pt}\circ\mathsf{pt}}\ \mathsf{zv}_\triangle^{\mathsf{pt}}\ ]}\right)_{n+1}$$

$$\pi\left(\lambda_\triangle^{\mathsf{sm}^{n+2}}\ t\right) \equiv \lambda_\triangle^{\mathsf{sm}^{n+1}}\ \pi t \qquad \pi\left(\mathsf{app}_\triangle^{\mathsf{sm}^{n+2}}\ f\ s\right) \equiv \mathsf{app}_\triangle^{\mathsf{sm}^{n+1}}\ \pi f\ s$$

$$\left(\lambda_\triangle^{\mathsf{sm}^{n+2}}\ t\right)_{n+2} \equiv \lambda_\triangle^{\mathsf{sm}^{n+1}}\ t^{\mathsf{d}} \qquad \left(\mathsf{app}_\triangle^{\mathsf{sm}^{n+2}}\ f\ s\right)_{n+2} \equiv \mathsf{app}_\triangle^{\mathsf{sm}^{n+1}}\ f^{\mathsf{d}}\ s.$$

As usual, the verification of many identities has been omitted.

Finally, we check that the truncation functors π preserve all the operations defined above. Therefore, we can define the untruncated operations $\triangle$ and $\blacksquare_\triangle$ on sm, with modal context extension $x :^{\triangle} A$ and modal Π-types, simply by acting levelwise on each sm^{n+1}. ◁

4.3.2 Pieces of the box modality

The box modality is not determined levelwise by operations on truncated diagrams, but we can still construct it using truncated data. We start with a truncated lock functor $\left(-, \blacksquare_{\Box_n}\right) : \mathcal{C} \to \mathcal{C}^{\Delta_n^{+\mathsf{op}}}$ that constructs a constant simplicial diagram:

$$\left(\gamma : \Gamma,\ \blacksquare_{\Box_{n+1}}\right)_{m+1} \equiv \Gamma \qquad \left(\gamma : \Gamma,\ \blacksquare_{\Box_{n+1}}\right)^b \equiv 1_\Gamma \qquad [\ \sigma,\ \blacksquare_{\Box_{n+1}}\ ]_{m+1} \equiv \sigma.$$

We the define four new operations, primarily $A_{\Box(n+1)}$ which is like a truncated version of $\Box$, taking the limit of a truncated diagram, but yielding a finite telescope rather than a type:

$$\frac{\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma \vdash_{\mathsf{dm}} A_{\Box(n+1)}\ \gamma\ \mathsf{tel}_\ell} \qquad \frac{\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} t\ \gamma : A\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{dm}} t_{\Box(n+1)} : A_{\Box(n+1)}\ \gamma} \qquad \frac{\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{\mathsf{pt}^A_{\Box_n} : \left(\gamma : \Gamma,\ \Box a : A_{\Box(n+1)}\ \gamma\right) \to \Gamma}$$

$$\frac{\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma,\ \Box a : A_{\Box(n+1)}\ \gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} \mathsf{zv}^A_{\Box_n}\ \gamma\ \Box a : A^{[\ \mathsf{pt}^A_{\Box_n}, \blacksquare_{\Box_n}\ ]}\ \gamma\ \Box a}$$

These will satisfy the inductively proven property that for $\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} t\ \gamma : A\ \gamma$:

$$\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} \mathsf{zv}^A_{\Box_n}\ \gamma\ \left(t_{\Box(n+1)}\ \gamma\right) \equiv t\ \gamma.$$

For sm^{-2}, the term $\mathsf{zv}^A_{\Box_{-2}}$ is trivial, since it lives in the terminal CwF. We also set:

$$A_{\Box(-1)}\ \gamma \equiv ()_{\mathsf{dm}} \qquad t_{\Box(-1)}\ \gamma \equiv [\ ]_{\mathsf{dm}} \qquad \mathsf{pt}^A_{\Box_{-2}} \equiv 1_\Gamma.$$

Note that, in general, since $\mathsf{zv}^A_{\Box_n}$ is a simplicial term, we may form its matching substitution:

$$\frac{\gamma : \Gamma,\ \blacksquare_{\Box_n} \vdash_{\mathsf{sm}^n} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma,\ \Box a : A_{\Box(n+1)}\ \gamma \vdash_{\mathsf{dm}} \left(\mathsf{zv}^A_{\Box_n}\right)_{\partial(n+1)}\ \gamma\ \Box a : A_{\partial(n+1)}\ \gamma}$$

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For sm^{n+1} we then inductively set:

$$\begin{aligned}
A_{\square(n+2)} &\equiv \big(a : \pi A_{\square(n+1)}\ \gamma,\ a' : A_{n+1}\ \gamma\ \big(\big(\mathsf{zv}^{\pi A}_{\square_n}\big)_{\partial(n+1)}\ \gamma\ a\big)\\
t_{\square(n+2)} &\equiv [\ t_{\square(n+1)},\ t_{n+1}\ ]\\
\mathsf{pt}^{A}_{\square_{n+1}} &\equiv \mathsf{pt}^{\pi A}_{\square_n} \circ \mathsf{pt}^{A_{n+1}}_{\mathsf{dm}}\\
\mathsf{zv}^{A}_{\square_{n+1}} &\equiv \langle\ \big(\mathsf{zv}^{\pi A}_{\square_n}\big)^{[\ \mathsf{pt}^{A_{n+1}}_{\mathsf{dm}},\, \blacksquare_{\square}\ ]}\ ,\ \big(\mathsf{zv}^{A}_{\square_{n+1}}\big)_{n+1}\ \rangle\\
\big(\mathsf{zv}^{A}_{\square_{n+1}}\big)_{n+1}\ \gamma\ [\ a,\ a'\ ] &\equiv a'.
\end{aligned}$$

The second line is well typed by the inductive hypothesis and makes the next case of the hypothesis clear. Also, note the pt substitution in the fourth line. The top-dimensional simplicial value of a boxed variable accesses the last component of the modal context extension, whereas lower dimensional simplicial values search further back in the linear context.

The untruncated $\blacksquare_{\square}$ similarly constructs a constant presheaf. Note that $(\Gamma,\ \blacksquare_{\square})^{\,\mathsf{D}} \equiv (\Gamma,\ \blacksquare_{\square})$, and $\rho_{\Gamma,\blacksquare_{\square}}$ is an identity; thus we will omit writing these. We now define:

$$\begin{aligned}
&\mathsf{q}^{\triangle\square \le 1_{\mathsf{sm}}} : 1_{\mathsf{sm}} \Rightarrow (-,\ \blacksquare_{\triangle\square})\\
&\big(\mathsf{q}^{\triangle\square \le 1_{\mathsf{sm}}}_{\Gamma}\big)_{m+1} \equiv \Gamma^{\mathbb{0}^{m+1}}.
\end{aligned}$$

We then use the constructions above to construct a modal type former:

$$\frac{\gamma : \Gamma,\ \blacksquare_{\square} \vdash_{\mathsf{sm}} A\ \gamma\ \mathsf{type}_{\ell}}{\gamma : \Gamma \vdash_{\mathsf{dm}} \square_{\mathsf{sm}}\ A\ \gamma\ \mathsf{type}_{\ell}} \qquad \frac{\gamma : \Gamma,\ \blacksquare_{\square} \vdash_{\mathsf{sm}} t\ \gamma : A\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{dm}} \square_{\mathsf{sm}}\ t\ \gamma : \square_{\mathsf{sm}}\ A\ \gamma} \qquad \frac{\gamma_{\text{-}1} : \blacksquare_{\triangle}\Gamma \vdash_{\mathsf{dm}} t\ \gamma_{\text{-}1} : \square\ A\ \gamma_{\text{-}1}}{\gamma : \Gamma \vdash_{\mathsf{sm}} \blacksquare^{A}_{\mathsf{sm}}\ t\ \gamma : A\ \big(\mathsf{q}^{\triangle\square \le 1_{\mathsf{sm}}}_{\Gamma}\ \gamma\big)}$$

(Recall from section 4.3.1 that $\blacksquare_{\triangle}\Gamma \equiv \Gamma_{\text{-}1}$.) In order to form these, we will take an ω-limit of sequences $A_{\square}$ or $t_{\square}$ obtained from the m-simplex levels of A or t:

$$\frac{\gamma : \Gamma,\ \blacksquare_{\square} \vdash_{\mathsf{sm}} A\ \gamma\ \mathsf{type}_{\ell}}{\gamma : \Gamma \vdash_{\mathsf{dm}} A_{\square}\ \mathsf{stel}^{\infty}_{\ell}} \qquad \frac{\gamma : \Gamma,\ \blacksquare_{\square} \vdash_{\mathsf{sm}} t\ \gamma : A\ \gamma\ \mathsf{type}_{\ell}}{\gamma : \Gamma \vdash_{\mathsf{dm}} t_{\square}\ \gamma : A_{\square}\ \gamma}$$

These are defined as follows:

$$\begin{aligned}
A^{\partial(m+1)}_{\square}\ \gamma &\equiv (\pi_m A)_{\square(m+1)}\ \gamma & t^{\partial(m+1)}_{\square}\ \gamma &\equiv (\pi_m t)_{\square(m+1)}\ \gamma\\
A^{m+1}_{\square}\ \gamma\ \square a &\equiv A_{m+1}\ \gamma\ \big(\big(\mathsf{zv}^{\pi_m A}_{\square_m}\big)_{\partial(m+1)}\ \gamma\ \square a\big) & t^{m+1}_{\square}\ \gamma &\equiv t_{m+1}\ \gamma.
\end{aligned}$$

We then define:

$$\square_{\mathsf{sm}}\ A \equiv \lim A_{\square} \qquad\qquad \square_{\mathsf{sm}}\ t \equiv \lim t_{\square}.$$

We define the eliminator by:

$$\pi_{n+1}\ \big(\blacksquare^{A}_{\mathsf{sm}}\ \mathfrak{a}\big)\ \gamma_{n+1} \equiv \mathsf{zv}^{\pi_{n+1}A}_{\square_{n+1}}\ \gamma_{n+1}{}^{\mathbb{0}^{n+1}}\ [\ \mathsf{res}^{\partial(n+1)}\ \gamma_{n+1}{}^{\mathbb{0}^{n+1}}\ \mathfrak{a},\ \mathsf{res}^{n+1}\ \gamma_{n+1}{}^{\mathbb{0}^{n+1}}\ \mathfrak{a}\ ]$$

and then check the computation laws. ◁

4.3.3 *The extended simplicial model*

Syntactically, the context $(\Gamma,\ \blacksquare_{\Diamond})$ is supposed to be *flat*. A naïve approach to modelling this would add initial objects to a discrete context. Instead we define an *extended simplicial model* sm_{+}, built from a copy of dm (the flat contexts) and sm (the non-flat contexts). Its objects, morphisms, types, and terms are generated inductively by the rules in Figure 11. Up to isomorphism, its underlying category $\mathcal{C}_{+}^{\Delta^{+\mathsf{op}}}$ is the '*collage*' or '*cograph*' of the functor $(-)_{\text{-}1}$; the rules of Figure 11 implement this by labelling everything based on its origin. We define the action of substitutions as follows,

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$$\frac{\Gamma\ \mathsf{ob}_{\mathsf{dm}}}{\mathsf{in}_{\mathsf{dm}}\ \Gamma\ \mathsf{ob}_{\mathsf{sm}_+}} \qquad \frac{\Gamma\ \mathsf{ob}_{\mathsf{sm}}}{\mathsf{in}_{\mathsf{sm}}\ \Gamma\ \mathsf{ob}_{\mathsf{sm}_+}}$$

$$\frac{\sigma : \Delta \to_{\mathsf{sm}} \Gamma}{\mathsf{in}_{\mathsf{sm}}\ \sigma : \mathsf{in}_{\mathsf{sm}}\ \Delta \to_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{sm}}\ \Gamma} \qquad \frac{\sigma : \Delta \to_{\mathsf{dm}} \Gamma}{\mathsf{in}_{\mathsf{dm}}\ \sigma : \mathsf{in}_{\mathsf{dm}}\ \Delta \to_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{dm}}\ \Gamma}$$

$$\frac{\sigma : \Delta \to_{\mathsf{dm}} \Gamma_{-1}}{\mathsf{in}_{\mathsf{fl}}\ \sigma : \mathsf{in}_{\mathsf{dm}}\ \Delta \to_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{sm}}\ \Gamma}$$

$$\frac{\gamma : \Gamma \vdash_{\mathsf{dm}} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \mathsf{in}_{\mathsf{dm}}\ \Gamma \vdash_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{dm}}\ A\ \gamma\ \mathsf{type}_\ell} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{sm}} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \mathsf{in}_{\mathsf{sm}}\ \Gamma \vdash_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{sm}}\ A\ \gamma\ \mathsf{type}_\ell}$$

$$\frac{\gamma : \Gamma \vdash_{\mathsf{dm}} t\ \gamma : A\ \gamma}{\gamma : \mathsf{in}_{\mathsf{dm}}\ \Gamma \vdash_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{dm}}\ t\ \gamma : \mathsf{in}_{\mathsf{dm}}\ A\ \gamma} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{sm}} t\ \gamma : A\ \gamma}{\gamma : \mathsf{in}_{\mathsf{sm}}\ \Gamma \vdash_{\mathsf{sm}_+} \mathsf{in}_{\mathsf{sm}}\ t\ \gamma : \mathsf{in}_{\mathsf{sm}}\ A\ \gamma}$$

Figure 11. The extended simplicial model.

discarding higher data in the flat case:

$$(\mathsf{in}_{\mathsf{dm}}\ A)^{\mathsf{in}_{\mathsf{dm}}\ \sigma} \equiv \mathsf{in}_{\mathsf{dm}}\ A^\sigma \qquad (\mathsf{in}_{\mathsf{sm}}\ A)^{\mathsf{in}_{\mathsf{sm}}\ \sigma} \equiv \mathsf{in}_{\mathsf{sm}}\ A^\sigma \qquad (\mathsf{in}_{\mathsf{sm}}\ A)^{\mathsf{in}_{\mathsf{fl}}\ \sigma} \equiv \mathsf{in}_{\mathsf{dm}}\ (A_{-1})^\sigma$$
$$(\mathsf{in}_{\mathsf{dm}}\ t)^{\mathsf{in}_{\mathsf{dm}}\ \sigma} \equiv \mathsf{in}_{\mathsf{dm}}\ A^\sigma \qquad (\mathsf{in}_{\mathsf{sm}}\ t)^{\mathsf{in}_{\mathsf{sm}}\ \sigma} \equiv \mathsf{in}_{\mathsf{sm}}\ A^\sigma \qquad (\mathsf{in}_{\mathsf{sm}}\ t)^{\mathsf{in}_{\mathsf{fl}}\ \sigma} \equiv \mathsf{in}_{\mathsf{dm}}\ (A_{-1})^\sigma\ .$$

Extension of contexts operates by passing under the inclusion:

$$(\mathsf{in}_{\mathsf{dm}}\ \Gamma,\ \mathsf{in}_{\mathsf{dm}}\ A) \equiv \mathsf{in}_{\mathsf{dm}}\ (\Gamma,\ A) \qquad [\ \mathsf{in}_{\mathsf{dm}}\ \sigma,\ \mathsf{in}_{\mathsf{dm}}\ t\ ] \equiv \mathsf{in}_{\mathsf{dm}}\ [\ \sigma,\ t\ ]$$
$$(\mathsf{in}_{\mathsf{sm}}\ \Gamma,\ \mathsf{in}_{\mathsf{sm}}\ A) \equiv \mathsf{in}_{\mathsf{sm}}\ (\Gamma,\ A) \qquad [\ \mathsf{in}_{\mathsf{sm}}\ \sigma,\ \mathsf{in}_{\mathsf{sm}}\ t\ ] \equiv \mathsf{in}_{\mathsf{sm}}\ [\ \sigma,\ t\ ]$$
$$[\ \mathsf{in}_{\mathsf{fl}}\ \sigma,\ \mathsf{in}_{\mathsf{dm}}\ t\ ] \equiv \mathsf{in}_{\mathsf{fl}}\ [\ \sigma,\ t\ ].$$

Note that in the last case, if we have $\mathsf{in}_{\mathsf{fl}}\ \sigma : \mathsf{in}_{\mathsf{dm}}\ \Delta \to \mathsf{in}_{\mathsf{sm}}\ \Gamma$ then $\sigma : \Delta \to \Gamma_{-1}$. To form the extension $[\ \mathsf{in}_{\mathsf{fl}}\ \sigma,\ s\ ] : \mathsf{in}_{\mathsf{dm}}\ \Delta \to \mathsf{in}_{\mathsf{sm}}\ (\gamma : \Gamma,\ a : A\ \gamma)$, we must give $\delta : \mathsf{in}_{\mathsf{dm}}\ \Delta \vdash_{\mathsf{sm}_+} s\ \delta : (\mathsf{in}_{\mathsf{sm}}\ A)^{\mathsf{in}_{\mathsf{fl}}\ \sigma}\ \delta$. Such an s has type $\mathsf{in}_{\mathsf{dm}}\ (A_{-1})^\sigma$ and must be of the form $\mathsf{in}_{\mathsf{dm}}\ t$. ◁

To define (non-modal) Π-types and universes in sm_+, we reduce to the respective constructs in dm and sm, depending on whether or not the context is flat:

$$\Pi^{\mathsf{sm}_+}\ (\mathsf{in}_{\mathsf{dm}}\ A)\ (\mathsf{in}_{\mathsf{dm}}\ B) \equiv \mathsf{in}_{\mathsf{dm}}\ \left(\Pi^{\mathsf{dm}}\ A\ B\right)$$
$$\Pi^{\mathsf{sm}_+}\ (\mathsf{in}_{\mathsf{sm}}\ A)\ (\mathsf{in}_{\mathsf{sm}}\ B) \equiv \mathsf{in}_{\mathsf{sm}}\ \left(\Pi^{\mathsf{sm}}\ A\ B\right)$$
$$\mathsf{Type}_\ell^{\mathsf{sm}_+} \equiv \begin{cases} \mathsf{in}_{\mathsf{dm}}\ \mathsf{Disc}_\ell & \text{for} \quad \mathsf{in}_{\mathsf{dm}}\ \Gamma \\ \mathsf{in}_{\mathsf{sm}}\ \mathsf{Type}_\ell^{\mathsf{sm}} & \text{for} \quad \mathsf{in}_{\mathsf{sm}}\ \Gamma. \end{cases}$$

The definitions of λ^{sm_+}, $\mathsf{app}^{\mathsf{sm}_+}$, $\mathsf{Code}^{\mathsf{sm}_+}$, and $\mathsf{El}^{\mathsf{sm}_+}$ are similar. For stability under substitution, we have to additionally consider flat substitutions; this works because everything agrees with its discrete version in dimension -1. In the rest of this section, we show how dm and sm^+ can be made into a model of all of dTT (except for the type-former △). ◁

4.3.4 Locks and keys

The definition of sm^+ is tailored to $🔒_\Diamond$. Putting this together with the $🔒_\triangle$ and $🔒_\Box$ defined on sm in Sections 4.3.1 and 4.3.2, we now define a 2-functor $[\![\ -\ ]\!] : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$, where $\mathcal{M}^{\mathsf{coop}}$ denotes

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the 2-category obtained by reversing both 1 and 2 cells. On modes, we have:

$$[\![\ \mathsf{dm}\ ]\!] \equiv \mathcal{C} \qquad\qquad [\![\ \mathsf{sm}\ ]\!] \equiv \mathcal{C}_+^{\Delta^{+\mathrm{op}}}.$$

To define this 2-functor on modalities, we extend the prior definitions of locks to sm_+:

$$\left(-,\ \mathbin{🔒}_\Delta^+\right) : \mathcal{C}_+^{\Delta^{+\mathrm{op}}} \to \mathcal{C} \qquad \left(-,\ \mathbin{🔒}_\square^+\right) : \mathcal{C} \to \mathcal{C}_+^{\Delta^{+\mathrm{op}}} \qquad \left(-,\ \mathbin{🔒}_\lozenge^+\right) : \mathcal{C} \to \mathcal{C}_+^{\Delta^{+\mathrm{op}}}$$

$$\left(\mathsf{in}_{\mathsf{sm}}\ \Gamma,\ \mathbin{🔒}_\Delta^+\right) \equiv \left(\Gamma,\ \mathbin{🔒}_\Delta\right) \qquad \left(\Gamma,\ \mathbin{🔒}_\square^+\right) \equiv \mathsf{in}_{\mathsf{sm}}\ \left(\Gamma,\ \mathbin{🔒}_\square\right) \qquad \left(\Gamma,\ \mathbin{🔒}_\lozenge^+\right) \equiv \mathsf{in}_{\mathsf{dm}}\ \Gamma$$

$$\left(\mathsf{in}_{\mathsf{dm}}\ \Gamma,\ \mathbin{🔒}_\Delta^+\right) \equiv \Gamma$$

$$[\ \mathsf{in}_{\mathsf{sm}}\ \sigma,\ \mathbin{🔒}_\Delta^+\ ] \equiv \sigma_{-1} \qquad [\ \sigma,\ \mathbin{🔒}_\square^+\ ]_{m+1} \equiv \mathsf{in}_{\mathsf{sm}}\ [\ \sigma,\ \mathbin{🔒}_\square\ ] \qquad [\ \sigma,\ \mathbin{🔒}_\lozenge^+\ ] \equiv \mathsf{in}_{\mathsf{dm}}\ \sigma.$$

$$[\ \mathsf{in}_{\mathsf{dm}}\ \sigma,\ \mathbin{🔒}_\Delta^+\ ] \equiv \sigma$$

$$[\ \mathsf{in}_{\mathsf{fl}}\ \sigma,\ \mathbin{🔒}_\Delta^+\ ] \equiv \sigma$$

We then define the evident composites:

$$\left(-,\ \mathbin{🔒}_{\Delta\square}^+\right) \equiv \left(-,\ \mathbin{🔒}_\Delta^+,\ \mathbin{🔒}_\square^+\right) \qquad\qquad \left(-,\ \mathbin{🔒}_{\Delta\lozenge}^+\right) \equiv \left(-,\ \mathbin{🔒}_\Delta^+,\ \mathbin{🔒}_\lozenge^+\right).$$

Finally, it is easy to check that $\left(-,\ \mathbin{🔒}_\square^+,\ \mathbin{🔒}_\Delta^+\right)$ and $\left(-,\ \mathbin{🔒}_\lozenge^+,\ \mathbin{🔒}_\Delta^+\right)$ define identity functors. It follows that we have a contravariantly functorial assignment:

$$\frac{\mu : p \to q}{[\![\ \mu\ ]\!] \equiv \left(-,\ \mathbin{🔒}_\mu^+\right) : [\![\ q\ ]\!] \to [\![\ p\ ]\!]}$$

To define this 2-functor on 2-cells, we define the *key* transformations. We have $\square \le \lozenge$, $\Delta\square \le 1_{\mathsf{sm}_+}$, and $1_{\mathsf{sm}_+} \le \Delta\lozenge$, which corresponds to the following natural transformations:

$$\mathbin{🔑}^{\square\le\lozenge} : \left(-,\ \mathbin{🔒}_\lozenge^+\right) \Rightarrow \left(-,\ \mathbin{🔒}_\square^+\right) \qquad \mathbin{🔑}^{\Delta\square\le 1_{\mathsf{sm}_+}} : 1_{\mathsf{sm}_+} \Rightarrow \left(-,\ \mathbin{🔒}_{\Delta\square}^+\right) \qquad \mathbin{🔑}^{1_{\mathsf{sm}_+}\le\Delta\lozenge} : \left(-,\ \mathbin{🔒}_{\Delta\lozenge}^+\right) \Rightarrow 1_{\mathsf{sm}_+}$$

$$\mathbin{🔑}_\Gamma^{\square\le\lozenge} \equiv \mathsf{in}_{\mathsf{fl}}\ 1_\Gamma \qquad \mathbin{🔑}_{\mathsf{in}_{\mathsf{sm}}\ \Gamma}^{\Delta\square\le 1_{\mathsf{sm}_+}} \equiv \mathsf{in}_{\mathsf{sm}}\ \mathbin{🔑}_\Gamma^{\Delta\square\le 1_{\mathsf{sm}}} \qquad \mathbin{🔑}_{\mathsf{in}_{\mathsf{sm}}\Gamma}^{1_{\mathsf{sm}}\le\Delta\lozenge} \equiv \mathsf{in}_{\mathsf{fl}}\ 1_{\Gamma_{-1}}$$

$$\mathbin{🔑}_{\mathsf{in}_{\mathsf{dm}}\ \Gamma}^{\Delta\square\le 1_{\mathsf{sm}}} \equiv \mathsf{in}_{\mathsf{fl}}\ 1_\Gamma \qquad \mathbin{🔑}_{\mathsf{in}_{\mathsf{dm}}\Gamma}^{1_{\mathsf{sm}}\le\Delta\lozenge} \equiv \mathsf{in}_{\mathsf{dm}}\ 1_\Gamma.$$

We also have $\mathbin{🔑}^{\Delta\square\le\Delta\lozenge} \equiv \mathbin{🔑}^{\Delta\square\le 1_{\mathsf{sm}_+}} \circ \mathbin{🔑}^{1_{\mathsf{sm}_+}\le\Delta\lozenge}$. These are contravariantly functorial:

$$\frac{\alpha : \mu \le \nu}{[\![\ \alpha\ ]\!] \equiv \mathbin{🔑}^\alpha : [\![\ \nu\ ]\!] \Rightarrow [\![\ \mu\ ]\!]}.$$

One checks whiskering identities to verify that $[\![\ -\ ]\!]$ defines a 2-functor $\mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$. ◁

4.3.5 Modal types

We are modelling two modal type formers, with their introduction and elimination rules:

$$\frac{\gamma : \Gamma,\ \mathbin{🔒}_\lozenge^+ \vdash_{\mathsf{sm}_+} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma \vdash_{\mathsf{dm}} \lozenge\ A\ \gamma\ \mathsf{type}_\ell} \qquad \frac{\gamma : \Gamma,\ \mathbin{🔒}_\lozenge^+ \vdash_{\mathsf{sm}_+} t\ \gamma : A\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{dm}} \lozenge\ t\ \gamma : \lozenge\ A\ \gamma} \qquad \frac{\gamma : \Gamma \vdash_{\mathsf{dm}} t\ \gamma : \lozenge\ A\ \gamma}{\gamma : \mathsf{in}_{\mathsf{dm}}\ \Gamma \vdash_{\mathsf{sm}_+} \blacklozenge^A\ t\ \gamma : A\ \gamma}$$

$$\frac{\gamma : \Gamma,\ \mathbin{🔒}_\square^+ \vdash_{\mathsf{sm}_+} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma \vdash_{\mathsf{dm}} \square\ A\ \gamma\ \mathsf{type}_\ell}. \qquad \frac{\gamma : \Gamma,\ \mathbin{🔒}_\square^+ \vdash_{\mathsf{sm}_+} t\ \gamma : A\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{dm}} \square\ t\ \gamma : \square\ A\ \gamma} \qquad \frac{\gamma : \Gamma,\ \mathbin{🔒}_\Delta^+ \vdash_{\mathsf{dm}} t\ \gamma : \square\ A\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{sm}_+} \blacksquare^A\ t\ \gamma : A\ [\ \mathbin{🔑}_\Gamma^{\Delta\square\le 1_{\mathsf{sm}_+}}\ \gamma\ ]}.$$

We have simplified the statement of $\blacklozenge$, using the fact that Γ flat means Γ is of the form $\mathsf{in}_{\mathsf{dm}}\ \Delta$, in which case we have $\left((\mathsf{in}_{\mathsf{dm}}\ \Delta),\ \mathbin{🔒}_\Delta^+\right) \equiv \Delta$ by definition. The key $\mathbin{🔑}_\Gamma^{\Delta\lozenge\ge 1_{\mathsf{sm}_+}}$ does not arise from a

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natural transformation and is only defined when $\Gamma \equiv \mathsf{in}_{\mathsf{dm}}\ \Delta$, in which case $\mathsf{key}^{\Delta\Diamond \geq 1_{\mathsf{sm}+}}_{\mathsf{in}_{\mathsf{dm}}\ \Delta} \equiv \mathsf{in}_{\mathsf{dm}}\ 1_\Delta$. Thus, this can be seen as a proof-relevant pattern match along flat.

The definition of the $\Diamond$ rules shuffle around discrete information and inclusions:

$$\Diamond\ (\mathsf{in}_{\mathsf{dm}}\ A) \equiv A \qquad \Diamond\ (\mathsf{in}_{\mathsf{dm}}\ t) \equiv t \qquad \blacklozenge^A\ t \equiv \mathsf{in}_{\mathsf{dm}}\ t.$$

For $\Box$, we fall back to our prior construction for the type former and introduction rule:

$$\Box\ (\mathsf{in}_{\mathsf{sm}}\ A) \equiv \Box_{\mathsf{sm}}\ A \qquad \Box\ (\mathsf{in}_{\mathsf{sm}}\ t) \equiv \Box_{\mathsf{sm}}\ t$$

while for the eliminator, we split on whether or not Γ is flat:

$$\blacksquare^{\mathsf{in}_{\mathsf{sm}}\ A}\ t \equiv \begin{cases} \mathsf{in}_{\mathsf{dm}}\ \left(\blacksquare^A_{\mathsf{dm}}\ t\right) & \text{for} \quad \mathsf{in}_{\mathsf{dm}}\ \Gamma \\ \mathsf{in}_{\mathsf{sm}}\ \left(\blacksquare^A_{\mathsf{sm}}\ t\right) & \text{for} \quad \mathsf{in}_{\mathsf{sm}}\ \Gamma \end{cases}$$

where the discrete case above is as follows:

$$\frac{\Gamma\ \mathsf{ob}_{\mathsf{dm}} \qquad \gamma : \Gamma \vdash_{\mathsf{dm}} t\ \gamma : \lim A_\Box\ \gamma}{\gamma : \Gamma \vdash_{\mathsf{dm}} \blacksquare^A_{\mathsf{dm}}\ t\ \gamma : A_{-1}\ \gamma}$$

It is defined by:

$$\blacksquare^{\mathsf{in}_{\mathsf{sm}}\ A}\ t \equiv \mathsf{res}^{-1}\ \gamma\ \mathfrak{a}$$

◁

4.3.6 Modal variables

We have the following rules for extending a context and substitution modally:

$$\frac{\mu : p \to q \qquad \Gamma\ \mathsf{ob}_{[\![q]\!]} \qquad \gamma : \Gamma, \fbox{μ}^+ \vdash_{[\![p]\!]} A\ \gamma\ \mathsf{type}_\ell}{(\gamma : \Gamma,\ a :^{\mu_+} A\ \gamma)\ \mathsf{ob}_{[\![q]\!]}} \qquad \frac{\sigma : \Delta \to_{[\![q]\!]} \Gamma \qquad \gamma : \Gamma, \fbox{μ}^+ \vdash_{[\![p]\!]} t\ \gamma : A\ \gamma}{[\ \sigma,\ t\ ]_{\mu_+} : \Delta \to_{[\![q]\!]} (\gamma : \Gamma,\ a :^{\mu_+} A\ \gamma)}$$

The case of Δ_+ is defined as follows, splitting on whether or not Γ is flat:

$$\left(\gamma : \mathsf{in}_{\mathsf{dm}}\ \Gamma,\ a :^{\Delta_+} A\ \gamma\right) \equiv \mathsf{in}_{\mathsf{dm}}\ (\gamma : \Gamma,\ a : A\ \gamma) \qquad [\ \mathsf{in}_{\mathsf{dm}}\ \sigma,\ t\ ]_{\Delta_+} \equiv \mathsf{in}_{\mathsf{dm}}\ [\ \sigma,\ t\ ]$$

$$\left(\gamma : \mathsf{in}_{\mathsf{sm}}\ \Gamma,\ a :^{\Delta_+} A\ \gamma\right) \equiv \mathsf{in}_{\mathsf{sm}}\ \left(\gamma : \Gamma,\ a :^{\Delta} A\ \gamma\right) \qquad [\ \mathsf{in}_{\mathsf{sm}}\ \sigma,\ t\ ]_{\Delta_+} \equiv \mathsf{in}_{\mathsf{sm}}\ [\ \sigma,\ t\ ]_\Delta.$$

Following this, the rest of the definitions say that the case of modal extension reduces to extension by a variable or term of modal type:

$$\left(\gamma : \Gamma,\ a :^{\Diamond_+} A\ \gamma\right) \equiv (\gamma : \Gamma,\ a : \Diamond\ A\ \gamma) \qquad \left(\gamma : \Gamma,\ a :^{\Delta\Diamond_+} A\ \gamma\right) \equiv \left(\gamma : \Gamma,\ a :^{\Delta_+} \Diamond\ A\ \gamma\right)$$

$$[\ \sigma,\ t\ ]_{\Diamond_+} \equiv [\ \sigma,\ t\ ] \qquad [\ \sigma,\ t\ ]_{\Delta\Diamond_+} \equiv [\ \sigma,\ t\ ]_{\Delta_+}$$

$$(\gamma : \Gamma,\ a :^{\Box_+} A\ \gamma) \equiv (\gamma : \Gamma,\ a : \Box\ A\ \gamma) \qquad \left(\gamma : \Gamma,\ a :^{\Delta\Box_+} A\ \gamma\right) \equiv \left(\gamma : \Gamma,\ a :^{\Delta_+} \Box\ A\ \gamma\right)$$

$$[\ \sigma,\ t\ ]_{\Box_+} \equiv [\ \sigma,\ t\ ] \qquad [\ \sigma,\ t\ ]_{\Delta\Box_+} \equiv [\ \sigma,\ t\ ]_{\Delta_+}.$$

Each of the context extension operations comes with a notion of parent maps and variables:

$$\frac{\gamma : \Gamma, \fbox{μ}^+ \vdash_{[\![p]\!]} A\ \gamma\ \mathsf{type}_\ell}{\mathsf{pt}^A_{\mu_+} : \left(\gamma : \Gamma,\ a :^{\mu_+} A\ \gamma\right) \to \Gamma} \qquad \frac{\gamma : \Gamma, \fbox{μ}^+ \vdash_{[\![p]\!]} A\ \gamma\ \mathsf{type}_\ell}{\gamma : \Gamma,\ a :^{\mu_+} A\ \gamma, \fbox{μ}^+ \vdash_{[\![p]\!]} \mathsf{zv}^A_{\mu_+}\ \gamma\ a : A^{[\ \mathsf{pt}^A_{\mu_+}, \fbox{μ}^+\ ]}\ \gamma\ a}$$

These are defined as follows:

$$\mathsf{pt}^A_{\Delta_+} \equiv \begin{cases} \mathsf{in}_{\mathsf{dm}}\ \mathsf{pt}^A_{\mathsf{dm}} & \text{for} \quad \mathsf{in}_{\mathsf{dm}}\ \Gamma \\ \mathsf{in}_{\mathsf{sm}}\ \mathsf{pt}^A_{\Delta} & \text{for} \quad \mathsf{in}_{\mathsf{sm}}\ \Gamma \end{cases} \qquad \mathsf{zv}^A_{\Delta_+} \equiv \begin{cases} \mathsf{zv}^A_{\mathsf{dm}} & \text{for} \quad \mathsf{in}_{\mathsf{dm}}\ \Gamma \\ \mathsf{zv}^A_{\Delta} & \text{for} \quad \mathsf{in}_{\mathsf{sm}}\ \Gamma. \end{cases}$$

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$$\mathsf{pt}^A_{\Diamond_+} \equiv \mathsf{pt}^{\Diamond A}_{\mathsf{dm}} \qquad \mathsf{pt}^A_{\Box_+} \equiv \mathsf{pt}^{\Box A}_{\mathsf{dm}} \qquad \mathsf{pt}^A_{\triangle\Diamond_+} \equiv \mathsf{pt}^{\Diamond A}_{\triangle_+} \qquad \mathsf{pt}^A_{\triangle\Box_+} \equiv \mathsf{pt}^{\Box A}_{\triangle_+}$$
$$\mathsf{zv}^A_{\Diamond_+} \equiv \blacklozenge^A\, \mathsf{zv}^{\Diamond A}_{\mathsf{dm}} \qquad \mathsf{zv}^A_{\Box_+} \equiv \blacksquare^A\, \mathsf{zv}^{\Box A}_{\mathsf{dm}} \qquad \mathsf{zv}^A_{\triangle\Diamond_+} \equiv \blacklozenge^A\, \mathsf{zv}^{\Diamond A}_{\triangle_+} \qquad \mathsf{zv}^A_{\triangle\Box_+} \equiv \blacksquare^A\, \mathsf{zv}^{\Box A}_{\triangle_+}. \quad \triangleleft$$

4.3.7 *Modal* Π*-types*

Finally, modal Π-types must satisfy the following rules:

$$\frac{\mu : p \to q \qquad \gamma : \Gamma,\ \blacklozenge^+_\mu \vdash_{[\![p]\!]} A\ \gamma\ \mathsf{type}_{\ell_0} \qquad \gamma : \Gamma,\ a :^{\mu+} A\ \gamma \vdash_{[\![q]\!]} B\ \gamma\ a\ \mathsf{type}_{\ell_1}}{\gamma : \Gamma \vdash_{[\![q]\!]} \Pi^{\mathsf{sm}+}_\mu\ A\ B\ \gamma\ \mathsf{type}_{\ell_0 \sqcup \ell_1}}$$

$$\frac{\gamma : \Gamma,\ a :^{\mu+} A\ \gamma \vdash_{[\![q]\!]} t\ \gamma\ a : B\ \gamma\ a}{\gamma : \Gamma \vdash_{[\![q]\!]} \lambda^{\mathsf{sm}+}_\mu\ t\ \gamma : \Pi^{\mathsf{sm}+}_\mu\ A\ B\ \gamma} \qquad \frac{\gamma : \Gamma \vdash_{[\![q]\!]} f\ \gamma : \Pi^{\mathsf{sm}+}_\mu\ A\ B\ \gamma \qquad \gamma : \Gamma,\ \blacklozenge^+_\mu \vdash_{[\![p]\!]} s\ \gamma : A\ \gamma}{\gamma : \Gamma \vdash_{[\![q]\!]} \mathsf{app}^{\mathsf{sm}+}_\mu\ f\ s\ \gamma : B^{[\ 1\Gamma, s\ ]_\mu}\ \gamma}$$

For $\triangle_+$ we define:

$$\Pi^{\mathsf{sm}+}_\triangle\ A\ (\mathsf{in}_{\mathsf{sm}}\ B) \equiv \mathsf{in}_{\mathsf{sm}}\ \left(\Pi^{\mathsf{sm}}_\triangle\ A\ B\right) \qquad \Pi^{\mathsf{sm}+}_\triangle\ A\ (\mathsf{in}_{\mathsf{dm}}\ B) \equiv \mathsf{in}_{\mathsf{dm}}\ \left(\Pi^{\mathsf{dm}}\ A\ B\right)$$
$$\lambda^{\mathsf{sm}+}_\triangle\ (\mathsf{in}_{\mathsf{sm}}\ t) \equiv \mathsf{in}_{\mathsf{sm}}\ \left(\lambda^{\mathsf{sm}}_\triangle\ t\right) \qquad \lambda^{\mathsf{sm}+}_\triangle\ (\mathsf{in}_{\mathsf{dm}}\ t) \equiv \mathsf{in}_{\mathsf{dm}}\ \left(\lambda^{\mathsf{dm}}\ t\right)$$
$$\mathsf{app}^{\mathsf{sm}+}_\triangle\ \left(\mathsf{in}_{\mathsf{sm}}\ f\right)\ s \equiv \mathsf{in}_{\mathsf{sm}}\ \left(\mathsf{app}^{\mathsf{sm}}_\triangle\ f\ s\right) \qquad \mathsf{app}^{\mathsf{sm}+}_\triangle\ \left(\mathsf{in}_{\mathsf{dm}}\ f\right)\ s \equiv \mathsf{in}_{\mathsf{dm}}\ \left(\mathsf{app}^{\mathsf{dm}}\ f\ s\right).$$

The other cases reduce to functions of a modal variable:

$$\Pi^{\mathsf{sm}+}_\Diamond\ A\ B \equiv \Pi^{\mathsf{sm}+}\ (\Diamond\ A)\ B \qquad \Pi^{\mathsf{sm}+}_{\triangle\Diamond}\ A\ B \equiv \Pi^{\mathsf{sm}+}_\triangle\ (\Diamond\ A)\ B$$
$$\Pi^{\mathsf{sm}+}_\Box\ A\ B \equiv \Pi^{\mathsf{sm}+}\ (\Box\ A)\ B \qquad \Pi^{\mathsf{sm}+}_{\triangle\Box}\ A\ B \equiv \Pi^{\mathsf{sm}+}_\triangle\ (\Box\ A)\ B$$

The cases of λ and app are similar. $\triangleleft$

4.4 *Semantics of dTT*

Since our syntax was presented as a Generalised Algebraic Theory, there is an immediate notion of *model*, namely an algebra for that theory (with algebraic syntax being the initial model). We now reformulate this in more familiar category-theoretic terms.

4.4.1 *Modal structure*

The general multimodal type theory MTT was presented algebraically in Gratzer et al. (2021). For a general mode 2-category $\mathcal{M}$, the starting point is a *modal context structure*, which is a 2-functor $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$, where $\mathcal{M}^{\mathsf{coop}}$ denotes reversal of both 1-cells and 2-cells. The image of a mode p is the category $\mathcal{C}_p$ of contexts and substitutions at that mode, and the image of a morphism $\mu : p \to q$ is the lock functor $\blacklozenge_\mu : \mathcal{C}_q \to \mathcal{C}_p$, which we write postfix, $\Gamma \mapsto \Gamma.\blacklozenge_\mu$. It is a *modal natural model* if each $\mathcal{C}_p$ is equipped with a morphism $\mathsf{pr}_p : \mathsf{Tm}_p \to \mathsf{Ty}_p$ in its presheaf category such that for any $\mu : p \to q$ the morphism $(\blacklozenge_\mu)^*(\mathsf{pr}_p)$ is representable. (In particular, taking $\mu = 1_p$, we see that each $\mathcal{C}_p$ is an ordinary natural model, hence a CwF.) This notion encapsulates all the rules for building contexts and substitutions from section 2.2 except for those that refer to flatness of contexts.

Now, specialising to our mode 2-category from section 2.1, the rules of section 2.2 say that the flat contexts form a full subcategory of $\mathcal{C}_{\mathsf{sm}}$ that contains the image of $\blacklozenge_\Diamond$. In our algebraic theory we take as primitive the derived rules that a context is flat if and only if it admits a substitution to $((), \blacklozenge_\Diamond)$, and in that case the latter substitution is unique. Thus, semantically, $1.\blacklozenge_\Diamond$ is subterminal and the flat contexts are the slice category $\mathcal{C}_{\mathsf{sm}}/1.\blacklozenge_\Diamond$. Since the unit of the adjunction $\blacklozenge_\Diamond \dashv \blacklozenge_\triangle$ is an identity, $\blacklozenge_\Diamond$ is fully faithful, and on its image it has $\blacklozenge_\Diamond$ as an inverse and therefore also a left adjoint. In addition, since flat contexts are fixed points (up to isomorphism) of $\blacklozenge_{\triangle\Diamond}$, when Γ is

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flat there is a bijection $\mathsf{Ty}(\Gamma) \cong \mathsf{Ty}(\Gamma \,.\, \fatbslash_{\triangle\Diamond})$, and so the modal comprehension $(\Gamma, x :^{\triangle\Diamond} A)$ is isomorphic to an ordinary one $(\Gamma, x : A')$. This justifies the special variable rule and key substitution. Thus, the following definition encapsulates the judgmental structure of our modal type theory.

Definition 19. A **dTT context structure** is a modal context structure $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ in the sense of Gratzer et al. (2021), where $\mathcal{M}$ is as in section 2.1, such that $1 \,.\, \fatbslash_{\Diamond}$ is subterminal and the slice category $\mathcal{C}_{\mathsf{sm}}/1 \,.\, \fatbslash_{\Diamond}$ is the replete image of the fully faithful functor $\fatbslash_{\Diamond} : \mathcal{C}_{\mathsf{dm}} \to \mathcal{C}_{\mathsf{sm}}$. A **dTT natural model** is a dTT context structure that is also a modal natural model.

Next, since our modalities are Fitch-style, their semantics follows Gratzer et al. (2022). This requires each functor $\fatbslash_{\mu}$ to *be* a parametric right adjoint and *have* a dependent right adjoint. However, since each μ other than $\Diamond$ and $\triangle\Diamond$ is already a right adjoint in $\mathcal{M}$, such $\fatbslash_{\mu}$ are also ordinary (hence parametric) right adjoints. And since $\fatbslash_{\Diamond}$ is an equivalence onto a slice category, the inverse of that equivalence is a parametric left adjoint of it. Thus, to justify our Fitch-style rules for modalities, it suffices to assume the following (recall we omit $\triangle$).

Definition 20. A **dTT modal model** is a dTT natural model such that the functors $\fatbslash_{\Diamond}$ and $\fatbslash_{\Box}$ have dependent right adjoints.

Theorem 21. The simplicial model of Sections 4.2 and 4.3 is a dTT natural model.

Proof. Of course, we use the extended simplicial model along with the discrete model. We showed explicitly in section 4.3 that this yields a modal context structure for our $\mathcal{M}$. In the notation of that section, the object $1 \,.\, \fatbslash_{\Diamond}$ referred to above is $\left((), \fatbslash^{+}_{\Diamond}\right) \equiv \mathsf{in}_{\mathsf{dm}}\ ()$. The definition of the category sm_{+} implies immediately that this object is subterminal and its slice category is the replete image (and even the literal image) of $\fatbslash^{+}_{\Diamond}$. Finally, in section 4.3.5 we verified the rules of Gratzer et al. (2022) for $\Diamond$ and $\Box$, which as shown in *loc. cit.* are equivalent to their being dependent right adjoints of $\fatbslash^{+}_{\Diamond}$ and $\fatbslash^{+}_{\Box}$. □ ◁

4.4.2 Telescopes

Definition 22. A modal natural model $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ **has telescopes** if each natural model $\mathcal{C}_p$ has telescopes as in definition 9, and in addition for each nonidentity $\mu : p \to q$ there are morphisms of polynomial functors $P_{\mathsf{tpr}_q} \circ P_{\fatbslash_{\mu}{}^{*}\mathsf{pr}_p} \to P_{\mathsf{tpr}_q}$ (the identity case being part of $\mathcal{C}_p$ having telescopes), satisfying the modal versions of the rules from section 2.3. It has **Π-telescopes** if $\mathcal{C}_p$ does for each p.

As in Sections 4.1.3 and 4.1.4, we can equip any modal natural model with telescopes and Π-telescopes, and interpret meta-abstracted types and telescopes automatically. ◁

4.4.3 Display and décalage

If $\mathcal{C}$ is a CwF with telescopes and $\Gamma \in \mathcal{C}$, we write $\mathsf{Tel} \mathbin{/\!\!/} \Gamma$ for the category whose objects are telescopes $\Theta \in \mathsf{Tel}(\Gamma)$ and whose morphisms are morphisms $(\Gamma \mid \Theta_1) \to (\Gamma \mid \Theta_2)$ in $\mathcal{C}/\Gamma$. Thus, it is equivalent to the full subcategory of $\mathcal{C}/\Gamma$ on objects of the form $\Gamma \mid \Theta$. We call this *the category of telescopes of* Γ. By substitution, it is strictly functorial in Γ, i.e. we have a functor $\mathsf{Tel} \mathbin{/\!\!/} - : \mathcal{C}^{\mathsf{op}} \to \mathcal{C}at$. Thus can regard it as an internal category in $\mathsf{Set}^{\mathcal{C}^{\mathsf{op}}}$.

In fact, $\mathsf{Tel} \mathbin{/\!\!/} \Gamma$ is itself a CwF. Its *'types'* in *'context'* $\Upsilon \in \mathsf{Tel} \mathbin{/\!\!/} \Gamma$ are meta-abstracted telescopes $\Gamma \vdash \Phi\ \ \mathsf{tel}_{\ell}\big/_{\upsilon : \Upsilon}$. These are equivalent (but not equal) to telescopes in an extended context, i.e. the elements of $\mathsf{Tel}(\Gamma \mid \Upsilon)$. Similarly, meta-abstracted partial substitutions $\Gamma \vdash \phi : \Phi$ are equivalent

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to terms $\Gamma \mid (\upsilon : \Upsilon) \vdash \phi : \Phi\ \upsilon$, and semantically to sections of such a projection, $(\Gamma \mid \Upsilon) \to (\Gamma \mid \Upsilon \mid \Phi)$ over $(\Gamma \mid \Upsilon)$. Comprehension is by telescope concatenation. Because $\Gamma \mid () \equiv \Gamma$, the CwF $\mathsf{Tel} /\!\!/ \Gamma$ has the following special property.

Definition 23. A CwF $\mathcal{C}$ is **strongly democratic** if every context is the comprehension of a unique type in the empty context.

Since all of the structure of $\mathsf{Tel} /\!\!/ \Gamma$ is strictly stable under substitution in Γ, these categories form an internal CwF in $\mathsf{Set}^{\mathcal{C}^{\mathrm{op}}}$. We call this the *internal telescope model* of $\mathcal{C}$ and denote it **Tel**. Recalling remark 5 and comparing to the discussion of the local theory in Altenkirch et al. (2024), we find that décalage can be described as an internal CwF morphism.

Definition 24. Let $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ be a dTT natural model with telescopes. We say it **has décalage** if it is equipped with a strict morphism of internal strongly democratic CwFs:

$$(-)^{\mathsf{D}} : (🔒_{\triangle\square})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}_{\mathsf{sm}}$$

in the category of presheaves over $\mathcal{C}_{\mathsf{sm}}$, together with an internal natural transformation evens from $(-)^{\mathsf{D}}$ to $(-)[\, 🗝^{\triangle\square \le 1_{\mathsf{sm}}} \,] : (🔒_{\triangle\square})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}_{\mathsf{sm}}$.

In particular, this structure includes functors $(-)^{\mathsf{D}} : \mathsf{Tel} /\!\!/ (\Gamma \centerdot 🔒_{\triangle\square}) \to \mathsf{Tel} /\!\!/ \Gamma$ and natural transformations evens consisting of maps $(\Gamma \mid \Upsilon^{\mathsf{D}}) \to (\Gamma \centerdot 🔒_{\triangle\square} \mid \Upsilon)$, all strictly stable under pullback. This, together with the strict preservation of empty contexts by the functor $(-)^{\mathsf{D}}$, yields the rules of section 2.3. The corresponding action on *'types'* (meta-abstracted telescopes) and *'terms'* (meta-abstracted partial substitutions) gives the rules of section 2.5.

We can phrase display similarly with an auxiliary internal CwF, starting with telescope display as in section 2.5. Recall from Shulman (2015), Kapulkin and Lumsdaine (2021) that from any CwF $\mathcal{C}$ we can construct a *'Sierpinski'* model $\mathcal{C}^2$. Its objects (contexts) are arbitrary morphisms $\gamma_{01} : \Gamma_1 \to \Gamma_0$ in $\mathcal{C}$, but its types are pairs of $A_0 \in \mathsf{Ty}(\Gamma_0)$ and $A_1 \in \mathsf{Ty}(\Gamma_1 \centerdot A_0[\, \gamma_{01} \,])$. Moreover, there is a strict CwF morphism $(-)_0 : \mathcal{C}^2 \to \mathcal{C}$ which preserves all type-formers. If $\mathcal{C}$ has Σ-types there is a functor $(-)_1 : \mathcal{C}^2 \to \mathcal{C}$ that sends a type (A_0, A_1) to $\Sigma(A_0[\, \gamma_{01} \,], A_1) \in \mathsf{Ty}(\Gamma_1)$, but this only preserves comprehension and Σ-types up to isomorphism. Thus it is a pseudo CwF morphism as in (Clairambault and Dybjer, 2011, Definition 10).

Now let $\mathcal{C}$ be a CwF, and apply this construction internally in $\mathsf{Set}^{\mathcal{C}^{\mathrm{op}}}$ to the internal telescope model **Tel**. We thus obtain another CwF $\mathbf{Tel}^2$ internal to presheaves, in which the *'contexts'* over Γ are morphisms of telescopes $\theta_{01} : \Theta_1 \to \Theta_0$ over Γ, and the *'types'* in such a *'context'* over Γ are pairs of two telescopes $\Upsilon_0 \in \mathsf{Tel}(\Gamma \mid \Theta_0)$ and $\Upsilon_1 \in \mathsf{Tel}(\Gamma \mid \Theta_1 \mid \Upsilon_0[\, \theta_{01} \,])$. This is no longer strongly democratic, but as always we have the strict CwF morphism $(-)_0 : \mathbf{Tel}^2 \to \mathbf{Tel}$ that preserves Σ-types (i.e. telescope concatenation), and the pseudo CwF-morphism $(-)_1 : \mathbf{Tel}^2 \to \mathbf{Tel}$. Moreover, in this case the latter is actually a strict CwF-morphism, because telescope concatenation is strictly associative.

Definition 25. Let $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ be a dTT natural model with telescopes. We say $\mathcal{C}$ **has telescope display** if it is equipped with:

(1) An internal pseudo CwF morphism $(-)^{\mathsf{d}} : (🔒_{\triangle\square})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}^2_{\mathsf{sm}}$ that preserves substitution, the empty context, and Σ-types strictly.

(2) An equality between the composite $(🔒_{\triangle\square})^*\mathbf{Tel}_{\mathsf{sm}} \xrightarrow{(-)^{\mathsf{d}}} \mathbf{Tel}^2_{\mathsf{sm}} \xrightarrow{(-)_0} \mathbf{Tel}_{\mathsf{sm}}$ and the key transformation $(-)[\, 🗝^{\triangle\square \le 1_{\mathsf{sm}}} \,] : (🔒_{\triangle\square})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}_{\mathsf{sm}}$. Since the latter is a strict morphism, that means that so is the former.

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(3) A strict internal CwF morphism $(-)^{\mathsf{D}} : (\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}_{\mathsf{sm}}$, and an isomorphism of pseudo CwF morphisms between $(-)^{\mathsf{D}}$ and the composite morphism $(\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm}} \xrightarrow{(-)^{\mathsf{d}}} \mathbf{Tel}^2_{\mathsf{sm}} \xrightarrow{(-)_1} \mathbf{Tel}$, that is the identity on underlying functors.

We have not assumed *a priori* in definition 25 that $\mathcal{C}$ has décalage, but it is actually included: the morphism $(-)^{\mathsf{D}}$ is of course the same as in definition 24, and the transformation $\mathsf{evens} : \Theta^{\mathsf{D}} \to \Theta$ from definition 24 arises in definition 25 as the image of Θ under the underlying functor of $(-)^{\mathsf{d}}$. The additional data in definition 25 beyond this is the 1-part of the action of $(-)^{\mathsf{d}}$ (the 0-part is determined by the composition with $(-)_0$ equaling $(-)[\text{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}}]$) making it a pseudo CwF morphism preserving substitution and Σ-types strictly, and the isomorphism on *'types'* (dependent telescopes) between $(-)^{\mathsf{D}}$ and the composite of $(-)^{\mathsf{d}}$ with $(-)_1$. But since the latter is to be a pseudo CwF transformation (see Castellan et al. 2017, Appendix B), and since $(-)^{\mathsf{D}}$ is strict and the underlying functor is the identity this just means that this isomorphism must coincide with the 1-part of the comprehension coherence isomorphism of $(-)^{\mathsf{d}}$ (the 0-part being the identity).

So all that remains is the 1-part of the action of $(-)^{\mathsf{d}}$ on meta-abstracted telescopes, preserving substitution and telescope concatenation, and coherence isomorphisms relating it to comprehension. Since the 1-part of the comprehension of $(\Upsilon, \Upsilon^{\mathsf{d}})$ in $\mathbf{Tel}^2$ is $(\Upsilon \mid \Upsilon^{\mathsf{d}})$, the comprehension isomorphisms are the pairing $\langle -, - \rangle$ together with ^{ev} and ^{od}.

Finally, we consider display of types. In some ways this is simpler, since we don't have to worry about rearranging between display and décalage; but in other ways it is more complicated, since we have to take account of extending dependent telescopes by types.

To start with, note that the internal telescope model $\mathbf{Tel}$ of any CwF has a *'sub-model'* $\mathbf{Tel}_1$ whose internal category of *'contexts'* is the same (telescopes), but whose internal presheaf of *'types'* consists of the *length-1 telescopes*, i.e. single types annotated by a modality. Unlike $\mathbf{Tel}$, it does not automatically have Σ-types.

Definition 26. Let $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ be a dTT natural model with telescopes. We say $\mathcal{C}$ **has type display** if it is equipped with:

(1) An internal strict CwF morphism $(-)^{\mathsf{d}} : (\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm},1} \to \mathbf{Tel}^2_{\mathsf{sm}}$.
(2) An equality between the composite $(\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm}} \xrightarrow{(-)^{\mathsf{d}}} \mathbf{Tel}^2_{\mathsf{sm}} \xrightarrow{(-)_0} \mathbf{Tel}_{\mathsf{sm}}$ and the key transformation $(-)[\text{🔑}^{\triangle\Box \le 1_{\mathsf{sm}}}] : (\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}_{\mathsf{sm}}$.
(3) If a length-1 telescope A is non-modal, then A^{d} is a single non-modal type.
(4) If a length-1 telescope A is nontrivially modal, then A^{d} is empty.

Note that, like definition 25, this definition includes décalage. Of course, when we have both telescope display and type display we want them to be compatible.

Definition 27. Let $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ be a dTT natural model with telescope display. We say it has **complete display** if the restriction of $(-)^{\mathsf{d}} : (\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm}} \to \mathbf{Tel}^2_{\mathsf{sm}}$ to $(\text{🔒}_{\triangle\Box})^*\mathbf{Tel}_{\mathsf{sm},1}$ is an internal strict CwF morphism such that items 3 and 4 of definition 26 hold, as do the rules from sections 2.4 and 2.5 for computing meta-abstracted décalage and display in terms of type display.

Finally, we add the compatibility conditions with type-formers:

Definition 28. Let $\mathcal{C} : \mathcal{M}^{\mathsf{coop}} \to \mathcal{C}at$ be a dTT natural model with telescopes and complete display. We say that **display respects Π-types** (respectively **universes**) if the relevant rules in Figure 8 hold.

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This completes the description of the abstract categorical semantics of the theory of section 2: it is a dTT natural model with telescopes and complete display that respects Π-types and universes. However, as noted in section 2, when telescopes are lists of types, as they almost always are, much of this structure can be deduced from the rest.

Theorem 29. Let $\mathcal{C}$ be a dTT natural model, with telescopes defined from types as in theorem 10, and with type display defined relative to these telescopes. Then there is a unique way to extend this type display on $\mathcal{C}$ to complete display.

Proof. The rules for computing telescope display and décalage in terms of type display uniquely determine those operations when telescopes are defined as lists of types. □

Theorem 30. The simplicial model of Sections 4.2 and 4.3 has type display, and hence complete display, which respects Π-types and universes.

Proof. The display operation that we constructed for the simplicial model in section 4.2, is a *'global'* operation that décalages the whole context:

$$\frac{\gamma : \Gamma \vdash_{\mathsf{sm}} A\ \gamma\ \mathsf{type}}{\gamma^+ : \Gamma^{\mathsf{D}},\ a : A^{\rho\Gamma}\ \gamma^+ \vdash_{\mathsf{sm}} A^{\mathsf{d}}\ \gamma^+\ a\ \mathsf{type}}.$$

(Note that this is only defined on the original simplicial model sm, not the extended one sm_+: indeed, décalage is not even defined on flat contexts.) But the (meta-abstracted version of the) operation we specified in the syntax of section 2 is a *'local'* one that only décalages part of the context, keeping the rest of it modally locked away:

$$\frac{\gamma : \Gamma,\ \fatbslash_{\triangle\Box},\ \upsilon : \Upsilon \vdash_{\mathsf{sm}} A\ \gamma\ \upsilon\ \mathsf{type}}{\gamma : \Gamma,\ \upsilon^+ : \Upsilon^{\mathsf{D}},\ a : A^{\fatbslash^{\triangle\Box \le 1_{\mathsf{sm}}}\ \gamma\ (\upsilon^+)^{\mathsf{ev}}} \vdash_{\mathsf{sm}} A^{\mathsf{d}}\ \gamma\ \upsilon^+\ a\ \mathsf{type}}$$

However, it is straightforward to obtain the latter from the former. In sm_+, a context of the form $(\gamma : \Gamma,\ \fatbslash_{\triangle\Box},\ \upsilon : \Upsilon)$ is not flat, hence lies essentially in sm so that décalage is defined on it. Furthermore, we already observed that $(\Gamma,\ \fatbslash_{\Box})^{\mathsf{D}} \equiv (\Gamma,\ \fatbslash_{\Box})$ since $\fatbslash_{\Box}$ lands in constant presheaves. Thus, when Υ is a telescope built out of types, we have:

$$(\gamma : \Gamma,\ \fatbslash_{\triangle\Box},\ \upsilon : \Upsilon)^{\mathsf{D}} \equiv \left(\gamma : \Gamma,\ \fatbslash_{\triangle\Box},\ \upsilon^+ : \Upsilon^{\mathsf{D}}\right)$$

and so the global operation yields as a special case:

$$\frac{\gamma : \Gamma,\ \fatbslash_{\triangle\Box},\ \upsilon : \Upsilon \vdash_{\mathsf{sm}_+} A\ \gamma\ \upsilon\ \mathsf{type}}{\gamma : \Gamma,\ \fatbslash_{\triangle\Box},\ \upsilon^+ : \Upsilon^{\mathsf{D}},\ a : A^{\rho}\ \gamma\ \upsilon^+ \vdash_{\mathsf{sm}_+} A^{\mathsf{d}}\ \gamma\ \upsilon^+\ a\ \mathsf{type}}.$$

Now we simply substitute along $\fatbslash^{\triangle\Box \le 1_{\mathsf{sm}}}$ to obtain the desired local rule. The necessary computation rules for décalage, Π-types, and universes follow immediately. □ ◁

4.4.4 *Display of ω-limits*

Finally, when we have both display and also ω-limits, it is reasonable to require the former to compute on the latter, in the following way. Suppose that $\Gamma,\ \fatbslash_{\triangle\Box} \mid \phi : \Phi \vdash_{\mathsf{sm}} \widetilde{\Upsilon}\ \phi\ \mathsf{stel}^{\infty}$, and we want to compute $\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \mathsf{lim}\ \widetilde{\Upsilon} \vdash \mathsf{lim}\ \widetilde{\Upsilon}^{\mathsf{d}}\ \phi\ \mathfrak{u}$. Then by definition, we have:

$$\Gamma,\ \fatbslash_{\triangle\Box} \vdash_{\mathsf{sm}} \Upsilon^{\partial n}\ \mathsf{tel}\big/_{\phi : \Phi} \qquad\qquad \Gamma,\ \fatbslash_{\triangle\Box} \vdash_{\mathsf{sm}} \Upsilon^{n}\ \mathsf{type}\big/_{\phi : \Phi,\ \partial\upsilon : \Upsilon^{\partial n}\ \phi}$$

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and therefore:

$$\Gamma \vdash_{\mathsf{sm}} (\Upsilon^{\partial n})^{\mathsf{d}}\ \ \mathsf{tel}_\ell \Big/_{\phi : \Phi^{\mathsf{D}},\, \partial\upsilon : \Upsilon^{\partial n}\, \phi^{\mathsf{ev}}}$$

$$\Gamma \vdash_{\mathsf{sm}} (\Upsilon^{n})^{\mathsf{d}}\ \ \mathsf{type}_\ell \Big/_{\phi : \Phi^{\mathsf{D}},\, \partial\upsilon : (\Upsilon^{\partial n})^{\mathsf{D}}\, \phi,\, \upsilon : \Upsilon^{n}\, \phi^{\mathsf{ev}}\, \partial\upsilon^{\mathsf{ev}}}$$

Weakening and substituting to the needed context $\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon}$, we have:

$$\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon} \vdash_{\mathsf{sm}} (\Upsilon^{\partial n})^{\mathsf{d}}\ \phi\ (\mathsf{res}^{\partial n}\ \mathfrak{u})\ \mathsf{tel}$$

$$\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon},\ \partial\upsilon : (\Upsilon^{\partial n})^{\mathsf{d}}\ \phi\ (\mathsf{res}^{\partial n}\ \mathfrak{u}) \vdash_{\mathsf{sm}} (\Upsilon^{n})^{\mathsf{d}}\ \phi\ \langle\, \mathsf{res}^{\partial n}\ \mathfrak{u}\,,\, \partial\upsilon\, \rangle\ (\mathsf{res}^{n}\ \mathfrak{u})\ \mathsf{type}$$

such that:

$$\begin{aligned}(\Upsilon^{\partial(n+1)})^{\mathsf{d}}\ \phi\ (\mathsf{res}^{\partial(n+1)}\ \mathfrak{u}) &\equiv \left(\partial\upsilon : \widetilde{\Upsilon}^{\partial n}, \upsilon : \widetilde{\Upsilon}^{n}\ \partial\upsilon\right)^{\mathsf{d}}\ \phi\ [\ \mathsf{res}^{\partial n}\ \mathfrak{u},\ \mathsf{res}^{n}\ \mathfrak{u}\ ] \\ &\equiv \left(\partial\upsilon : (\widetilde{\Upsilon}^{\partial n})^{\mathsf{d}}\ \phi\ (\mathsf{res}^{\partial n}\ \mathfrak{u}),\ \upsilon : (\widetilde{\Upsilon}^{n})^{\mathsf{d}}\ \phi\ \langle\, \mathsf{res}^{\partial n}\ \mathfrak{u}\,,\, \partial\upsilon\, \rangle\ (\mathsf{res}^{n}\ \mathfrak{u})\right).\end{aligned}$$

Thus, these data form another infinite telescope, which we denote:

$$\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon} \vdash_{\mathsf{sm}} \widetilde{\Upsilon}^{\mathsf{d}}\ \phi\ \mathfrak{u}\ \mathsf{stel}^{\infty}$$

$$(\Upsilon^{\mathsf{d}})^{\partial n}\ \phi\ \mathfrak{u} \equiv (\Upsilon^{\partial n})^{\mathsf{d}}\ \phi\ (\mathsf{res}^{\partial n}\ \mathfrak{u})$$

$$(\Upsilon^{\mathsf{d}})^{n}\ \phi\ \mathfrak{u}\ \partial\upsilon \equiv (\Upsilon^{n})^{\mathsf{d}}\ \phi\ \langle\, \mathsf{res}^{\partial n}\ \mathfrak{u}\,,\, \partial\upsilon\, \rangle\ (\mathsf{res}^{n}\ \mathfrak{u})$$

Definition 31. In a CwF with telescopes, type display, and ω-limits, we say that **display respects ω-limits** if:

$$\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon} \vdash_{\mathsf{sm}} \lim \widetilde{\Upsilon}^{\mathsf{d}}\ \phi\ \mathfrak{u} \equiv \lim(\widetilde{\Upsilon}^{\mathsf{d}}\ \phi\ \mathfrak{u})$$

$$\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon} \vdash_{\mathsf{sm}} (\mathsf{res}^{\partial n})^{\mathsf{d}}\ \phi\ \mathfrak{u} \equiv \mathsf{res}^{\partial n}\ \phi\ \mathfrak{u}$$

$$\Gamma,\ \phi : \Phi^{\mathsf{D}},\ \mathfrak{u} : \lim \widetilde{\Upsilon} \vdash_{\mathsf{sm}} (\mathsf{res}^{n})^{\mathsf{d}}\ \phi\ \mathfrak{u} \equiv \mathsf{res}^{n}\ \phi\ \mathfrak{u}$$

where in the last two equations, the left-hand side is a restriction relative to $\widetilde{\Upsilon}$, and on the right-hand side it is relative to $\widetilde{\Upsilon}^{\mathsf{d}}$.

Theorem 32. Display respects ω-limits in the simplicial model.

Proof. This holds essentially by construction of ω-limits therein, plus passing across the translation between different forms of display from theorem 30. □ ◁

4.5 Semantics of semi-simplicial types

Finally, we construct semantics for the displayed coinductive types of section 3.3, including SST. As with most coinductive definitions, they are terminal coalgebras – in this case, for a *copointed* endofunctor – and can be constructed by a sequential limit.

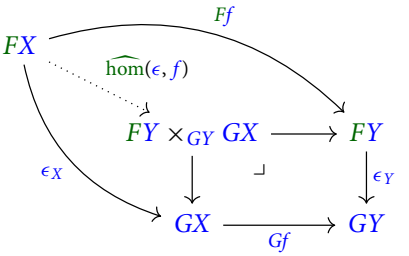
4.5.1 Terminal coalgebras for copointed endofunctors

Definition 33. A **copointed endofunctor** of a category $\mathcal{C}$ is a functor $F : \mathcal{C} \to \mathcal{C}$ together with a natural transformation $\epsilon : F \to 1_{\mathcal{C}}$. A **coalgebra** for a copointed endofunctor is an object X with a morphism $x : X \to FX$ such that the composite $X \xrightarrow{x} FX \xrightarrow{\epsilon_X} X$ is the identity. A **terminal coalgebra** is a terminal object of the category of coalgebras.

To construct copointed terminal coalgebras, we cannot simply consider the limit of the tower $\cdots \to F^3\mathbb{1} \to F^2\mathbb{1} \to F\mathbb{1} \to \mathbb{1}$ as in the unpointed case; we have to incorporate ϵ.

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Definition 34. (c.f. [Shulman 2019, Definition 8.6]). Given a natural transformation $\epsilon : F \to G$ and a morphism $f : X \to Y$ in the domain of F and G, we write $\widehat{\hom}(\epsilon, \ )f$ for the gap map in the following pullback, assuming that the pullback exists:



If the domain and codomain of F and G have a notion of *'fibration'*, we say that ϵ is a **Quillen pre-fibration** if whenever f is a fibration, so is $\widehat{\hom}(\epsilon, f)$.

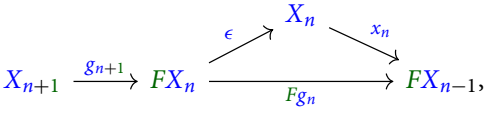
Lemma 35. In a category of telescopes, consider the fibrations to be the morphisms isomorphic to a dependent projection of some telescope. Then the transformation $\mathsf{evens} : (-)^{\mathsf{D}} \to (-)[\, 🔑^{\triangle\square \le 1_{\mathsf{sm}}} \,]$ from definition 24 is a Quillen pre-fibration.

Proof. Given a dependent projection $(\Theta \mid \Upsilon) \to \Theta$ in $\mathsf{Tel} \,/\!\!/\, (\Gamma \,.\, 🔒_{\triangle\square})$, the gap map is isomorphic to the dependent projection $(\Theta^{\mathsf{D}} \mid \Upsilon[\, 🔑^{\triangle\square \le 1} \,] \mid \Upsilon^{\mathsf{d}}) \to (\Theta^{\mathsf{D}} \mid \Upsilon[\, 🔑^{\triangle\square \le 1} \,])$ of Υ^{d}. □

Theorem 36. Suppose $\mathcal{C}$ is a category with a terminal object and a notion of fibration that is stable under pullback, and that F is a copointed endofunctor of $\mathcal{C}$ such that $\epsilon : F \to 1_{\mathcal{C}}$ is a Quillen pre-fibration. Suppose also that $\mathcal{C}$ has limits of inverse ω-sequences of fibrations, and that F preserves these limits. Then there is a terminal F-coalgebra.

Proof. We will construct a sequence of objects X_n, with fibrations $g_{n+1} : X_{n+1} \to X_n$ and morphisms $x_{n+1} : X_{n+1} \to FX_n$, such that $\epsilon_{X_n} \circ x_{n+1} = g_{n+1}$ and $Fg_n \circ x_{n+1} = x_n \circ g_{n+1}$. Let $X_0 = \mathbb{1}$, the terminal object, and let $X_1 = FX_0 = F\mathbb{1}$, with x_1 the identity.

Now, assume the data constructed up to level $n > 0$; we want X_{n+1} to be the *universal* object equipped with g_{n+1} and x_{n+1} satisfying the desired equations. The usual approach (e.g. the dual of Kelly 1980) is to take the equaliser:



but this does not make it evident that g_{n+1} is a fibration. Instead, we take the pullback:

$$\begin{array}{ccc} X_{n+1} & \xrightarrow{x_{n+1}} & FX_n \\ {\scriptstyle g_{n+1}}\downarrow & \lrcorner & \downarrow{\scriptstyle \widehat{\hom}(\epsilon,\, g_n)} \\ X_n & \xrightarrow[(1,x_n)]{} & X_n \times_{X_{n-1}} FX_{n-1}. \end{array} \tag{37}$$

The commutativity of this square says that $Fg_n \circ x_{n+1} = x_n \circ g_{n+1}$ and $\epsilon_{X_n} \circ x_{n+1} = g_{n+1}$. And by assumption, $\widehat{\hom}(\epsilon,\ g_n)$ is a fibration, hence so is its pullback g_{n+1}.

Now let X_∞ be the limit of the ω-sequence of fibrations:

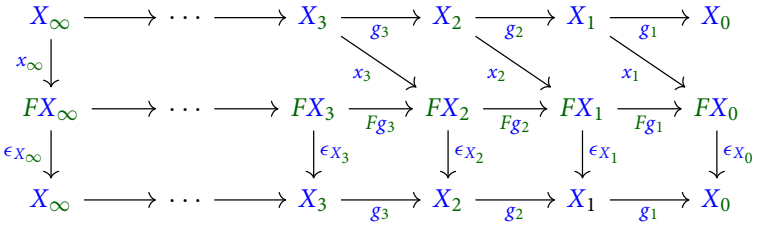
$$X_\infty \cdots \xrightarrow{g_{n+1}} X_n \xrightarrow{g_n} \cdots \xrightarrow{g_2} X_1 \xrightarrow{g_1} X_0 = \mathbb{1}.$$

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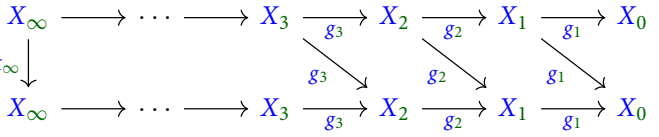
Since F preserves ω-limits, FX_∞ is the limit of the corresponding sequence:

$$FX_\infty \cdots \xrightarrow{Fg_{n+1}} FX_n \xrightarrow{Fg_n} \cdots \xrightarrow{Fg_2} FX_1 \xrightarrow{Fg_1} FX_0 = \mathbb{1}.$$

The morphisms x_n and ϵ_{X_n} form fence diagrams:

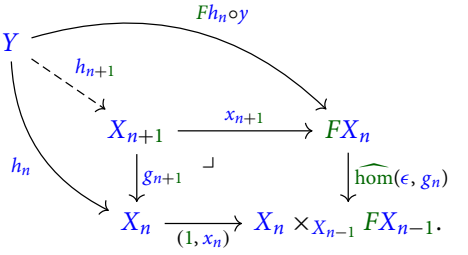


composed of the parallelograms $Fg_n \circ x_{n+1} = x_n \circ g_{n+1}$ from our construction, and naturality squares $\epsilon_{X_n} \circ Fg_{n+1} = g_{n+1} \circ \epsilon_{X_{n+1}}$. The former induces a map of limits $x_\infty : X_\infty \to FX_\infty$, while by naturality the latter induces ϵ_{X_∞}. The universal property of limits implies that $\epsilon_{X_\infty} \circ x_\infty$ is induced by the composite fence, and since $g_{n+1} = \epsilon_{X_n} \circ x_{n+1}$ this is:



which induces the identity 1_{X_∞}. Thus, X_∞ is an F-coalgebra.

Now suppose $y : Y \to FY$ is another F-coalgebra. We construct inductively maps $h_n : Y \to X_n$ such that $x_{n+1} \circ h_{n+1} = Fh_n \circ y$ and $g_{n+1} \circ h_{n+1} = h_n$. We start with $h_0 : Y \to X_0 = \mathbb{1}$ the unique morphism, and $h_1 : Y \to X_1 = FX_0$ the composite $Y \xrightarrow{y} FY \xrightarrow{Fh_0} FX_0$. Then we induce h_{n+1} by the universal property of the pullback defining X_{n+1}:



This is valid because using the inductive assumptions about h_n and h_{n-1}, we have:

$$\epsilon_{X_n} \circ Fh_n \circ y = h_n \circ \epsilon_Y \circ y = h_n \qquad Fg_n \circ Fh_n \circ y = F\left(g_n \circ h_n\right) \circ y = Fh_{n-1} \circ y = x_n \circ h_n,$$

and the two triangles relating to h_{n+1} show that it has the necessary properties.

Now, the equations $g_{n+1} \circ h_{n+1} = h_n$ imply there is an induced map $h_\infty : Y \to X_\infty$, such that $x_\infty \circ h_\infty$ is induced by the composites $x_{n+1} \circ h_{n+1}$. But $x_{n+1} \circ h_{n+1} = Fh_n \circ y$, and the morphisms Fh_n induce the limit map Fh_∞. Thus, h_∞ is an F-coalgebra morphism.

Finally, suppose $k : Y \to X_\infty$ is any F-coalgebra morphism, so we have $x_\infty \circ k = Fk \circ y$. Then k is uniquely determined by the maps $k_n : Y \to X_n$, and we have $x_{n+1} \circ k_{n+1} = Fk_n \circ y$. But this equation implies by induction that $k_n = h_n$ for all n, hence $k = h_\infty$. □ ◁

4.5.2 Displayed coinductive types

Let $\mathcal{C}$ be a dTT natural model with all the structure from section 4.4. We will apply theorem 36 in Tel $/\!\!/ (\Gamma \,.\, \blacksquare_{\triangle\square} \mid \Phi)$, in which the fibrations are the morphisms isomorphic to the dependent

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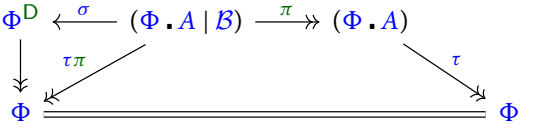
projection from some telescope. In the presence of levels, the objects of $\mathsf{Tel} /\!\!/ (\Gamma \,.\, \triangle\!\!\square \mid \Phi)$ are telescopes at *any* level.

Suppose given the input data for a displayed coinductive type, consisting of:

$$\Gamma \in \mathcal{C} \qquad \Phi \in \mathsf{Tel}_{\ell_0}(\Gamma \,.\, \triangle\!\!\square) \qquad A \in \mathsf{Ty}_{\ell_1}(\Gamma \,.\, \triangle\!\!\square \mid \Phi) \qquad \mathcal{B} \in \mathsf{Tel}_{\ell_2}(\Gamma \,.\, \triangle\!\!\square \mid \Phi \,.\, A)$$

$$\sigma \in \mathsf{PSub}_{\ell_0}\left((\Gamma \,.\, \triangle\!\!\square \mid \Phi \,.\, A \mid \mathcal{B}),\ \Phi^{\mathsf{d}}\right).$$

Categorically, this yields a *'display polynomial'*. Omitting the base context $\Gamma \,.\, \triangle\!\!\square$ for conciseness and indicating fibrations with $\twoheadrightarrow$, this looks like:



Here the left vertical map is a fibration because it is isomorphic to the dependent projection: $(\Gamma \,.\, \triangle\!\!\square \mid \Phi \mid \Phi^{\mathsf{d}}) \to (\Gamma \,.\, \triangle\!\!\square \mid \Phi)$. We then have a copointed endofunctor of $\mathsf{Tel} /\!\!/ (\Gamma \,.\, \triangle\!\!\square \mid \Phi)$:

$$\frac{\Gamma,\ \triangle\!\!\square \vdash_{\mathsf{sm}} X\ \ \mathsf{tel}_{\ell}\big/_{\phi:\Phi}}{\Gamma,\ \triangle\!\!\square \vdash_{\mathsf{sm}} FX\ \ \mathsf{tel}_{\ell \sqcup \ell_1 \sqcup \ell_2}\big/_{\phi\,:\,\Phi,\ x\,:\,X\,\phi}}$$

$$FX \equiv \Big(\!\!\Big(\ a : A\ \phi,\ x' : (b : \mathcal{B}\ \phi\ a) \to X^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ x\ \Big)\!\!\Big)_{\phi\,:\,\Phi,\ x\,:\,X\,\phi}$$

Here the $\to$ denotes a Π-telescope and X^{d} denotes meta-abstracted telescope display. Note that FX is meta-abstracted over Φ extended by X, so it lies in $\mathsf{Tel} /\!\!/ (\Gamma \,.\, \triangle\!\!\square \mid \Phi \mid X)$ rather than $\mathsf{Tel} /\!\!/ (\Gamma \,.\, \triangle\!\!\square \mid \Phi)$. The actual *endo*functor of $\mathsf{Tel} /\!\!/ (\Gamma \,.\, \triangle\!\!\square \mid \Phi)$ is thus

$$\widetilde{F}(X) \equiv (X \mid FX).$$

Lemma 38. The projection $(X \mid FX) \to X$ defines a Quillen pre-fibration $\epsilon : \widetilde{F} \to 1$.

Proof. Suppose given a fibration over $X \in \mathsf{Tel}_{\ell_0}(\Gamma,\ \triangle\!\!\square)$, meaning a dependent telescope:

$$\Gamma,\ \triangle\!\!\square \vdash_{\mathsf{sm}} Y\ \ \mathsf{tel}_{\ell_1}\big/_{\phi\,:\,\Phi,\ x\,:\,X\,\phi}\ .$$

Then we have:

$$\Gamma,\ \triangle\!\!\square \vdash_{\mathsf{sm}} \Big(\!\!\Big(\ x\ :\ X\ \phi \mid y\ :\ Y\ \phi\ x\ \Big)\!\!\Big)_{\phi\,:\,\Phi}\ \ \mathsf{tel}_{\ell_0 \sqcup \ell_1}\big/_{\phi\,:\,\Phi}$$

which we write as XY for conciseness. Then by definition, we have:

$$F(XY) \equiv \Big(\!\!\Big(\ a : A\ \phi,\ z' : (b : \mathcal{B}\ \phi\ a) \to (XY)^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ [\ x \mid y\ ]\ \Big)\!\!\Big)_{\phi\,:\,\Phi,\ x\,:\,X\,\phi,\ y\,:\,Y\,\phi\,x}.$$

To simplify this, note that the computation rules for telescope display give

$$(XY)^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ [\ x \mid y\ ] \equiv \Big(x' : X^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ x \mid Y^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ \langle\ x\ ,x'\ \rangle\ y\Big),$$

and therefore the rules for computing Π-telescopes give

$$\begin{aligned}(b : \mathcal{B}\ \phi\ a) \to (XY)^{\mathsf{d}}\ &\langle\ \phi\ ,\sigma\ a\ b\ \rangle\ [\ x \mid y\ ] \\ &\equiv \Big(\delta : (b : \mathcal{B}\ \phi\ a) \to X^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ x\ \mid\ \epsilon : (b : \mathcal{B}\ \phi\ a) \to Y^{\mathsf{d}}\ \langle\ \phi\ ,\sigma\ a\ b\ \rangle\ \langle\ x\ ,\delta\ b\ \rangle\ y\Big).\end{aligned}$$

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Now when δ is paired with $a : A\ \phi$, it yields FX. Thus, the gap map $XY \mid F(XY) \to XY \mid F(X)$ is the dependent projection from the telescope:

$$\left(\!\!\left((b : \mathcal{B}\ \phi\ a) \to Y^{\mathsf{d}}\ \langle\, \phi\,, \sigma\ a\ b \,\rangle\ \langle\, x\,, \delta\ b \,\rangle\ y \right)\!\!\right)_{\phi : \Phi,\, x : X\ \phi,\, y : Y\ \phi\ x,\, a : A\ \phi,\ \delta : (b : \mathcal{B}\ \phi\ a) \to X^{\mathsf{d}}\ \langle\, \phi\,, \sigma\ a\ b \,\rangle\ x} \tag{39}$$

and thus a fibration. □

Lemma 40. The endofunctor $\widetilde{F}$ preserves inverse limits of ω-sequences of fibrations.

Sketch of proof. This follows from the η-rules for inverse limits, together with the fact that display also preserves inverse limits. □

Therefore, by theorem 36, there is a terminal $\widetilde{F}$-coalgebra, which is a type (since ω-limits are types). This is our candidate for the displayed coinductive type. Its construction produces a tower of fibrations g_n, i.e. a sequence of finite telescopes:

$$\phi : \Phi \vdash_{\mathsf{sm}} X^{\partial n}\ \phi\ \mathsf{tel}_\ell \qquad \phi : \Phi \vdash_{\mathsf{sm}} X^{\partial 0}\ \phi \equiv ()$$

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi \vdash_{\mathsf{sm}} X^{n}\ \phi\ \partial x\ \mathsf{tel}_\ell \qquad \phi : \Phi \vdash_{\mathsf{sm}} X^{\partial(n+1)}\ \phi \equiv (\partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x)$$

We choose the level of $X^{\partial 0}$ to be $\ell \equiv \ell_0 \sqcup \ell_1$; then the other $X^{\partial n}$ and X^n are also at level ℓ.

The object X_{n+1} in section 4.5.1 corresponds to the telescope:

$$(\phi : \Phi,\ \partial x : X^{\partial(n+1)}\ \phi) = (\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x).$$

Each morphism $x_{n+1} : X_{n+1} \to \widetilde{F}X_n$ such that $\epsilon \circ x_{n+1} = g_{n+1}$ then corresponds to a term:

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x \vdash_{\mathsf{sm}} \xi_n : F(X^{\partial n})\ \phi\ \partial x.$$

By definition of F, ξ_n is equivalent to two terms:

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x \vdash_{\mathsf{sm}} h_n\ \phi\ \partial x\ x : A\ \phi$$

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x,\ b : \mathcal{B}\ \phi\ (h_n\ \phi\ \partial x\ x) \vdash_{\mathsf{sm}} t_n\ \phi\ \partial x\ x\ b : (X^{\partial n})^{\mathsf{d}}\ \langle\, \phi\,, \sigma\ (h_n\ \phi\ \partial x\ x)\ b \,\rangle\ \partial x.$$

The equation $Fg_{n+1} \circ x_{n+2} = x_{n+1} \circ g_{n+2}$ means that:

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x,\ x' : X^{n+1}\ \phi\ [\ \partial x, x\ ] \vdash_{\mathsf{sm}} h_{n+1}\ \phi\ [\ \partial x, x\ ]\ x' \equiv h_n\ \phi\ \partial x\ x$$

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x,\ x' : X^{n+1}\ \phi\ [\ \partial x, x\ ],\ b : \mathcal{B}\ \phi\ (h_n\ \phi\ \partial x\ x) \vdash_{\mathsf{sm}} t_{n+1}\ \phi\ [\ \partial x, x\ ]\ x'\ b \equiv [\ t_n\ \phi\ \partial x\ x\ b,\ s_n\ \phi\ \partial x\ x\ x'\ b\ ]$$

for some term:

$$\phi : \Phi,\ \partial x : X^{\partial n}\ \phi,\ x : X^n\ \phi\ \partial x,\ x' : X^{n+1}\ \phi\ [\ \partial x, x\ ],\ b : \mathcal{B}\ \phi\ (h_n\ \phi\ \partial x\ x) \vdash_{\mathsf{sm}} s_n\ \phi\ \partial x\ x\ x'\ b : (X^n)^{\mathsf{d}}\ \langle\, \phi\,, \sigma\ (h_n\ \phi\ \partial x\ x)\ b \,\rangle\ \langle\, \partial x\,, t_n\ \phi\ \partial x\ x\ b \,\rangle\ x \equiv (X^n)^{\mathsf{d}}\ \langle\, \phi\,, \sigma\ (h_{n-1}\ \phi\ \partial x)\ b \,\rangle\ \langle\, \partial x\,, t_n\ \phi\ \partial x\ x\ b \,\rangle\ x$$

Inspecting the actual construction, we start with $X_0 = X^{\partial 0} = ()$ and $X^0 = F(X^{\partial 0}) = F() = A$. By induction, the functions h_n then all just project to X^0. For the rest, combining eqs. (37) and (39), we find that X^{n+1} is defined by:

$$X^{n+1}\ \phi\ [\ \partial x,\ x\ ] \equiv (b : \mathcal{B}\ \phi\ (h_{n-1}\ \phi\ \partial x)) \to (X^n)^{\mathsf{d}}\ \langle\, \phi\,, \sigma\ (h_{n-1}\ \phi\ \partial x)\ b \,\rangle\ \langle\, \partial x\,, t_n\ \phi\ \partial x\ x\ b \,\rangle\ x$$

and we have $s_n\ \phi\ \partial x\ x\ x'\ b \equiv x'\ b$. In particular, each X^n is just a single type. Therefore, $(X^{\partial n}, X^n)$ is an infinite telescope and our displayed coinductive type is its limit:

$$\widetilde{X} = (X^{\partial n}, X^n) \qquad \mathsf{dCoind}\ [\ \Phi, A, \mathcal{B}, \sigma\ ]\ \phi \equiv \lim\left(\widetilde{X}\ \phi\right).$$

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Example 41. Recall that for semi-simplicial types SST, we have $\Phi \equiv ()$, with $A \equiv \mathsf{Type}$ and $\mathcal{B}\ a \equiv (x : \mathsf{El}\ a)$. Therefore, in this case we have:

$$\begin{aligned}
X^0 &\equiv \mathsf{Type} \\
X^0\ A_0 &\equiv \mathsf{El}\ A_0 \to \mathsf{Type}^{\mathsf{d}}\ A_0 \\
&\equiv \mathsf{El}\ A_0 \to \mathsf{El}\ A_0 \to \mathsf{Type} \\
X^2\ A_0\ A_1 &\equiv (a_1 : A_0) \to (\!|\ A \to A \to \mathsf{Type}\ |\!)_{A\,:\,\mathsf{Type}^{\mathsf{d}}\ A_0\ (A_1\ a_1)}\ A_1 \\
&\equiv (a_1 : A_0) \to (\!|\ (x : A)(x' : A'\ x)(y : A)(y' : A'\ y) \to \mathsf{Type}^{\mathsf{d}}\ (A''\ x\ y)\ |\!)_{A\,:\,\mathsf{Type},\,A'\,:\,\mathsf{Type}^{\mathsf{d}}\ A,\,A''\,:\,A \to A \to \mathsf{Type}}\ A_0\ (A_1\ a_1)\ A_1 \\
&\equiv (a_{001} : A_0)(a_{010} : A_0)(a_{011} : A_1\ a_{001}\ a_{010}) \\
&\qquad (a_{100} : A_0)(a_{101} : A_1\ a_{001}\ a_{100})(a_{110} : A_1\ a_{010}\ a_{100}) \to \mathsf{Type}
\end{aligned}$$

This suggests that in general, $X^{\partial n}$ will be the type of $(n-1)$-truncated semi-simplicial types, while $X^n\ A$ will be the type of ways to extend such an A to an n-truncated one, i.e. the types of indexed families of n-simplices. We will prove this formally in section 4.5.5.

It remains to give dCoind the structure from section 3.3. A type $C \in \mathsf{Ty}(\Gamma \,.\, \text{🔒}_{\triangle\square} \mid \Phi)$ is an $\widetilde{F}$-coalgebra when it has a section of the projection $\widetilde{F}(C) = (C \mid F(C)) \to C$, that is:

$$\Gamma,\ \text{🔒}_{\triangle\square} \mid (\phi : \Phi), x : C \vdash_{\mathsf{sm}} c : (a : A\ \phi,\ x' : (b : \mathcal{B}\ \phi\ a) \to X^{\mathsf{d}}\ \langle\, \phi \,,\, \sigma\ a\ b \,\rangle\ x).$$

This is equivalent to giving its components, which we abstract over x:

$$\Gamma,\ \text{🔒}_{\triangle\square} \mid (\phi : \Phi) \vdash_{\mathsf{sm}} h : C \to A\ \phi$$

$$\Gamma,\ \text{🔒}_{\triangle\square} \mid (\phi : \Phi) \vdash_{\mathsf{sm}} t : (x : C)(b : \mathcal{B}\ \phi\ (h\ x)) \to X^{\mathsf{d}}\ \langle\, \phi \,,\, \sigma\ (h\ x)\ b \,\rangle\ x$$

This is the structure of head and tail from section 3.3; thus dCoind admits these. Furthermore, to give any $\Theta \in \mathsf{Tel}(\Gamma \,.\, \text{🔒}_{\triangle\square} \mid \Phi)$ an $\widetilde{F}$-coalgebra structure is equivalent to giving $\Upsilon \equiv (\Phi \mid \Theta)$ the premises of the corecursor, where $\zeta : (\Phi \mid \Theta) \to \Phi$ is the dependent projection. In the next section we show that for more general $\zeta : \Upsilon \to \Phi$, the premises of the corecursor still equip Υ with *'enough of an $\widetilde{F}$-coalgebra structure'* to model the corecursor. ◁

4.5.3 *Terminal generalised coalgebras*

Let F be a copointed endofunctor of a category $\mathcal{C}$, where $\mathcal{C} \subseteq E$ is a full subcategory.

Definition 42. An object $Y \in E$ is a **generalised F-coalgebra** if it is equipped with, for any $X \in \mathcal{C}$ and $h : Y \to X$, a specified morphism $\overline{h} : Y \to FX$ such that $\epsilon_X \circ \overline{h} = h$ and for any $g : X \to Z$ in $\mathcal{C}$, we have $Fg \circ \overline{h} = \overline{g \circ h}$.

More abstractly, F induces a copointed endofunctor F^* of $\mathsf{Set}^{\mathcal{C}}$ by precomposition, and Y is a generalised F-coalgebra if the functor $E(Y, -) : \mathcal{C} \to \mathsf{Set}$ is an F^*-coalgebra.

Lemma 43. If $Y \in \mathcal{C}$, then generalised F-coalgebra structures on Y are bijective to ordinary F-coalgebra structures.

Proof. This follows immediately from the above observation and the Yoneda lemma. □

Of course, a **morphism of generalised F-coalgebras** is a morphism $f : Y \to Z$ such that for any $h : Z \to X \in \mathcal{C}$ we have $\overline{h} \circ f = \overline{h \circ f}$.

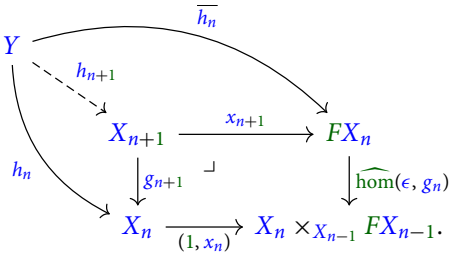
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Lemma 44. If $Y \in E$ is a generalised F-coalgebra and $x : X \to FX$ is an F-coalgebra in $\mathcal{C}$, then a morphism $f : Y \to X$ is a generalised F-coalgebra morphism if and only if $\overline{f} = x \circ f$.

Proof. If it is a generalised F-coalgebra map, then taking $h = 1_X$ in $\overline{h} \circ f = \overline{h \circ f}$ we get $x \circ f = \overline{f}$. On the other hand, if $x \circ f = \overline{f}$ then for any $h : X \to X'$ in $\mathcal{C}$ we have $\overline{h} \circ f = x' \circ h \circ f = Fh \circ x \circ f = Fh \circ \overline{f} = \overline{h \circ f}$, as desired. □

Theorem 45. Let $\mathcal{C}$ and F be as in theorem 36, and let $\mathcal{C}$ be a full subcategory of E such that the embedding preserves the terminal object and the inverse limits of ω-sequences of fibrations. Then the terminal F-coalgebra constructed in theorem 36 is also a terminal generalised F-coalgebra.

Proof. Indeed, the proof of theorem 36 really only uses the generalised F-coalgebra structure. Specifically, let $Y \in E$ be a generalised F-coalgebra. We construct inductively maps $h_n : Y \to X_n$ such that $x_{n+1} \circ h_{n+1} = \overline{h_n}$ and $g_{n+1} \circ h_{n+1} = h_n$. We start with $h_0 : Y \to X_0 = \mathbb{1}$ the unique morphism (since $\mathbb{1} \in \mathcal{C}$ is also terminal in E), and $h_1 = \overline{h_0} : Y \to X_1 = FX_0$. Then we induce h_{n+1} by the universal property of the pullback defining X_{n+1}:



This is valid because using the inductive assumptions about h_n and h_{n-1} and the properties of generalised coalgebras, we have: $\epsilon_{X_n} \circ \overline{h_n} = h_n$ and $Fg_n \circ \overline{h_n} = \overline{g_n \circ h_n} = \overline{h_{n-1}} = x_n \circ h_n$, and the two triangles relating to h_{n+1} show that it has the necessary properties.

Now $g_{n+1} \circ h_{n+1} = h_n$ yields an induced map $h_\infty : Y \to X_\infty$, such that $x_\infty \circ h_\infty$ is induced by $x_{n+1} \circ h_{n+1}$. But $x_{n+1} \circ h_{n+1} = \overline{h_n}$, and the morphisms Fh_n induce the limit map Fh_∞, so $x_\infty \circ h_\infty = \overline{h_\infty}$. Thus, by lemma 44, h_∞ is an F-coalgebra morphism.

Finally, suppose $k : Y \to X_\infty$ is any F-coalgebra morphism, so we have $x_\infty \circ k = \overline{k}$. Then k is uniquely determined by the maps $k_n : Y \to X_n$, and we have $x_{n+1} \circ k_{n+1} = \overline{k_n}$. But this equation implies by induction that $k_n = h_n$ for all n, hence $k = h_\infty$. □ ◁

4.5.4 The general corecursor

Suppose $\Upsilon \in \mathsf{Tel}(\Gamma \bullet \blacksquare_{\triangle\square})$ has the structure of the premises of the corecursor from section 3.3:

$$\Gamma, \blacksquare_{\triangle\square} \mid \upsilon : \Upsilon \vdash_{\mathsf{sm}} \zeta\ \upsilon : \Phi$$
$$\Gamma, \blacksquare_{\triangle\square} \mid \upsilon : \Upsilon \vdash_{\mathsf{sm}} h\ \upsilon : A\ (\zeta\ \upsilon)$$
$$\Gamma, \blacksquare_{\triangle\square} \mid \upsilon : \Upsilon \mid y : \mathcal{B}(\zeta\,\upsilon, h) \vdash_{\mathsf{sm}} \tau\ \upsilon\ y : \Upsilon^{\mathsf{d}}\ \upsilon$$
$$\Gamma, \blacksquare_{\triangle\square} \mid \upsilon : \Upsilon \mid y : \mathcal{B}(\zeta\,\upsilon, h) \vdash_{\mathsf{sm}} \zeta^{\mathsf{d}}\ \langle\, \upsilon\,,\, \tau\ \upsilon\ y \,\rangle = \sigma\ (\zeta\ \upsilon)\ (h\ \upsilon)\ y$$

Then ζ makes it an object of the slice category $(\mathsf{Tel} /\!\!/ (\Gamma \bullet \blacksquare_{\triangle\square}))/\Phi$. We will apply theorem 45 to the full subcategory $\mathsf{Tel} /\!\!/ (\Gamma \bullet \blacksquare_{\triangle\square} \mid \Phi) \subseteq (\mathsf{Tel} /\!\!/ (\Gamma \bullet \blacksquare_{\triangle\square}))/\Phi$. To that end, we give Υ the structure of a generalised $\widetilde{F}$-coalgebra as follows. Suppose $X \in \mathsf{Tel}(\Gamma \bullet \blacksquare_{\triangle\square} \mid \Phi)$, and suppose we have a

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map $g : \Upsilon \to X$ in $(\mathsf{Tel} /\!\!/ (\Gamma \centerdot \fatsemi_{\triangle\square}))/\Phi$, which is to say:

$$\Gamma,\ \fatsemi_{\triangle\square} \mid \upsilon : \Upsilon \vdash_{\mathsf{sm}} g\ \upsilon : X\ (\zeta\ \upsilon).$$

We want to lift g to FX, which is to say we want to give:

$$\Gamma,\ \fatsemi_{\triangle\square} \mid (\upsilon : \Upsilon) \vdash_{\mathsf{sm}} h\ \upsilon : A\ (\zeta\ \upsilon)$$

$$\Gamma,\ \fatsemi_{\triangle\square} \mid (\upsilon : \Upsilon), (b : \mathcal{B}\ \phi\ (h\ x)) \vdash_{\mathsf{sm}} t\ \upsilon\ b : X^{\mathsf{d}}\ \langle\ \zeta\ \upsilon\ ,\sigma\ (h\ (\zeta\ \upsilon)\ (g\ \upsilon))\ b\ \rangle\ (g\ \upsilon)$$

But such an h is exactly part of the structure of Υ, while we can define:

$$t\ \upsilon\ b \equiv g^{\mathsf{d}}\ \upsilon\ (\tau\ \upsilon\ b).$$

The final equation in the structure of Υ makes this well-typed, and functoriality is immediate from that of ^{d}. Thus Υ is a generalised $\widetilde{F}$-coalgebra, hence admits a unique generalised $\widetilde{F}$-coalgebra morphism to C. This is a generalized $\widetilde{F}$-coalgebra map $\Upsilon \to (\Phi \mid X)$ over ζ, which is precisely the right type of corec, and lemma 44 gives it the correct computation rules. ◁

4.5.5 *Correctness of semi-simplicial types*

Finally, we will show that when SST is constructed as in section 4.5.2, it yields a *'classifier'* of Reedy fibrant semi-simplicial types in the classical sense. We begin by constructing such a classifier category-theoretically, and then show that this construction coincides with the one in section 4.5.2. We will assume some familiarity with the classical notions of Reedy fibrant diagrams as in Kapulkin and Lumsdaine (2021). For all this section, we fix a particular level ℓ.

4.6 Ordered direct categories

Definition 46. A **direct category** is one such that the relation *'there is a nonidentity arrow from x to y'* on its objects is well-founded. A **sieve** in a (direct) category is a full subcategory J such that if $f : y \to x$ and $x \in J$, then $y \in J$. An **ordered direct category** is a *finite* direct category together with (1) a total ordering on its objects such that if $f : x \to y$ then $x \leq y$, and (2) such that for all objects x, the set of arrows with codomain x has a linear order such that $f \circ g \leq f$ for any composable f, g (hence in particular 1_x is the greatest element).

An **ordered presheaf** on a direct category is a finite presheaf together with a linear order on the finite set $\sum_{x \in I} H(x)$ such that $H(f)(h) < h$ whenever the left-hand side makes sense.

An ordered direct category is the opposite of a (finite) *'ordered inverse category'* as in (Kapulkin and Lumsdaine, 2021, Definition 3.17), plus the object ordering (to order the variables in the classifying context). An ordered presheaf is a *'finite extension'* $\emptyset \hookrightarrow H$ as in (Kapulkin and Lumsdaine, 2021, Definition 3.10).

Example 47. Let Δ_n be the subcategory of Δ^+ containing the objects $\langle k \rangle$ with $0 \leq k \leq n$. For fixed l we give these morphisms Campion's ordering, namely the usual ordering of binary numbers. Then Δ_n is an ordered direct category.

For $x \in I$ we write $\partial\!\lrcorner_x$ for the sub-presheaf of the representable $\lrcorner_x$ consisting of nonidentity morphisms, i.e. $\partial\!\lrcorner_x(y) = \{f \in I(y,x) \mid f \neq 1_x\} = \{f \in I(y,x) \mid y \prec x\}$.

If I is a finite direct category and H is a finite presheaf on it, there is a new finite direct category $I \oplus H$, called the **collage** of H, which contains I as a full subcategory, together with one new object $\star$ such that $I(x, \star) = H(x)$ for all $x \in I$. Note that $\partial\!\lrcorner_\star$ restricted to I coincides with H. Moreover, I and H are ordered if and only if $I \oplus H$ is. Moreover, if I is an ordered direct category of finite height with x its object of greatest rank, then $I \cong (I \setminus \{x\}) \oplus \partial\!\lrcorner_x$. Thus, we can treat this as an induction principle for ordered direct categories. ◁

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4.7 Classifying contexts

As our first use of this sort of induction, we construct by simultaneous induction:

(1) For each ordered direct category I, a context Γ^I. This will be the classifying context of Reedy fibrant I-types at level ℓ.
(2) For each ordered presheaf H on I, a telescope $\Gamma^I \vdash_{\mathsf{sm}} \Theta^H\ \mathsf{tel}_\ell$.
(3) For each map of ordered presheaves $\alpha : H \to H'$ (not necessarily order-preserving) on I, a partial substitution $\Gamma^I \vdash_{\mathsf{sm}} \theta^\alpha : \Theta^{H'} \to \Theta^H$, varying functorially.
(4) For each object $x \in I$, a type $\Gamma^I \mid \Theta^{\partial \pitchfork x} \vdash_{\mathsf{sm}} B^x\ \mathsf{type}_\ell$.
(5) For each $h \in H(x)$, inducing by the Yoneda lemma a map $\beta_h : \partial \pitchfork_x \subseteq \pitchfork_x \to H$, a term $\Gamma^I \mid \Theta^H \vdash_{\mathsf{sm}} b^h : B^x[\ \theta^{\beta_h}\ ]$, such that $b^h[\ \theta^\alpha\ ] = b^{\alpha(h)}$ for any $\alpha : H \to H'$.
(6) For each sieve $J \subseteq I$, a telescope $\Gamma^J \vdash_{\mathsf{sm}} \Gamma^{J,I}\ \mathsf{tel}_{\mathsf{lsuc}\,\ell}$ such that $\Gamma^I \cong \left(\Gamma^J \mid \Gamma^{J,I}\right)$ and for all the structure in 2–5, the action of the weakening substitution $\Gamma^I \cong \left(\Gamma^J \mid \Gamma^{J,I}\right) \to \Gamma^J$ corresponds to left Kan extension along the inclusion $J \hookrightarrow I$.

For item 1, we inductively use item 2 and set:

$$\Gamma^\emptyset \equiv () \qquad\qquad \Gamma^{I \oplus H} \equiv \left(\Gamma^I,\ A_\star : \Theta^H \to \mathsf{Type}_\ell\right).$$

For item 2, we argue inductively on the linear ordering of H. If H is empty, we set:

$$\Theta^\emptyset \equiv ().$$

Otherwise, $H = (H \setminus \{h\}) \cup \{h\}$ where $h \in H(x)$ is the last element in the ordering; the condition on the ordering ensures that $H \setminus \{h\}$ is still an (ordered) presheaf. By the Yoneda lemma, h induces a map $\beta_h : \partial \pitchfork_x \subseteq \pitchfork_x \to H \setminus \{h\}$, hence by item 3 a substitution $\Gamma^I \vdash_{\mathsf{sm}} \theta^{\beta_h} : \Theta^{H \setminus \{h\}} \to \Theta^{\partial \pitchfork x}$. Thus, inductively using item 4 as well, we can define:

$$\Theta^H = \left(\Theta^{H \setminus \{h\}},\ a_h : B^x[\ \theta^{\beta_h}\ ]\right).$$

We similarly construct item 3 by induction on H (the domain of α). The case when H is empty is trivial. Otherwise, we inductively have $\theta^{\alpha \setminus \{h\}} : \Theta^{H'} \to \Theta^{H \setminus \{h\}}$, and to extend the codomain to Θ^H it suffices to give a term in context $\Gamma^I \mid \Theta^{H'}$ of type:

$$B^x[\ \theta^{\beta_h}\ ][\ \theta^{\alpha \setminus \{h\}}\ ] \equiv B^x[\ \theta^{\beta_h} \circ \theta^{\alpha \setminus \{h\}}\ ] \equiv B^x[\ \theta^{\beta_{\alpha(h)}}\ ].$$

For this we can pick $b^{\alpha(h)}$, using item 5 inductively. Functoriality follows inductively.

For item 4, note that the slice category I/x is a sieve in I containing x. Then it suffices to define B^x in the case of I/x, since it can then be weakened to I using item 6. In this case we have $I/x = (I/x \setminus \{x\}) \oplus \partial \pitchfork_x$, so the last variable in $\Gamma^{I/x}$ is $A_x : \Theta^{\partial \pitchfork x} \to \mathsf{Type}_\ell$. Thus, we can define $\Gamma^{I/x} \mid y : \Theta^{\partial \pitchfork x} \vdash_{\mathsf{sm}} B^x$ to be $\Gamma^{I/x \setminus \{x\}}, A_x : \Theta^{\partial \pitchfork x} \to \mathsf{Type}_\ell \mid y : \Theta^{\partial \pitchfork x} \vdash_{\mathsf{sm}} A_x\ y$.

For item 5, it suffices to deal with the case when h is the last element in the ordering of H, since otherwise we can weaken from the sub-presheaf of all elements $\leq h$ to all of H, using item 3 for the inclusion of this sub-presheaf. But in this case, the last variable in Θ^H is $a_h : B^x[\ \theta^h\ ]$, so we can take $b^h \equiv a_h$. Functoriality follows immediately, as does stability under weakening from initial segments for all the data.

For item 6 we induct on I. A sieve in $I \oplus H$ could be either J or $J \oplus H$ for some sieve J in I, depending on whether it contains $\star$. (If it contains $\star$, it must contain all objects y such that $H(y) \neq \emptyset$, so H must be left Kan extended from J.) In these two cases, we define:

$$\Gamma^{J,\, I \oplus H} \equiv \left(\Gamma^{J,I},\ A_\star : \Theta^H \to \mathsf{Type}_\ell\right)$$
$$\Gamma^{J \oplus H,\, I \oplus H} \equiv \Gamma^{J,I} \quad \text{weakened to } \Gamma^{J \oplus H}.$$

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This completes the construction. In particular, item 3 implies that re-ordering the elements of a presheaf H modifies Θ^H only up to isomorphism. ◁

4.8 The classifying context is classifying

We first construct a *'universal'* diagram over Γ^I. Specifically, in any CwF, we construct simultaneously:

(1) For each ordered direct category I, a Reedy type B of shape I and level ℓ over Γ^I in the sense of (Kapulkin and Lumsdaine, 2021, Definition 3.22).
(2) For each ordered presheaf H on I, the object Θ^H is the H-weighted limit of B constructed by (Kapulkin and Lumsdaine, 2021, Lemma 3.11). In particular, $\Theta^{\partial\downarrow x}$ is the matching object $M_X B$.
(3) The maps θ^α are the functorial action of these limits.
(4) The type $\Gamma^I \mid \Theta^{\downarrow x} \vdash_{\mathsf{sm}} B^x$ is the object $B(x)$ with its fibration to $M_x B = \Theta^{\partial\downarrow x}$.
(5) The elements b^x are the projections from the weighted limit Θ^H.

The interesting case is item 1, where we weaken the Reedy I-type B over Γ^I to $\Gamma^{I\oplus H}$ and then must extend it to a Reedy $(I \oplus H)$-type by giving a type over the matching object $\Theta^{\partial\downarrow\star}$. But $\downarrow_\star = H$ (weakened to $I \oplus H$), and so we can use $\mathsf{El}\, A^H$ where A^H is the newly added variable in $\Gamma^{I\oplus H}$. The other parts follow essentially tautologically.

Lastly, suppose C is a Reedy I-type over any context Δ, and suppose that it is *'ℓ-small'* in the sense that each type $C(x)$ over the matching object $M_x C$ is classified by a specified map into the universe $M_x C \to \mathsf{Type}_\ell$. We show by induction on I that it is classified by a unique map $c : \Delta \to \Gamma^I$ such that $B[\, c\,] \equiv C$. This is trivial when I is empty. Assuming it to be true for I, if C is a Reedy $(I \oplus H)$-type over Δ, and its restriction to an I-type is classified by a map $c : \Delta \to \Gamma^I$, then to extend this to a map into $\Gamma^{I\oplus H}$ we must give a term in context Δ of type $\Theta^H[\, c\,] \to \mathsf{Type}_\ell$. But substitution preserves matching objects, so $\Theta^H[\, c\,]$ is the matching object $M_\star C$, and thus this is exactly the data extending C to a Reedy $(I \oplus H)$-type. ◁

4.9 Display and décalage of classifying contexts

Since Γ^I, Θ^H, B^x, and so on are concrete syntactic objects (for any fixed I, H, x and so on), the computation rules completely determine display and décalage on them. We can characterise the results as follows.

Let 2 denote the interval category $(0 \xrightarrow{\xi} 1)$. Then if I is a direct category, so is $2 \times I$ where $x \prec y$ yields $(0, x) \prec (1, x) \prec (0, y) \prec (1, y)$. If I is ordered, we order $2 \times I$ as follows. The morphisms in $2 \times I$ with codomain $(0, x)$ are bijective to those in I with codomain x, so we inherit that ordering. And the morphisms in $2 \times I$ with codomain $(1, x)$ are two copies of those in I with codomain x indexed by 1_1 and ξ respectively, so we order them where $g \prec f$ yields $(\xi, g) \prec (1_1, g) \prec (\xi, f) \prec (1_1, f)$.

The projection $p : 2 \times I \to I$ induces by precomposition from any presheaf H on I a presheaf p^*H on $2 \times I$. Each element $h \in H(x)$ then induces two elements p^*H in $H((0, x))$ and $H((1, x))$; we denote these $(0, h)$ and $(1, h)$ respectively for clarity. If H is ordered, we induce an ordering on p^*H by ordering each element of $H((0, x))$ before the corresponding element of $H((1, x))$ (which is necessary, since ξ maps the latter to the former).

The inclusion $i_0 : I \to 2 \times I$ defined by $i_0(x) = (0, x)$ is a sieve. Left Kan extending along this inclusion takes a presheaf H on I to a presheaf $(i_0)_! H$ on $2 \times I$ that is supported on objects $(0, x)$ and has exactly the same elements, hence inherits an ordering as well.

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Now we prove by simultaneous induction:

(1) For any I, we have $(\Gamma^I)^{\mathsf{D}} \equiv \Gamma^{2\times I}$.
(2) For any presheaf H on I, $(\Theta^H)^{\mathsf{D}} \equiv \Theta^{p^*H}$ and $\gamma' : \Gamma^{2\times I} \vdash_{\mathsf{sm}} \Theta^{(i_0)_! H}\ \gamma' \equiv \Theta^H\ \gamma'^{\mathsf{ev}}$.
(3) This identification is functorial in maps $\alpha : H \to H'$.
(4) For each $x \in I$, we have:

$$\gamma' : \Gamma^{2\times I} \vdash_{\mathsf{sm}} \Theta^{\partial\!\downarrow(0,x)}\ \gamma' \equiv \Theta^{\partial\!\downarrow x}\ \gamma'^{\mathsf{ev}}$$
$$\gamma' : \Gamma^{2\times I},\ y : \Theta^{\partial\!\downarrow(0,x)}\ \gamma' \vdash_{\mathsf{sm}} B^{(0,x)}\ \gamma'\ y \equiv B^x\ \gamma'^{\mathsf{ev}}\ y$$
$$\partial\!\downarrow_{(1,x)} = p^*\partial\!\downarrow_x \cup \{(\xi, 1_x)\}$$
$$\gamma' : \Gamma^{2\times I} \vdash_{\mathsf{sm}} \Theta^{\partial\!\downarrow(1,x)}\ \gamma' \equiv \left(y' : (\Theta^{\partial\!\downarrow x})^{\mathsf{D}}\ \gamma',\ a_{(0,x)} : B^x\ \gamma'^{\mathsf{ev}}\ y'^{\mathsf{ev}}\right)$$
$$\gamma' : \Gamma^{2\times I},\ y' : \Theta^{\partial\!\downarrow(1,x)}\ \gamma' \vdash_{\mathsf{sm}} B^{(1,x)}\ \gamma'\ y' \equiv (B^x)^{\mathsf{d}}\ \gamma'\ y'$$

(5) For each $h \in H(x)$, we have:

$$\gamma' : \Gamma^{2\times I},\ y' : \Theta^{p^*H} \vdash_{\mathsf{sm}} b^{(0,h)}\ \gamma'\ y' \equiv b^h\ \gamma'^{\mathsf{ev}}\ y'^{\mathsf{ev}}$$
$$\gamma' : \Gamma^{2\times I},\ y' : \Theta^{p^*H} \vdash_{\mathsf{sm}} b^{(1,h)}\ \gamma'\ y' \equiv (b^h)^{\mathsf{d}}\ \gamma'\ y'$$

For the inductive step of item 1, we have:

$$2 \times (I \oplus H) = (2 \times I) \oplus (i_0)_! H \oplus (p^*H \cup \{(\xi, 1_\star)\}).$$

Thus, using item 2, we have:

$$\begin{aligned}\Gamma^{2\times(I\oplus H)} &\equiv \left(\gamma' : \Gamma^{(2\times I)},\ A_{(0,\star)} : \Theta^{(i_0)_! H} \to \mathsf{Type}_\ell,\ A_{(1,\star)} : \Theta^{(p^*H \cup \{(\xi,1_\star)\})} \to \mathsf{Type}_\ell\right)\\ &\equiv \left(\gamma' : (\Gamma^I)^{\mathsf{D}},\ A_{(0,\star)} : \Theta^H\ \gamma'^{\mathsf{ev}} \to \mathsf{Type}_\ell,\ A_{(1,\star)} : (y' : (\Theta^H)^{\mathsf{D}}\ \gamma') \to A_{(0,\star)}\ y'^{\mathsf{ev}} \to \mathsf{Type}_\ell\right)\\ &\equiv \left(\gamma : \Gamma^I,\ A_\star : \Theta^H\ \gamma \to \mathsf{Type}_\ell\right)^{\mathsf{D}}.\end{aligned}$$

The other cases are similar. We can likewise show $\Gamma^{I,2\times I} \equiv (\Gamma^I)^{\mathsf{d}}$, the isomorphism $\Gamma^{2\times I} \cong \left(\Gamma^I \mid \Gamma^{I,2\times I}\right)$ coinciding with the evens/odds pairing $(\Gamma^I)^{\mathsf{D}} \cong \left(\Gamma^I \mid (\Gamma^I)^{\mathsf{d}}\right)$. ◁

4.10 Discrete fibrations

The isomorphism $\Gamma^I \cong \left(\Gamma^J \mid \Gamma^{J,I}\right)$ ensures that if $J \subseteq I$ is a sieve, we have a weakening substitution $\Gamma^I \to \Gamma^J$. But more generally, we can expect to induce a context substitution from any discrete fibration. Even more generally, we can get a *partial* substitution from a *'dependent'* discrete fibration, in the following sense.

Definition 48. If $i : J \hookrightarrow I$ is the inclusion of a sieve in a direct category, a **co-section** of it is a discrete fibration $p : I \to J$ such that $p \circ i = 1_J$. In this case, if H is a presheaf on I and K a presheaf on J, a morphism $H \to K$ over p is a **relative isomorphism** if it induces a bijection $\sum_{y\in I} H(y) \to \sum_{y\in J} K(y)$.

The prototypical relative isomorphism is $\partial\!\downarrow_x \to \partial\!\downarrow_{p(x)}$ for any $x \in I$. Now we define and prove inductively:

(1) For any co-section $p : I \to J$ of a sieve $i : J \hookrightarrow I$ in an ordered direct category, a partial substitution $\Gamma^J \vdash_{\mathsf{sm}} \gamma^p : \Gamma^{J,I}$.
(2) For any order-preserving relative isomorphism $H \to K$, we have $\Theta^H[\ \gamma^p\ ] \equiv \Theta^K$.

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(3) For $x \in I$, we have $B^x[\ \gamma^p\ ] \equiv B^{p(x)}$.
(4) For $h \in H(x)$, and $\alpha : H \to K$ an order-preserving relative iso, $b^x[\ \gamma^p\ ] \equiv b^{\alpha(x)}$.

For item 1, as before a sieve in $I \oplus H$ can be either J or $J \oplus H$ for a sieve J in I. In the latter case, we have $\Gamma^{J\oplus H,\, I\oplus H} = \Gamma^{J,I}$ weakened to $\Gamma^{J\oplus H}$, and a co-section of $J \oplus H \hookrightarrow I \oplus H$ is determined by a co-section p of $J \subseteq I$; thus we can similarly weaken γ^p. In the former case, a co-section $I \oplus H \to J$ is determined by a co-section $p : I \to J$ with $x \in J$ and a relative isomorphism $H \to \partial\!\downarrow_x$. Since $\Gamma^{J,\, I\oplus H} = \left(\Gamma^{J,I},\ A_\star : \Theta^H \to \mathsf{Type}_\ell\right)$, to extend γ^p as desired it suffices to give a term of type $\Gamma^J \vdash_{\mathsf{sm}} \Theta^H[\ \gamma^p\ ] \to \mathsf{Type}_\ell$. But using item 2 inductively, this is equal to $\Gamma^J \vdash_{\mathsf{sm}} \Theta^{\partial\downarrow_x} \to \mathsf{Type}_\ell$, so we can use the variable A_x in Γ^J.

We can then prove items 2–4 by inducting on the ordering of H and K. ◁

4.11 Categorical coning

Let $J \subseteq I$ be a sieve in a direct category that contains the bottom object, which we denote $\langle 0\rangle$. Let I^+ be I augmented by a new morphism $\zeta_x : \langle 0\rangle \to x$ for all objects $x \in I \setminus J$. (The notation is somewhat abusive, since the construction depends on J as well as I.) We define $f \circ \zeta_x = \zeta_y$ for all $f : x \to y$; note that $x \in I \setminus J$ implies $y \in I \setminus J$ since J is a sieve. If I is ordered, we order I^+ by placing ζ_x *before* all other morphisms with codomain x. Note that J is still a sieve in I^+. Similarly, for a presheaf H on I, let H^+ denote the presheaf on I^+ consisting of H plus by a new element $\zeta_H \in H(\langle 0\rangle)$, such that $H^+(\zeta_x)(h) = \zeta_H$ for all $h \in H(x)$. If H is ordered, we order H^+ by putting ζ_H first.

We now inductively prove:

(1) For any sieve $J \subseteq I$ in an ordered direct category, we have $\Gamma^{J,I^+} \equiv (z : B^{\langle 0\rangle}) \to \Gamma^{J,I}$ (meaning a Π-telescope).
(2) In addition, for any H on I, if we transfer Θ^H and Θ^{H^+} across the isomorphisms:

$$\Gamma^I \cong \left(\gamma : \Gamma^J \mid \delta : \Gamma^{J,I}\ \gamma\right)$$
$$\Gamma^{I^+} \cong \left(\gamma : \Gamma^J \mid \delta : \Gamma^{J,I^+}\ \gamma\right) \equiv \left(\gamma : \Gamma^J \mid \delta : (z : B^{\langle 0\rangle}\ \gamma) \to \Gamma^{J,I}\ \gamma\right)$$

to get $\tilde\Theta^H$ and $\tilde\Theta^{H^+}$, then we have: $\tilde\Theta^{H^+}\ \gamma\ \delta \equiv \left(z : B^{\langle 0\rangle}\ \gamma,\ \tilde\Theta^H\ \gamma\ \ (\delta\ z)\right)$.

Both are straightforward, using the inductive definition of Π-telescopes, Γ^I, and Θ^H. ◁

4.12 Correctness of semi-simplicial types

Recall that our definition of semi-simplicial types SST_ℓ is as a displayed coinductive type with $\Phi \equiv ()$, $A \equiv \mathsf{Type}_\ell$, and $\mathcal{B}\ a \equiv (x : \mathsf{El}\ a)$. Therefore, the construction in section 4.5.2 simplifies as follows:

$$\vdash_{\mathsf{sm}} X^{\partial n}\ \mathsf{tel} \qquad \partial x : X^{\partial n} \vdash_{\mathsf{sm}} X^n\ \partial x\ \mathsf{tel} \qquad \vdash_{\mathsf{sm}} X^{\partial 0} \equiv () \qquad \vdash_{\mathsf{sm}} X^{\partial(n+1)} \equiv (\partial x : X^{\partial n},\ x : X^n\ \partial x)$$

$$\vdash_{\mathsf{sm}} X^0 \equiv \mathsf{Type}_\ell \qquad \partial x : X^{\partial n},\ x : X^n\ \partial x \vdash_{\mathsf{sm}} h_n\ \partial x\ x : \mathsf{Type}_\ell \qquad x : X^0 \vdash_{\mathsf{sm}} h_0\ x \equiv x$$

$$\partial x : X^{\partial n},\ x : X^n\ \partial x,\ x' : X^{n+1}\ [\ \partial x, x\ ] \vdash_{\mathsf{sm}} h_{n+1}\ [\ \partial x,\ x\ ]\ x' \equiv h_n\ \partial x\ x$$

$$\partial x : X^{\partial n},\ x : X^n\ \partial x,\ b : \mathsf{El}\ (h_n\ \partial x\ x) \vdash_{\mathsf{sm}} t_n\ \partial x\ x\ b : (X^{\partial n})^{\mathsf{d}}\ \partial x$$

$$x : X^0,\ b : \mathsf{El}\ (h_0\ x) \vdash_{\mathsf{sm}} t_0\ x\ b \equiv [\ ]$$

$$\partial x : X^{\partial n},\ x : X^n\ \partial x,\ x' : X^{n+1}\ [\ \partial x, x\ ],\ b : \mathsf{El}\ (h_n\ \partial x\ x) \vdash_{\mathsf{sm}} t_{n+1}\ [\ \partial x, x\ ]\ x'\ b \equiv [\ t_n\ \partial x\ x\ b,\ x'\ b\ ]$$

$$\partial x : X^{\partial n},\ x : X^n\ \partial x \vdash_{\mathsf{sm}} X^{n+1}\ [\ \partial x,\ x\ ] \equiv (b : \mathsf{El}\ (h_{n-1}\ \partial x)) \to (X^n)^{\mathsf{d}}\ \langle\ \partial x\ ,\ t_n\ \partial x\ x\ b\ \rangle\ x$$

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We will prove inductively that:

$$X^{\partial n} \equiv \Gamma^{\Delta_{n-1}} \quad \text{and} \quad X^n \equiv \Theta^{\partial\!\!\downarrow\langle n\rangle} \to \mathsf{Type}_\ell.$$

This will imply that $\mathsf{SST} = \lim_n X^{\partial n}$ is a classifying context for all of Δ. The claim about X^n clearly inductively implies the claim about $X^{\partial n}$. Also it is easy to show inductively that $h_n \equiv B^{\langle 0\rangle}$. So it remains to say something useful about t_n.

Let I_n be the subcategory of $\mathbb{2} \times \Delta_n$ containing all the objects except $(1, \langle n\rangle)$, and let $J_n = \{0\} \times \Delta_n$ regarded as a sieve in I_n. The central fact is the following.

Lemma 49. For any n, there is a co-section $q_n : {I_n}^+ \to J_n$.

Proof. On objects, let $q_n((1, \langle k\rangle)) = (0, \langle k+1\rangle)$ for $0 \le k < n$. A morphism $(1, \langle k\rangle) \to (1, \langle l\rangle)$ is a length $l+1$ sequence with $k+1$ $\mathbb{1}$s, and we augment it by another $\mathbb{1}$ on the *right* to get a length $l+2$ sequence with $k+2$ $\mathbb{1}$s, hence a morphism $(0, \langle k+1\rangle) \to (0, \langle l+1\rangle)$. A morphism $(0, \langle k\rangle) \to (1, \langle l\rangle)$ is also a length $l+1$ sequence with $k+1$ $\mathbb{1}$s, but this time we augment it by a $\mathbb{0}$ on the right to get a length $l+2$ sequence with $k+1$ $\mathbb{1}$s, hence a morphism $(0, \langle k\rangle) \to (0, \langle l+1\rangle)$. Finally, we send the new morphism $\zeta_{(1,\langle l\rangle)}$ to the sequence of $l+1$ $\mathbb{0}$s followed by one $\mathbb{1}$. Functoriality is easy to check. And to see that it is a discrete fibration, we observe that any binary sequence of length $l+2$ with a positive number of $\mathbb{1}$s must be of exactly one of these three forms: a positive number of $\mathbb{1}$s followed by a $\mathbb{1}$, a positive number of $\mathbb{1}$s followed by a $\mathbb{0}$, or a sequence of $\mathbb{0}$s followed by a $\mathbb{1}$. □

Evidently q_{n+1} restricts to q_n as we shrink the categories. Thus, we also get a relative isomorphism $\partial\!\!\downarrow^+_{(1,\langle n\rangle)} \to \partial\!\!\downarrow_{(0,\langle n+1\rangle)}$ over q_n. Now note that if we abstract over b, the type of t_n matches that of γ^{q_n}. Thus, we can now prove by simultaneous induction that:

(1) $X^{\partial n} \equiv \Gamma^{\Delta_{n-1}}$.
(2) $X^n \equiv \Theta^{\partial\!\!\downarrow\langle n\rangle} \to \mathsf{Type}_\ell$.
(3) $h_n \equiv B^{\langle 0\rangle}$.
(4) $t_n \equiv \gamma^{q_n}$.

We have already remarked that items 1 and 3 are easy, and the base cases of items 2 and 4 are likewise trivial. For the induction step of item 2, we have:

$$\begin{aligned} X^{n+1}\ [\ \partial x,\ x\ ] &\equiv (b : \mathsf{El}\ (h_{n-1}\ \partial x)) \to (X^n)^{\mathsf{d}}\ \langle\ \partial x\ ,\ t_n\ \partial x\ x\ b\ \rangle\ x \\ &\equiv (b : B^{\langle 0\rangle}\ \partial x) \to \Theta^{\partial\!\!\downarrow(1,\langle n\rangle)}\ \langle\ \partial x\ ,\ t_n\ \partial x\ x\ b\ \rangle\ x \to \mathsf{Type}_\ell \\ &\equiv \Theta^{\partial\!\!\downarrow^+_{(1,\langle n\rangle)}}\ [\ \gamma^{q_n}\ ] \to \mathsf{Type}_\ell \\ &\equiv \Theta^{\partial\!\!\downarrow(1,\langle n+1\rangle)}\ [\ \partial x,\ x\ ] \to \mathsf{Type}_\ell. \end{aligned}$$

Finally, the induction step of item 4 follows from the definition of γ^p and the inductive hypothesis of item 2. This completes the proof of the correctness of SST. ◁

5. Conclusion and Future Work

In this paper we have made two main contributions. First, we have described *Displayed Type Theory (dTT)*, a new kind of type theory that incorporates (unary) internal parametricity but guarded by a modality, and showed that any model of dependent type theory with countable Reedy limits can be lifted to a model of dTT using augmented semi-simplicial diagrams. Because the latter are diagrams on an *inverse* category, their type theory is more closely related to that of the original model, and indeed the original model sits inside our model of dTT at the discrete mode. In particular, unlike other internally parametric type theories, dTT is compatible with

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classical axioms such as excluded middle and choice, as long as they are formulated at the discrete mode (or under the modality $\Diamond$), and can be used as an internal logic to reason about arbitrary $(\infty, 1)$-toposes.

Secondly, inside dTT we have introduced a notion of *displayed coinductive type*, where the output of a destructor can be a parametricity *'computability witness'* of the input, and showed that as a particular case of this notion we can define a type of *semi-simplicial types*. This yields a new approach to the long-standing open problem of representing infinitely coherent higher structures in type theory. Relative to other approaches, ours has the advantage that semi-simplicial types are defined (not postulated) as a simple instance of a type-former with natural introduction and elimination rules, i.e. a categorical universal property. While it remains to be seen how much can actually be done in practice with our definition, early indications of its utility are promising.

There are a number of directions for future work suggested by our results.

5.1 Computation and implementation

We conjecture that dTT, including the type SST of section 3.1 and the displayed coinductive types of section 3.3, satisfies canonicity, normalisation, and decidable typechecking. Indeed, while we have presented dTT only as a Generalised Algebraic Theory, all the equations are directed and there are no obvious stuck terms. Therefore, it should be possible to implement a proof assistant for dTT.

Some evidence in this direction is that, as mentioned in section 3, the second author has implemented a prototype proof assistant called `Narya` with a mode that makes its type theory similar to dTT. The major differences are that there are no modalities (instead, typechecking display clears the entire context), there are symmetries, and the computation rules for display only hold up to definitional isomorphism. Although `Narya` includes a normalisation algorithm, this algorithm is not yet proved correct; but the related proof of canonicity in Altenkirch et al. (2024) provides additional positive evidence for our conjecture.

5.2 Modal internal parametricity

We expect that most applications of ordinary internal parametricity have modal versions that can be proven in dTT (or a higher-ary version of it, as discussed below). In addition to the traditional *'free theorems'*, it would be interesting to investigate this for the proof of the pentagon identity for the smash product in Cavallo (2021), since such a proof would then apply internally to any $(\infty, 1)$-topos.

5.3 Computing diamond

Our intended semantic model strictly validates computation rules for diamond, such as $\Diamond\ (A \to B) \equiv \Diamond\ A \to \Diamond\ B$. The syntax of dTT could be augmented with similar computation rules for diamond, by extending the definition of diamond to telescopes and meta-abstractions to handle open terms under binders.

5.4 Higher category theory

We have *defined* a type of semi-simplicial types in dTT, but such a definition is not an end in itself; it is intended as a tool for developing a theory of higher categories and other higher structures. We hope that our corecursion principle of SST, and the availability of other displayed coinductive types for predicates and structures on them, should make the development of such a theory feasible in dTT. We sketched some initial ideas in sections 3.2 and 3.4, but much remains to be done.

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5.5 Elementary models

Our model construction in section 4 using ω-limits is probably not the only way to construct models of dTT. We conjecture there are *'realisability'* models of dTT in which SST is a classifier for *'uniform'* semi-simplicial types. If so, displayed coinductive types might be useful to include in a definition of elementary $(\infty, 1)$-topos.

5.6 Higher-ary dTT

The parametricity of dTT is *unary*, meaning A^{d} depends on one copy of A; but parametricity can be n-ary for any natural number n. We expect that higher-ary versions of dTT can be defined and modeled by a straightforward modification of this paper, using higher-ary semi-cubical types in place of augmented semi-simplicial types. In this case the binary numbers $\mathbb{B}$ become base-$(n+1)$ numbers.

5.7 Symmetries

In non-modal internal parametricity and Higher Observational Type Theory, it appears necessary to include a *'symmetry'* operation, which in our notation would have type $A^{\mathsf{dd}}\ a_{00}\ a_{01}\ a_{10} \to A^{\mathsf{dd}}\ a_{00}\ a_{10}\ a_{01}$. The absence of symmetry in dTT is a significant simplification; e.g. it means that $\Delta^{+\mathsf{op}}$ is a strict inverse category, allowing the explicit model construction in section 4.2. However, it also leads to certain limitations, e.g. without symmetry it is unclear how to give a corecursion principle for $\mathsf{SST}^{\mathsf{d}}$. It should be possible to add symmetries to dTT, but in the presence of symmetry it is unclear whether it is possible for display to compute definitionally on type-formers. However, it should work to use either the interval-based style of Bernardy and Moulin (2012), Bernardy (2015), Moulin (2016) or the *'observational'* style of Altenkirch et al. (2024).

5.8 Unimode dTT

We have formulated dTT with two modes, but intuitively the discrete mode is unnecessary, as the dm-types are embedded in the sm-types by $\triangle$. Thus, it should be possible to formulate a single-moded version of dTT, similarly to spatial/cohesive type theory Shulman (2018) and synthetic guarded domain theory Gratzer et al. (2021).

5.9 Conjectural syntax

One may also consider other kinds of generalised inductive and coinductive types, especially when taking a more *'synthetic'* approach to higher structures in dTT, using sm-types directly as augmented semi-simplicial objects instead of SST.

Firstly, regarding display as analogous to paths in homotopy type theory suggests *displayed inductive types* as analogues of higher inductive types, where the constructors generate displayed elements. As an example, we can construct the simplicial cone of any type:

```
data C (A :^△□ Type) : Type where
  ι : A → C A
  σ : (x : A) → (C A)^d (ι x)

f : C A → B
f (ι x) = ?_ι : B
f^d (ι x) (σ x) = ?_σ : B^d ?_ι
```

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Secondly, regarding both display and paths as modalities suggests more general *modal inductive types*, whose constructors can land in modal versions of the type. For instance, a $\Diamond$-modal constructor adds a (-1)-simplex without any higher simplices above it. In this way we can construct the free-living simplices:

```
data Δ⁻¹ : Type where
  ⋆ : ◇ Δ⁻¹

f : Δ⁻¹ → A
◇ f ⋆ = ?⋆ : ◇ A

Δ : ℕ → Type
Δ zero = C Δ⁻¹
Δ (suc n) = C (Δ n)
```

In both cases, we rely on the computation of ^{d} and $\Diamond$ to directly give induction principles. For example, the pattern match $\Diamond\ f\ a$ requires that $\Diamond$ of a function type compute to a function type. This is an improvement on the treatment of higher inductive types in Univalent Foundations Program (2013) using `ap` (a compound expression defined by path induction). In particular, we conjecture that in dTT, these displayed and modal inductive types can be made fully computational. ◁

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